HyperBlock III DC3.0

User Manual

EU Version

(V1.4)

HyperStrong Technology Co., Ltd

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Preface

Objectives

This technical document applies to the following equipment (20Ft Battery Energy Storage System):

- HSL3702-05015EU
- HSL3702-04597EU
- HSL3702-04179EU
- HSL3702-03761EU
- HSL3702-03343EU

Symbol Conventions

The symbols that may be found in this document are defined as below.

Symbol	Description
DANGER	Indicates a hazard with a high level of risk which, if not avoided, will result in death or serious injury.
WARNING	Indicates a hazard with a medium level of risk which, if not avoided, could result in death or serious injury.
CAUTION	Indicates a hazard with a low level of risk which, if not avoided, could result in minor or moderate injury.
NOTICE	Indicates a potentially hazardous situation which, if not avoided, could result in equipment damage, data loss, performance deterioration, or unanticipated results. NOTICE is used to address practices not related to personal injury.

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NOTE	Supplements the important information in the main text. NOTE is used to address information not related to personal injury, equipment damage, and environment deterioration.

Change History

Updates between document versions are cumulative. Therefore, the latest document version contains all updates made to previous versions.

Issue 1.4 (2026-07-06)

- Simplified the BA controller diagram
- Full chapter description upgrade for fire protection system

Issue 1.3 (2026-04-15)

Revised symbol description, changed from Q1 to Q11.

Issue 1.2 (2026-03-20)

Changed the Item UPS as “Optional”

Issue 1.1 (2026-03-09)

Updated the water radiator information and add a description of the auxiliary power supply scheme.

Issue 1.0 (2026-02-27)

Initial commercial release.

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1 Safety Instructions

1.1 Safety Signs

Safety signs on the energy storage system equipment contain essential information for safe operations. Manual removal or damage is strictly prohibited.

Table 1-1 shows common safety signs.

Signs	Explanation
	It indicates that the area contains fire-hazardous materials, and pyrotechnics are strictly prohibited in this location.
	It indicates that the equipment in the area shall not be disassembled.
	It indicates that the area contains fire-hazardous materials, and smoking is prohibited in this location.
	It indicates that the device poses a danger of entrapment of body parts and that inserting body parts into the machine is prohibited.
	It indicates that the area prohibits the stacking of items.
	It indicates that all personnel operating the equipment shall wear protective gloves to prevent electric shock or static electricity.
	It indicates that the equipment needs to be grounded. The equipment shall be grounded through the grounding point, ensuring regular and safe operation.
	It indicates an immediate hazardous situation that, if not avoided, will result in serious personal injury or death.

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	It indicates a flammable substance that, if not avoided, could result in severe injury or death.
	It is used to convey safety warning information about equipment or the environment. It may lead to equipment damage, data loss, performance degradation, or other unpredictable results if not avoided.
	It indicates that cables have been placed in this area and you need to be aware of the cables underneath your feet when servicing or passing through the site to prevent tripping.
	It indicates a risk of electric shock from this equipment or device. Operators should take care while operating this equipment or device to avoid electrocution.
	It indicates that the machinery of the equipment may pose a risk of injury and that care should be taken when operating the equipment to prevent injury from the machinery.
	Do not dispose of this product as household waste.
	Read the instructions before performing any operation on the product.
	It is required to wear goggles.
	Ear protection shall be worn.
	In case of contact with the eyes, rinse the eyes immediately with running water or normal saline and seek medical help in time.
	Live parts! Please do not touch them until 5 minutes after disconnecting from the power sources.
 	Emergency Exit:It indicates an emergency safety exit in the direction of the arrow of movement and that you can escape in the order given in the event of an emergency danger.

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1.2 General Safety Precautions

The safety precautions outlined in this document provide essential guidance for selecting measuring and test equipment during the installation, operation, and maintenance of the equipment.

1.2.1 Statement

Before transporting, storing, installing, operating, using, or maintaining the equipment, carefully read this document, adhere strictly to the instructions provided, and follow all the safety instructions indicated on the equipment and within this document.

For the purpose of this document:

- "Equipment" refers to the products, software, components, spare parts, and/or services related to this document.
- "The Company" refers to the manufacturer (producer), seller, and/or service provider of the equipment.
- "You" refers to the entity that transports, stores, installs, operates, uses, and/or maintains the equipment. The Danger, Warning, Caution, and Notice statements included in this document are supplementary to, but do not encompass, all necessary safety precautions. Operators shall also comply with relevant international, national, and regional safety standards.
- The Company shall not be liable for any consequences resulting from violations of safety requirements or safety standards concerning the design, production, and usage of the equipment.
- The equipment shall be operated in an environment that meets the design specifications. Use in non-compliant environments may result in faulty, malfunctioning, or damaged, which are not covered under warranty. The Company shall not be liable for any property loss, personal injury, or even death caused thereby.
- Ensure compliance with all applicable laws, regulations, standards, and specifications during transportation, storage, installation, operation, use, and maintenance.
- Do not perform reverse engineering, decomplication, disassembly, adaptation, implantation, or other derivative operations on the equipment software
- Do not study the internal implementation logic of the equipment, obtain the source code of the equipment software, violate intellectual property rights, or disclose any of the performance test results of the equipment software.
- During the storage phase of the energy storage system, the customer should implement safety measures to mitigate unforeseen circumstances.

The Company shall not be liable for any of the following circumstances or their consequences:

- Equipment damage caused force majeure, including but not limited to earthquakes, floods, volcanic eruptions, debris flows, lightning strikes,

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fires, wars, armed conflicts, typhoons, hurricanes, tornadoes, and other extreme weather conditions.

- Operation of equipment beyond the conditions specified in this document.
- Installation or usage in environments that do not comply with international, national, or regional standards.
- Failure to follow the operation instructions, operation and maintenance manual, or safety precautions on the product and in the document.
- Unauthorized removal, modification of the product, or alteration of software code.
- Equipment damage resulting from improper storage conditions that deviate from the requirements specified in the product document.
- Failure to prepare materials and tools that comply with local laws, regulations, and related standards.
- Negligence, gross negligence, intentional breach, improper operations, or other third-party actions causing equipment damage.
- Unauthorized change to the application of battery energy storage system.
- Battery stored beyond the maximum storage period.
- Failure to charge promptly, resulting in capacity loss or irreversible damage to the battery.
- Installation or operation by unqualified personnel.
- Mixing the batteries supplied by the Company with other batteries, including mixing with different brands or batteries with incompatible capacities, causing accelerated capacity degradation.

1.2.2 Safety Precautions

- During installation, overhaul, or maintenance, place clear warning signs or safety caution tape near the operating area. Prominent signs shall also be displayed at the front and rear switches of the energy storage system to prevent accidental closures.
- Technicians responsible for installation and maintenance shall be professionally trained, fully understand all safety precautions, be familiar with the composition and working principle, and master correct operation methods.
- Only qualified professionals are authorized to perform equipment removal and overhaul operations.
- Replacement and modification of equipment or components (including software) shall be carried out by personnel authorized by HyperStrong.
- Ensure the escape routes remain fully accessible during the maintenance and overhaul activities.
- After completing maintenance, overhaul, and other operations, remove all tools and restore the site to its original state for safe and regular equipment operation.
- During the storage, shall establish safety measures to prevent unforeseen incidents.

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1.3 Personal Safety

DANGER	Please wear protection gear before operations to prevent electric shocks or burns.
DANGER	Equipment operates in high voltage. Any accidental contact may result in a fatal shock hazard. It is essential to have at least two personnel presents during maintenance or operation for safety.
DANGER	When the equipment is powered on, it is strictly prohibited to move, open, or dismantle the electric equipment container door or touch any internal components. Before performing maintenance operations, ensure the equipment has been safely powered off.
DANGER	Avoid contact cords or operating equipment during storms.

1.4 Electrical Safety

CAUTION	After performing AC and DC double-side disconnect operations on the energy storage system, wait at least 15 minutes before opening the front door to perform maintenance, overhaul, or other functions on the energy storage system.
NOTICE	When the energy storage system is electrically connected for a test run or other operations, use high-quality measuring equipment with a range and accuracy that meets the site requirements to ensure that the electrical parameters meet the requirements. Ensure that measuring equipment is correctly connected and used according to standards to prevent arcs and other hazards.
DANGER	Even if all external circuits are disconnected, high voltage may still exist between the positive and negative terminals of the energy storage battery pack. If measurements shall be conducted under charged conditions, use dedicated instruments, and ensure that only qualified professionals perform the tests.
WARNING	If the energy storage system operates abnormally or requires maintenance or an, shut down the equipment and confirm that the energy storage system completely stops running before proceeding with any maintenance or repair work.

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Grounding

The following requirements apply to equipment that shall be grounded:

- During installation, it shall be grounded first.
- During removal, the ground wire shall be the last item disconnected.
- Before operations, check the electrical connections to ensure that the equipment is reliably grounded. Operate the equipment without a grounding conductor is strictly prohibited.
- The equipment shall remain permanently connected to a protective earth. Destruction of the grounding conductor is prohibited. (HyperStrong uses TN-S protective earthing system)

1.5 Operational Safety

NOTE	Read all equipment information and instructions carefully and completely before operation.
NOTICE	Before operation, please make sure that all cables are correctly connected. During operation, please pay attention to the normal operation of the equipment fan. Do not block the vent.
DANGER	Do not place small metal devices such as crank pins, screws, etc. next to connectors, slots, sockets, or circuits to prevent short circuits.
CAUTION	If the equipment gets damaged, please do not repair it yourself. Contact HyperStrong to assign professional personnel to repair. Do not modify the internal equipment of the battery system privately to avoid danger.

1.6 Electrostatic Protection

Accumulation of static electricity can result in electric shocks, fire, explosion, failure and damage of electronic devices, and other safety hazards. The energy storage system contains circuit boards and other static-sensitive components. To mitigate the risks associated with static electricity, operators shall implement static protection measures, minimize static electricity generation, accelerate the dissipation of static electricity, and neutralize static electricity.

NOTICE	The prevention methods include but are not limited to: • During replacement, all equipment devices that have not yet been installed should be retained in bags with anti-static shielding, and temporarily removed equipment devices should be placed on foam pads with anti-static features. • Do not touch soldering points, pins, or exposed circuits.

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EMC Related Cautions

The system’s emission characteristics make it suitable for use in industrial areas, meaning Class A electromagnetic environment. If used in a residential environment, where Class B is typically required, this equipment may not provide adequate protection to radio-frequency communication services. User may need to implement mitigation measures, such as relocating or re-orienting the equipment.

Mains power quality should conform to the standards of a specialized industrial distribution network. For users requiring uninterrupted operation during or after power mains interruptions, it is recommended to connect the system's information storage components to an uninterruptible power supply (UPS) or a backup power source.

Some portable RF communications equipment such as high-power walkie-talkie may degrade the electromagnetic performance of the system. If the essential system performance is lost or impaired due to Electromagnetic disturbances, maintain a safe distance between high-power transmitters and the system, and contact customer service to address the issue.

1.7 Lifting and Transportation Safety

- When walking on top of the equipment, strictly follow the procedures for working at heights.
- Wear appropriate personal protective equipment (PPE) as required, including safety helmets, work clothing, safety shoes, and gloves.
- When using lifting machinery, adhere to all relevant safety operation guidelines. Do not engage in improper lifting practices.
- Protect the equipment during handling and storage. Prevent it from falling, tumbling, or being subjected to excessive pressure.
- Batteries transported shall be securely fastened to prevent severe vibration, impact, or compression.
- Drivers and transporters shall undergo appropriate training. For those handling hazardous materials, training should include emergency response procedures for fire prevention, spill containment, and emergency contact protocols. They shall be familiar with the items being transported and understand the correct handling procedures.

1.8 Installation and Wiring Safety

- During the entire mechanical and electrical installation process, strictly adhere to the relevant standards and requirements of the project location.
- Electrical wiring for the system shall only be operational after inspection and verification. Ensure warning signs are prominently displayed during the process, such as "Personnel Working: Do Not Close the Gate" or "High-Voltage Danger.".

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- Ensure that electrical wiring equipment perforations are sealed properly after installation. The sealing shall be tight, secure, uniform in thickness, and have a neat appearance.
- Operate carefully at the installation site to prevent electric shock battery short circuit.
- Wear appropriate protective equipment during installation, including helmets, work clothes, shoes, gloves.
- Pay attention to the positive and negative direction of the battery to avoid reverse installation which can lead to battery leakage, overheating, and other issues.
- Before wiring, disconnect the power supply to eliminate the risk of electric shock.
- When wiring, ensure obstacles are avoided to prevent damage to equipment or interference with operations

1.9 Operation and Maintenance Safety

- Schedule maintenance activities during favorable weather conditions. Avoid conducting maintenance in rain, snow, typhoons, or other adverse weather.
- Personnel shall wear appropriate Personal Protective Equipment (PPE) before beginning operations, including helmets, insulated gloves, long-sleeved coveralls, and insulated shoes.
- Keep an up-to-date emergency contact list on site and ensure communication devices are operational.
- During the operation and maintenance, at least two staff members should be present on site. Confirm the maintenance tasks in writing beforehand and strictly follow the outlined operational procedures.
- After completing maintenance or overhaul, securely store the tools and restore the site to its original condition to ensure the equipment can operate as intended.
- If the ambient temperature of the battery cabin is lower than 0°C, the air conditioner will automatically switch to heating mode to preheat the cabin. In case of abnormalities, contact HyperStrong professional team for assistance.

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2 Transportation and Storage

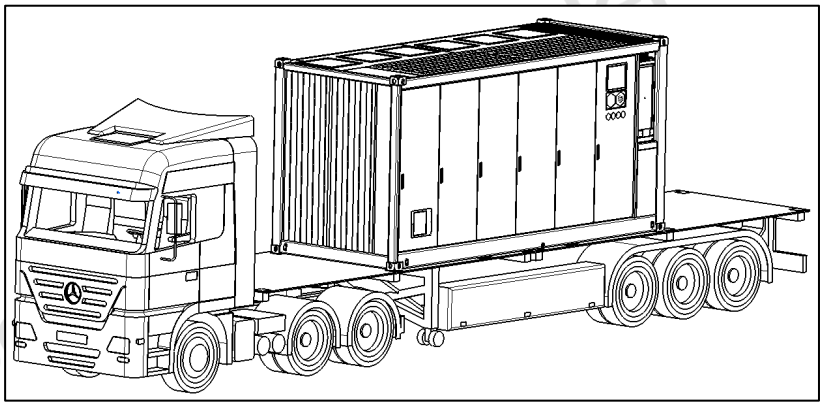
2.1 Cautions

⚠ CAUTION	If the ESS fails to be transported and stored in accordance with the requirements specified in this manual, the warranty may be voided.

2.2 Type of Transportation

- BESS containers can be transported by road, sea, and rail. With an integrated and user-friendly lifting design, ESS containers are easy to be handled during transportation. However, air transportation is not permitted, and specific guidelines for rail transportation are not available currently.
- The BESS container can be transported domestically by truck.

Figure 2-1 Schematic Diagram of BESS Container Transportation



🕮 NOTE	In most cases, when transported by truck, the total weight of the truck, including the BESS container, may exceed the limitations allowed by the general roadways. In such instances, an overweight permit may be required in the shipping region

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2.3 Transportation Requirements

The equipment in the BESS has been installed and securely fixed inside the container before shipment, allowing the container to be lifted and transported as a single unit.

- The applicable national safety procedures and regulations shall be complied with during the operating processes of loading, unloading, and transportation for BESS containers in the country where the project is located.
- The BESS containers and all handling equipment used in the operating processes shall be properly maintained
- All personnel involved in loading, unloading and bolting shall be trained accordingly, especially safety training.
- The mechanical parameters such as external dimension, gross weight of each battery prefabrication pod shall be clarified and verified during the entire operating processes of loading, unloading, and transportation.

The following requirement shall be fulfilled prior to the transportation of BESS containers:

- All cabinet doors within the battery container shall be securely locked.
- A suitable crane or lifting device shall be selected based on site-specific conditions with adequate load capacity, boom length, and radius of rotation.
- When lifting the ESS container, it is recommended to use a single crane.
- Additional towing or traction equipment may be required when moving across inclined surfaces or slopes.
- The intended route shall be cleared of obstructions such as overhanging trees, power lines, or other impediments.
- The transportation and movement of ESS containers should be conducted under favorable weather conditions whenever practicable.
- Warning signs or cordons shall be set up and formed to restrict access to the lifting and transportation area and prevent accidents.
- When transported by land, ropes shall be used to secure the corner fittings on the top of the BESS container to the vehicle and ensure that the ESS container remains stable and the inclination angle is minimized during transportation.

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2.4 Storage Requirements

- In order to prevent possible condensation or the bottom soaking from rainwater in the rainy season. The BESS should be stored on higher place.
- It is necessary to raise the container base to acclimatize site-specific conditions. The specific height should be reasonably determined based on the geological and meteorological conditions of the site.

- Store the BESS on dry, flat, and stable ground with sufficient load-bearing capacity and free of vegetation.
- Ensure the container doors and all internal equipment are securely locked.
- Storage environment temperature: -35~60℃.
- Long-term storage is not recommended because it may cause a decrease in battery capacity. Even when stored at the recommended storage temperature, irreversible capacity fading will still occur over time. The longer the system is stored, the greater of capacity will fade. Refer to the technical protocol for specific capacity fade rates.
- The relative humidity should be between 0~95%, without condensation.
- Protect the air inlet and outlet of the BESS to prevent rain water, sand, and dust from entering the container.
- Perform periodic inspections.
  - Check the container and the inner equipment for damage at least once per month.
  - For containers stored for more than six months, open the door and inspect the interior to ensure there is no visible condensation. Verify that the container and internal equipment are undamaged.
  - Inspect the system after powering it on and initiating startup procedures.
- If required, request professional assistance for testing before installation.
- For BESS systems stored for more than six months under the above conditions, a charge and discharge cycle shall be performed to bring the system SOC to 30% The storage period begins from the date of departure from the HyperStrong factory

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3 BESS Product

3.1 Product Introduction

The BESS utilizes lithium iron phosphate battery cells known for their high safety and long cycle life. The system is housed in an outdoor cabin, designed for modularity, easy installation, and straightforward maintenance.

A battery cell rated 3.2V/314Ah is configured into a 1P104S pack. Four packs are combined to form a rack, configured as 1P416S. Each BESS can accommodate up to 12 sets of racks,

- 1 set of system control and distribution cabinet (SCDC).
- 1 set of thermal management system (TMS).
- 1 set of fire suppression system (FSS).

The battery system adopts a multi-level modular design, consisting of battery cell-module-rack-system. It connects to the SCDC via power cables.

3.1.1 Product Parameters

Table 3-1 Product Parameters

Model (Type A)	**HSL3702 -03343EU**	**HSL3702 -03761EU**	**HSL3702 -04179EU**	**HSL3702 -04597EU**	**HSL3702 -05015EU**
Model (Type B)	**HSL3702-03343EU-S**	**HSL3702-03761EU-S**	**HSL3702-04179EU-S**	**HSL3702-04597EU-S**	**HSL3702-05015EU-S**
DC Side					
Battery Type	LFP-314Ah				
Configuration	8P416S	9P416S	10P416S	11P416S	12P416S
Rated Energy(kWh)	3343	3761	4179	4597	5015
Rated Voltage ($V_{d.c.}$)	1331.2				
Voltage Range ($V_{d.c.}$)	1123.2-1497.6				
Charge and Discharge Power (kW)	835	940	1044	1149	1253

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Rated Charge and Discharge Current (A)	628	707	785	864	942
General Parameters					
Dimensions (L*W*H)	6058*2438*2896 mm				
Weight (kg)	31600	34200	36800	39400	42000
IP Rating	Pack-IP65 / PDU-IP65 / Container-IP55				
Fluorine Mass	1094 kg				
Carbon Mass	6013 kg				
Operating Humidity	0~100%RH(Non-condensing)				
Operating Temperature (℃)	-30~55				
Storage Temperature (℃)	-35~60				
Altitude (m)	≤3000				
Auxiliary Power	3L/N/PE, 400/230Vac 50/60Hz				
Protective Class	I				
Communication Interfaces	Ethernet (Optical Fiber), RS485, Dry Contact				
Communication Protocols	Modbus TCP, Modbus RTU				
Heat Transfer	Liquid-cooling & Liquid-heating				
Coolant Type and Design Pressure	Ethylene glycol solution@2.0Bar				
Output Short-Circuit Ratings	82kA @1.3ms	92.25kA @1.3ms	102.5kA @1.3ms	112.75kA @1.3ms	123kA @1.3ms
Fire Suppression System	Temperature sensing; Smoke sensing; Combustible gas detection; Cabin level aerosols (electric starting)				
	Ventilation system				
Certifications and Compliance	IEC62477-1, IEC 60730-1 Appendix H, IEC61000-6-2/4, NFPA 69, UN3536, GR-63-CORE zone3				
Optional	UPS / Explosion Venting Panel / Water Fire Protection				

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3.1.2 Product Appearance

Figure 3-1 Description of Product Appearance



3.1.3 Salt Spray Resistance and Seismic Performance

The total thickness of the BESS container paint film shall meet the requirements of C5 level, and the overall neutral salt spray resistance time shall reach 1440h and above after three spraying and three drying processes in accordance with ISO12944-6 standards.

The seismic design code for BESS Container complies with the requirements of GR-63-CORE, and the structural strength of container frame is suitable for seismic zone 3.

In order to address the diverse seismic design requirements worldwide, the most provision of design codes listed in the following table shall be applied. However, the precise structural and foundation designs will depend on the specific project conditions, including soil condition, rock types, ground motion parameters and relevant local laws and regulations.

Table 3-2 Seismic Design Codes by Region

Regions	Design Codes
US/Canada/Mexico	IEEE Std 693-2018
Europe	IEC 60068-3-3-2019
China Domestic	GB 50260-2013
Japan	JEAG 5003-2019
Australia/New Zealand	SAA AS 1170.4/SNZ NZS 1170.5
Chile (for South America)	Nch2369-2003

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3.2 Structural Appearance and Internal Layout of ESS

3.2.1 Appearance of BESS Container

Figure 3-2 Appearance Schematic of Energy Storage System

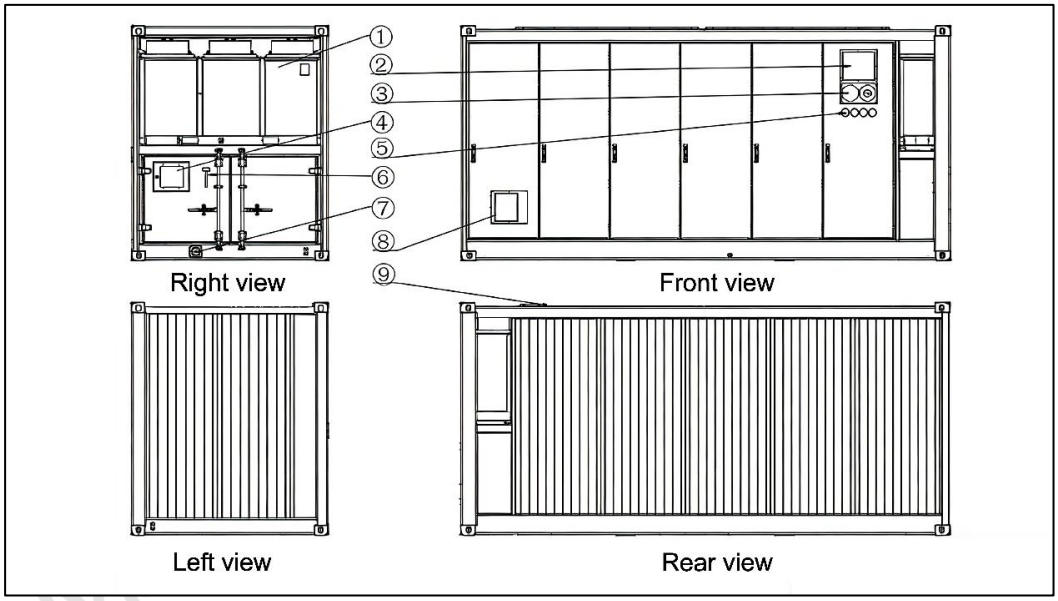


Table 3-3 Related Part Number and Part Name of Figure 5-2

No.	Name
1	Liquid-cooling System
2	Explosion-proof Flammable Gas Exhaust Fan
3	Sound and Strobe
4	Emergency Stop Button（red button） and Fire Buttons Box
5	Indicator
6	Safety Hammer
7	Waterborne Fire Protection Interface
8	Air Inlet Louvers for Fire Suppression System
9	Explosion Venting Panel (Optional)

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⚠ CAUTION	• The nameplate contains important parameter information related to the ESS, which should be protected during transportation, installation, maintenance, overhaul and other operations. Destruction or removal is strictly prohibited. • Emergency Stop Button (red button): In the event of an emergency, press this button to disconnect the electrical connections inside the ESS container.

3.2.2 Indicator Lights

Figure 3-3 Position, Appearance and Function of Indicator Lights

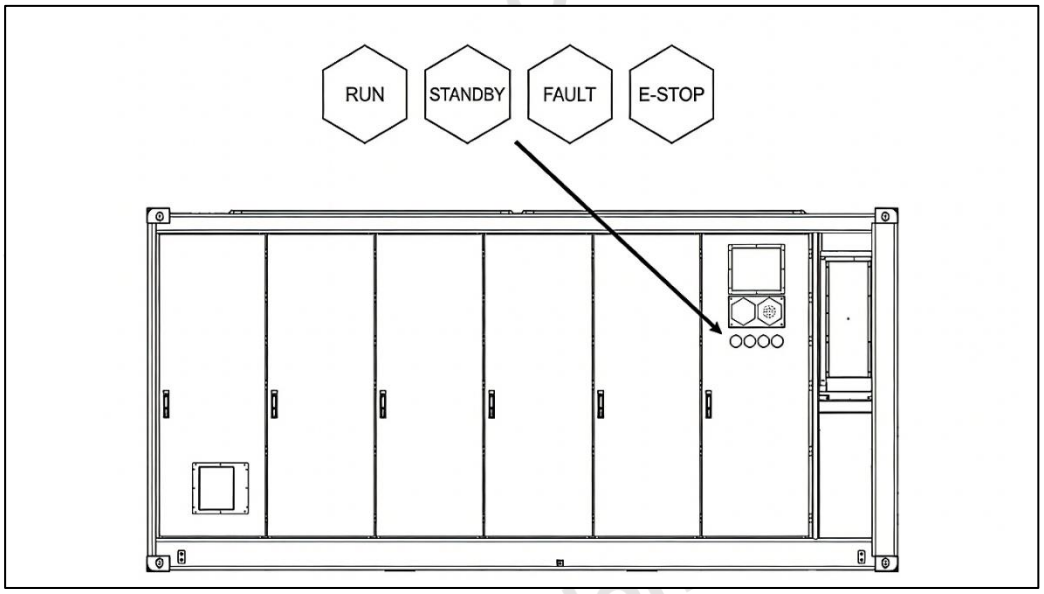


Table 3-4 Meaning of Indicator Lights

Code	Name	Indicator/ Button Color	Function Declaration
HL1	RUN		The system is in charging or discharging operation.
HL2	STANDBY		The system is in a shutdown state but can be started.
HL3	FAULT		There is a fault in the system and it is out of service.

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HL4	E-STOP		When this lamp light is on, it indicates that the system is in an emergency shutdown state. At this time, the system cannot start and operate normally. To restart the system, please follow the following sequence: 1. Ensure the Emergency Stop button is not in the locked state; 2. Check if the system Emergency Stop fault has been rectified or cleared; 3. Confirm that there is no actual Emergency Stop fault, and reset the system status; 4. Verify that the system has returned to the normal stop state.

3.2.3 Dimensions of BESS Container

The external dimensions are shown in the figure below.

Figure 3-4 Energy Storage System Container Dimensions



3.2.4 Internal Design of BESS

The internal arrangement of the energy storage system is shown in the figure below.

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Figure 3-5 Internal Schematic of the BESS

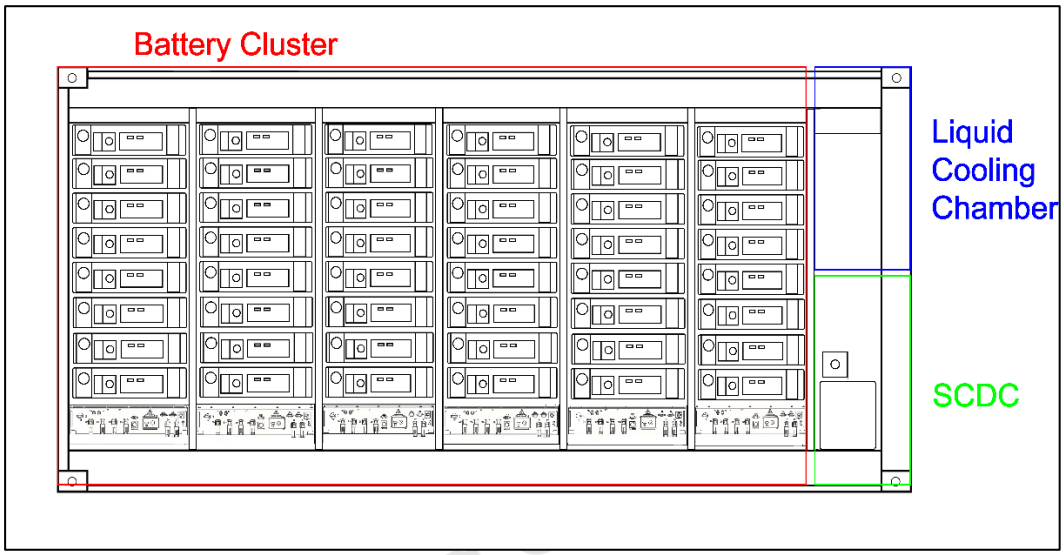


Figure 3-6 Internal Equipment Composition

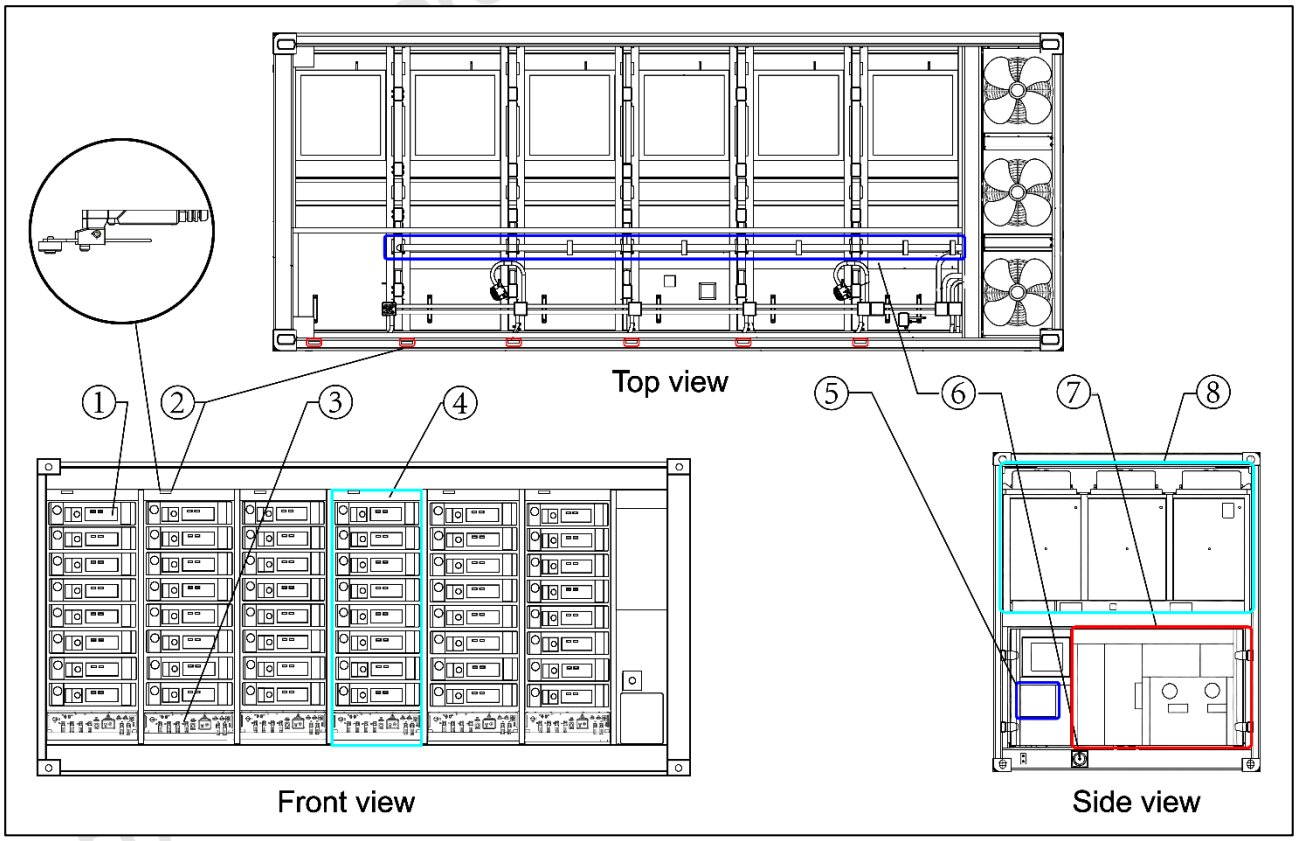


Table 3-5 Related Part Number and Part Name

No.	Name
1	Pack
2	Travel Switch

3	PDU
4	Rack
5	Fire Control Panel
6	Water Fire Protection System
7	System Control and Distribution Cabinet
8	Liquid-cooled Unit

NOTE	The above figures are for reference only, please refer to the actual product for details

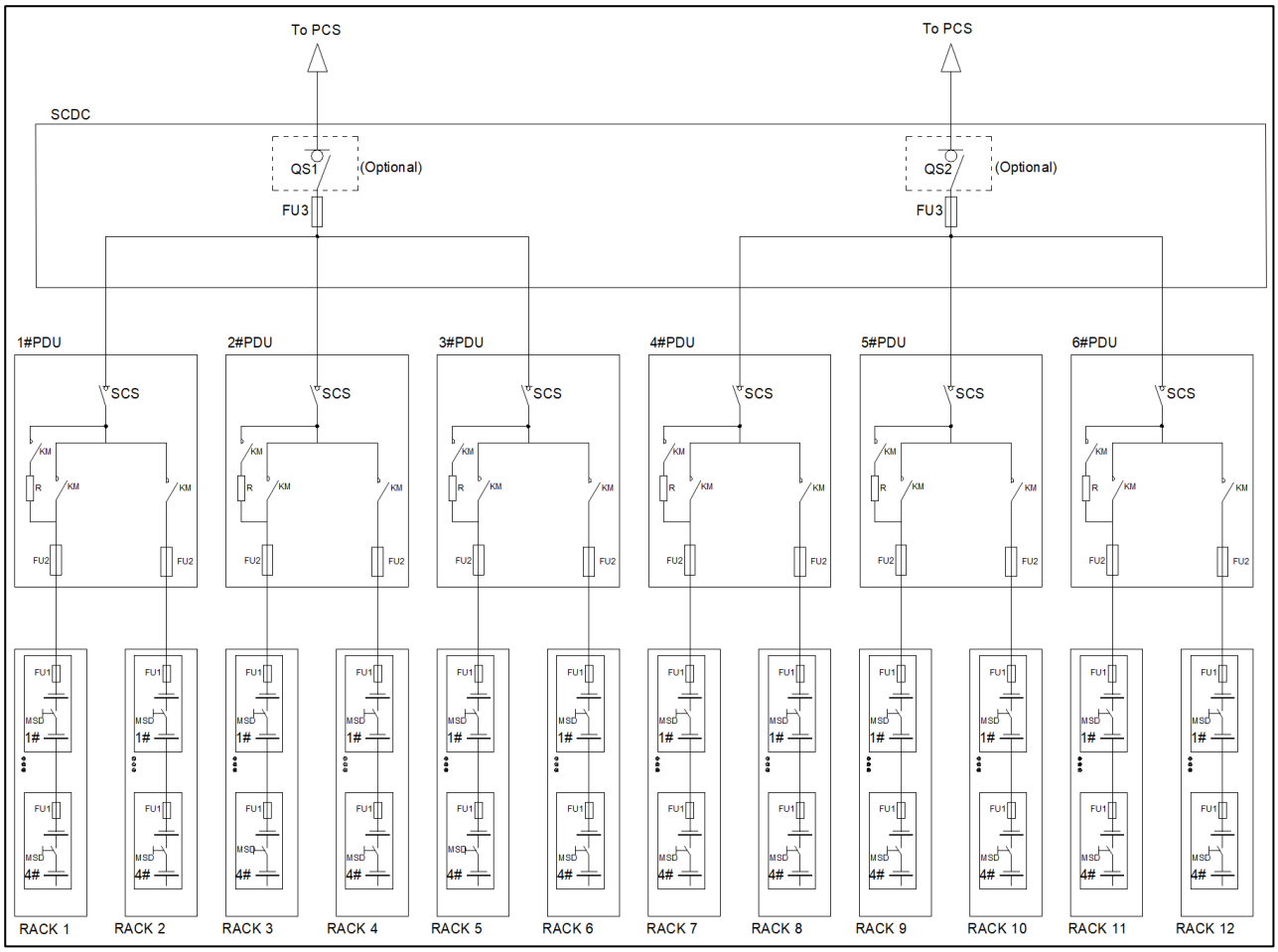
3.3 Primary Topology of BESS

The primary circuit protection device of BESS mainly consists of isolation switch, surge protector, and fuse. The container consists of two branches, each containing six battery clusters(racks). Within each battery cluster, there are four battery packs, every two battery clusters(racks) are connected to one PDU.

A schematic diagram of the product is shown in the figure below.

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Figure 3-7 Schematic Diagram of Product-12 racks (for reference only)



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3.4 Power Supply Topology of BESS

There are two options available for the auxiliary power supply of the battery compartment:

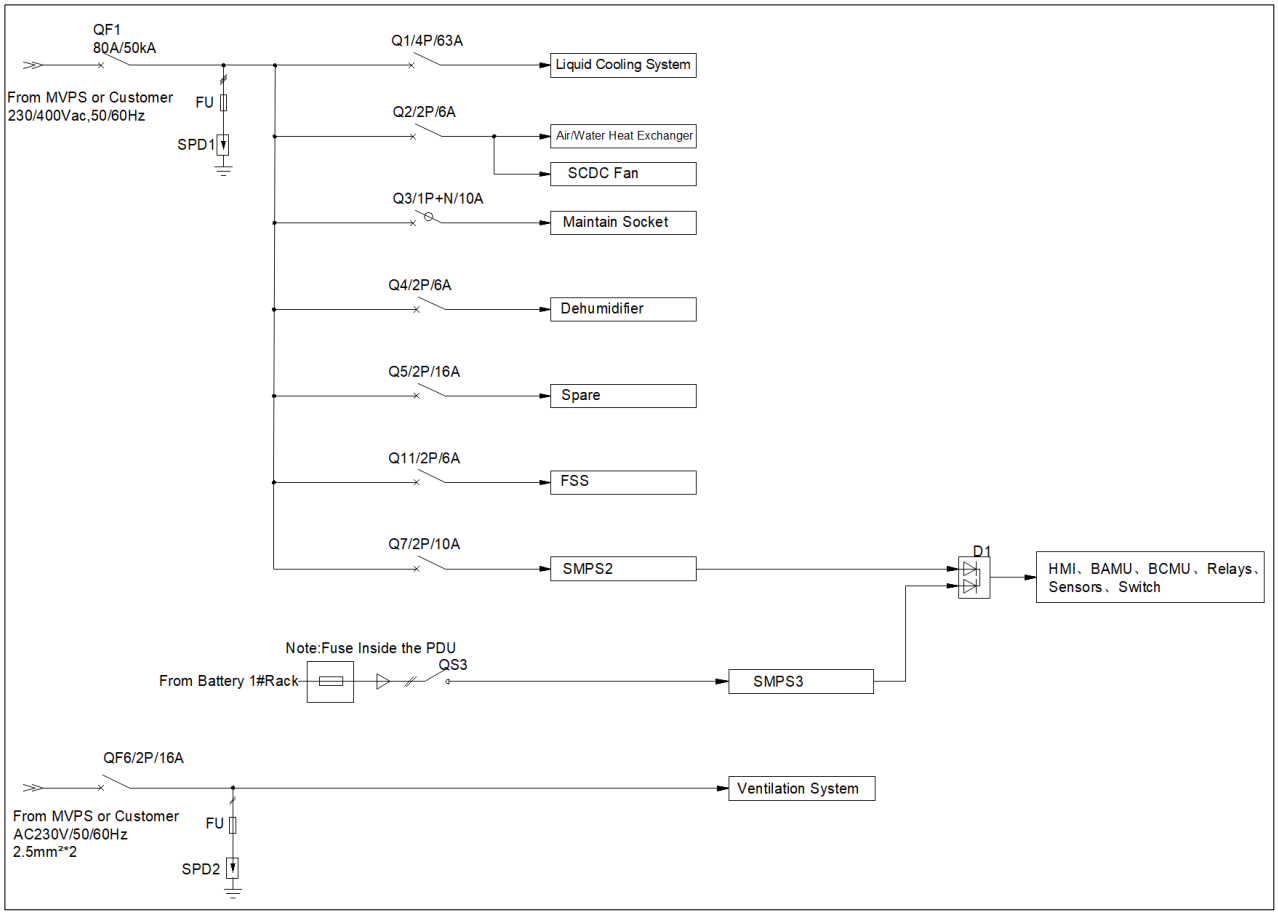
- Standard DC Power Take-off Scheme (Battery Power Version)
- Optional UPS Scheme (UPS Version)

3.4.1 Standard DC Power Take-off Scheme

The auxiliary power supply requires one three-phase power supply (230V/400Vac, 50/60Hz) and one single-phase power supply (230Vac, 50/60Hz). The single-phase power supply must be capable of providing uninterrupted power for 2 hours.

The power supply circuit is shown in the following figure.

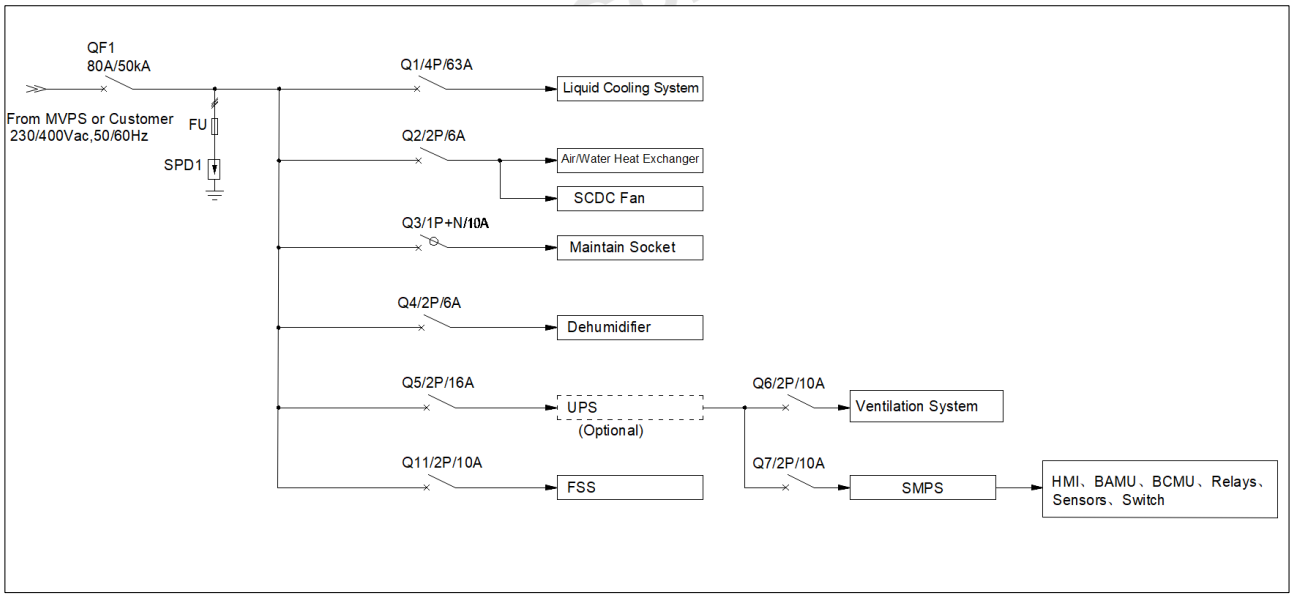
Figure 3-8 Power Supply Topology Diagram-Battery Power Version (for reference only)



3.4.2 Optional UPS Scheme (UPS Version)

The auxiliary power supply requires one three-phase power supply (230V/400Vac, 50/60Hz) for optional UPS (UPS Version).

Figure 3-9 Power Supply Topology Diagram- UPS Version (for reference only)



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3.5 Communication Topology of BESS

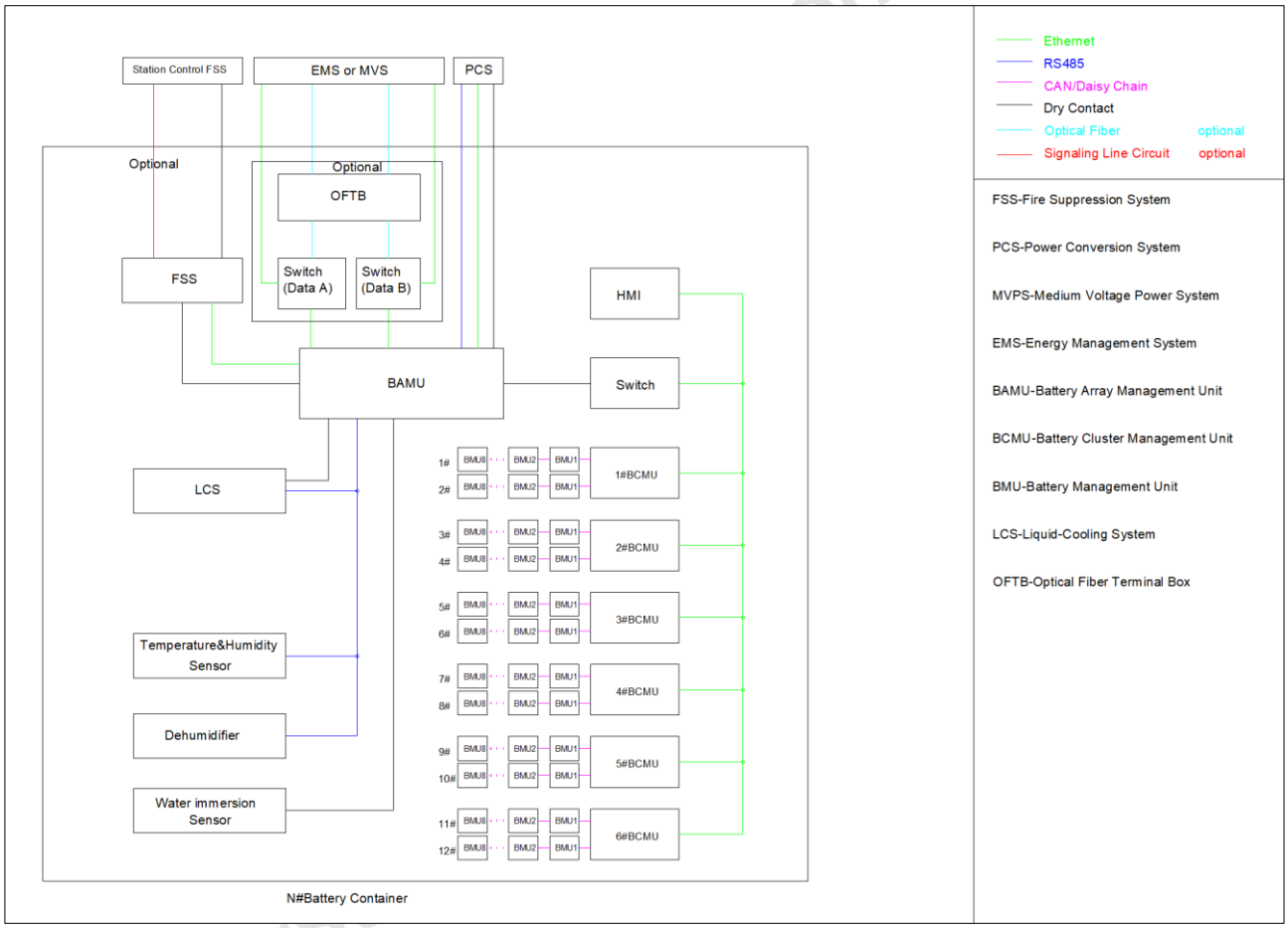
The secondary topology of this product consists of BMU, BCMU, and BAMU. The communication modes between these components are:

- BMU and BCMU: Daisy chain.
- BCMU and BAMU: Ethernet.

Depending on whether the UPS is optional, the communication topology diagram also has the following two forms.

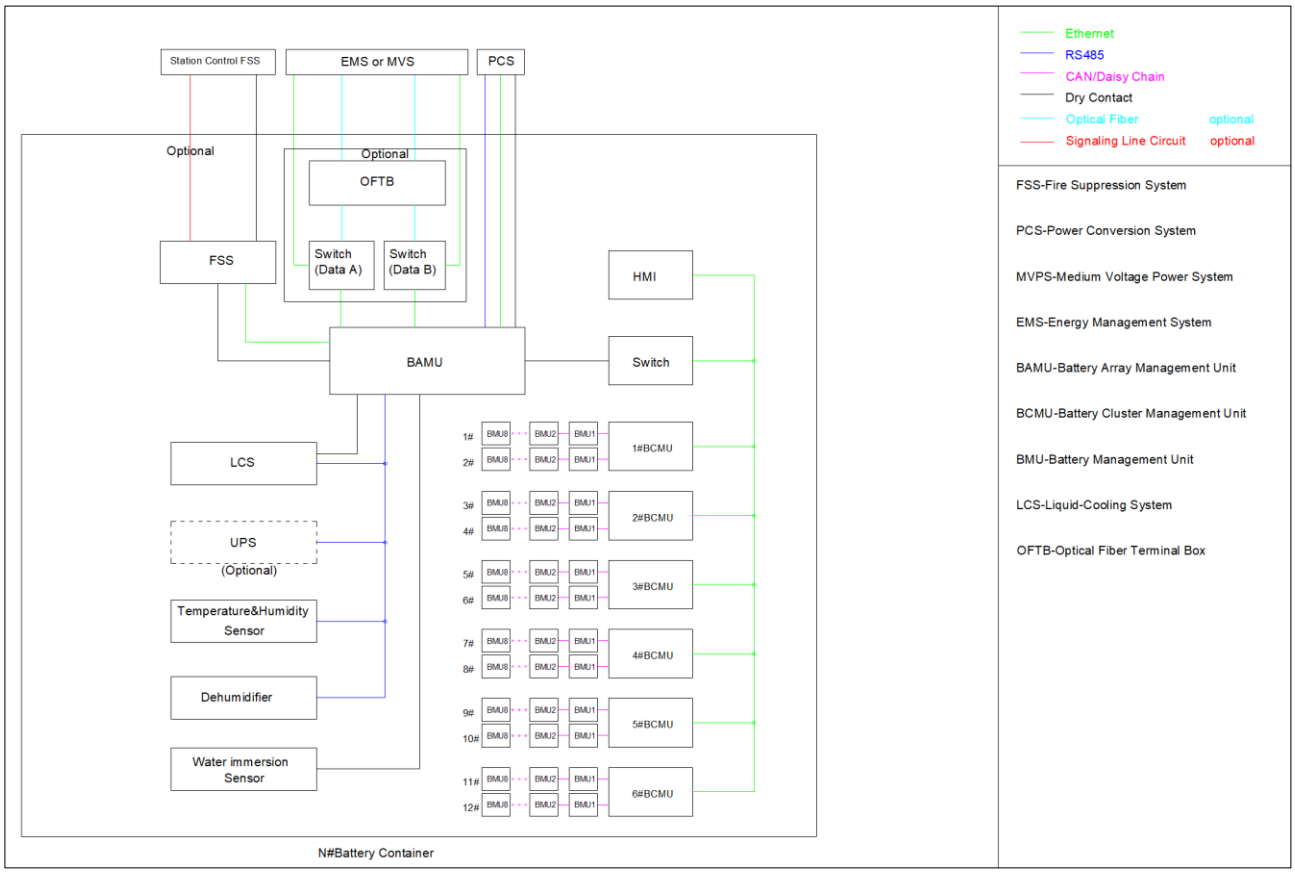
The communication topology diagram is shown in the following figures.

Figure 3-10 System Communication Topology Diagram -12 Racks- Battery Power Version (for reference only)



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Figure 3-11 System Communication Topology Diagram -12 Racks-UPS Version (for reference only)



3.6 Rack

The rack adopts frame structure design, with the packs and PDU are fixed by bolts. A single column of 8 packs shares one PDU. The wiring diagram below actually shows two racks.

The battery rack wiring diagram is shown in the following figure: Battery Rack Writing Diagram.

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Figure 3-12 Battery Rack Wiring Diagram (for reference only)

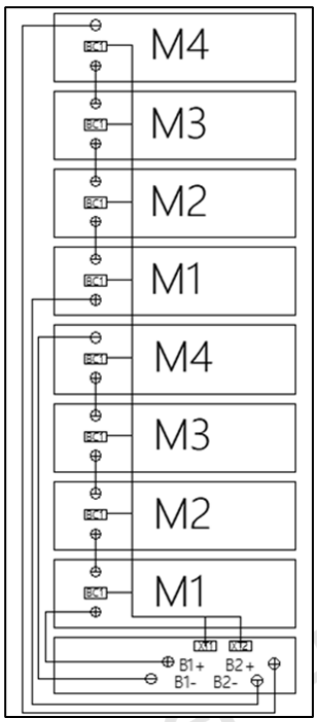


Table 3-6 Rack Parameters

No.	Specifications	Parameters
1	Cell Type	LFP314Ah
2	Cell Quantity	416
3	Nominal voltage	1331.2Vd.c.
4	Rated capacity	314Ah
5	Total Storage Energy	417.996kWh
6	Rated/Max.Charge/Discharge Power	208.998kW
7	Connecting Type	(104S) 4S
8	Upper Limit Charging Voltage	1497.6Vd.c.
9	Final Discharge Voltage	1123.2Vd.c.
10	Rack Maximum Short-circuit Current	10.25kA(@1.3ms)
11	Nominal Current	78.5A
12	Recommend Current	78.5A
13	Operating Temperature	-20℃ to 55℃
14	Storage temperature [℃]	-20℃-45℃ (one month) 0℃-35℃ (six months)
15	Dimension	H1528.5±5, W1001±10, L2195±15 (mm)
16	Weight	≤5400kg

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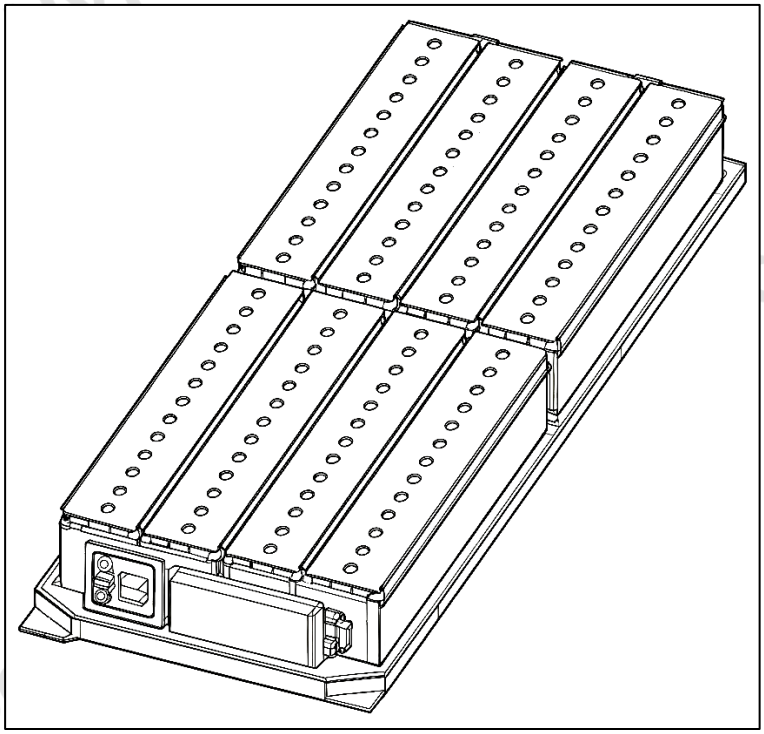
17	Discharge Cut-off Voltage [V]	1123.2
18	Enclosure Protection (IP)	IP65 for PDU, IP65 for Battery Packs
19	Heat Transfer	Liquid-cooling & Liquid-heating
20	Liquid Coolant Type	Ethylene Glycol Solution
21	Altitude	≤ 3000m

3.7 Pack

The pack is designed as a battery box with a well-organized structural layout, attractive appearance, and excellent heat dissipation performance. Each 1P104S battery pack consists of eight 1P13S battery modules, collection harness, BMU, related electrical components and structural components, etc., please refer to the right part of the figure below.

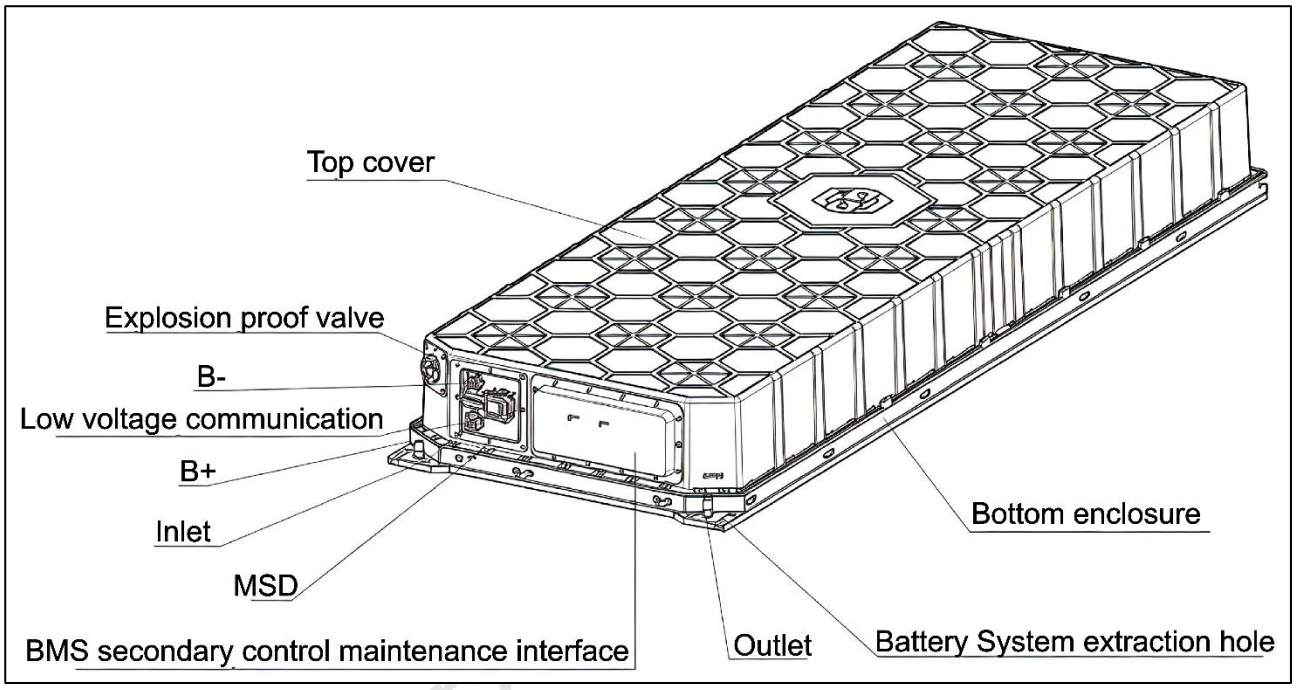
The battery pack is shown in the following figure： Battery Pack Diagram.

Figure 3-13 Battery Pack(Internal View)



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Figure 3-14 Battery Pack Appearance



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Table 3-7 Pack Parameters

No.	Specifications	Parameters	Remarks
1	Cell Type	LFP-314Ah	
2	Configuration	104S1P	
3	Rated Capacity	314Ah	
4	Rated Energy	104.449kWh for 314Ah cells	0.25P/0.5P, 25℃±2℃, BOL
5	Rated Voltage	332.8Vdc	
6	Voltage Range	265.2~379.6Vdc (>0℃)	
7	Recommended Voltage Range	280.8~374.4Vdc	Cell: 2.7~3.6Vdc
8	Charge and Discharge Rate	≤0.25P	
9	Operating Temperature	Charge： 0~60℃ Discharge：-20~60℃	Cell level, suggest operation with thermal management
10	Storage Temperature	-35℃-60℃	
11	IP Rating	IP65	
12	Dimensions	790±5mm(W)*2190±15mm(D)*245±5mm(H)	
13	Weight	650(1±1.5%) kg	
14	Cooling/Heating Method	Liquid cooling & liquid heating	
15	Voltage Acquisition	104 ports	
16	Temperature Acquisition	59-ports	48 for cell, 8 for external output busbars, and 3 for aviation socket temperature
17	SOC at Shipment	≤30%SOC	

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NOTE Parameter values are for reference only, please refer to the actual project specifications for precise details.

3.8 Power Distribution Unit (PDU)

The Power Distribution Unit (PDU) is a control unit for distributing battery energy, containing BCMU, positive and negative contactors, fan cooling circuits, and high and low voltage connectors. All contactors can receive control commands from the BCMU, which can collect individual cell voltages

and temperatures reported by BMUs within each battery pack in the rack. After data collection and analysis, the BCMU reports the data to the BAMU. The PDU connects the high and low voltage circuits in series with each battery pack, serving as the input and output interface for the power and control circuits of the battery rack.

The appearance diagram of PDU is shown in Figure Appearance Diagram of PDU.

Table 3-8 Specifications and Parameters of PDU

No.	Specifications	Parameters	Remarks
1	Maximum Operating Voltage	1500VDC	
2	Maximum Continuous Current	100A/branch	
3	Dimensions (W*D*H)	790±2mm(W)*750±2mm(D) *176.6±2mm(H)	
4	Weight	40±5kg	
5	Power Supply	24VDC	External Power Supply
6	Communication	Ethernet, Daisy Chain, CAN	Ethernet for Communication with the BAMU
7	Display Function	Power and control indicators	

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Figure 3-15 Multi-view drawings of PDU

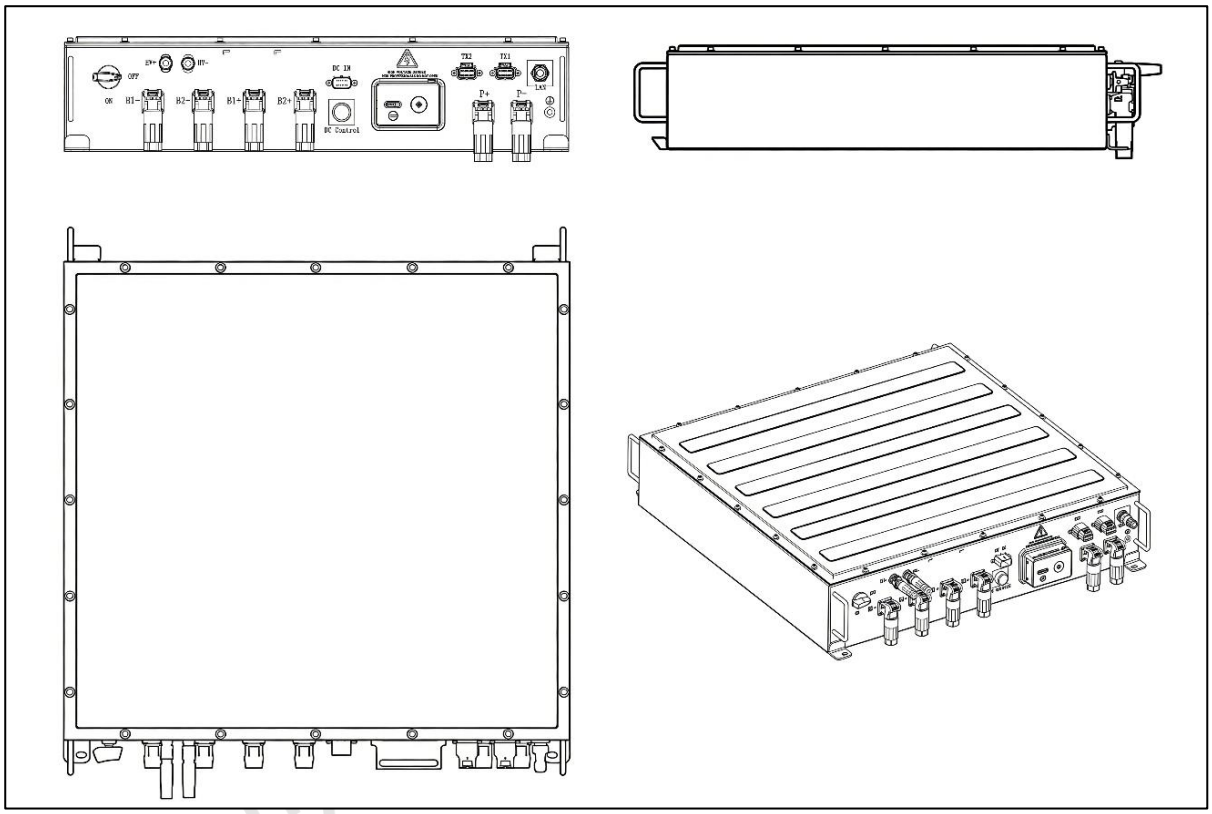


Figure 3-16 Front panel drawing of PDU

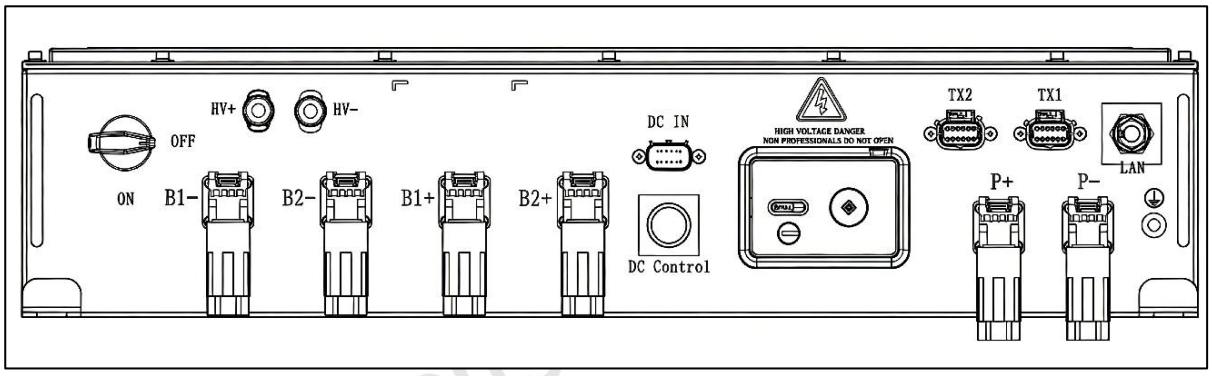


Table 3-9 PDU front panel plug description

Name	Function Declaration
OFF	The DC power output switch off
ON	The DC power output switch on
HV+	Positive pole of the DC power output
HV-	Negative pole of the DC power output
B1-	Negative pole of 1 # battery rack
B2-	Negative pole of 2 # battery rack

B1+	Positive pole of 1 # battery rack
B2+	Positive pole of 2 # battery rack
HL	24V Button indicator light
DC IN	24VPower in
DC Control	24V Button indicator light
TX2	BCMU COM port of 2 # battery rack
TX1	BCMU COM port of 1 # battery rack
P+	Positive pole of PDU DC output
P-	Negative pole of PDU DC output

3.9 System Control and Distribution Cabinet (SCDC)

The power distribution equipment integrated with the control system and the auxiliary system has the functions of controlling the whole energy storage container and distributing power to the auxiliary system, as shown in the figure 5-13 System Control and Distribution Cabinet.

- The SCDC contains the control part and power distribution system. The power distribution part provides power supply for thermal management, BMS and other systems. The control power supply is equipped with a dual-circuit redundant design. In the event that the BESS loses external auxiliary power, Battery Cluster (Rack) #1 or UPS can be utilized to maintain the operation of key equipment.
- Indoor temperature and humidity status information and fire protection status information, and upload the above data to the background.
- Try to ensure the load balance of each phase in the whole distribution system through reasonable distribution; each branch has a complete protection function, and the micro-break opening and closing status of key branches can be collected and uploaded to the background.
- Connect the BMS in the container to the switch in the SCDC, and view and control the BMS system information through the touch screen in the SCDC.
- The BCMU is connected to the BAMU in the SCDC through a network cable, forming the entire BMS system. The BAMU can be connected to the station-level EMS and other equipment via Ethernet or optical fiber to receive operating or shutdown instructions.

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Figure 3-17 System Control and Distribution Cabinet

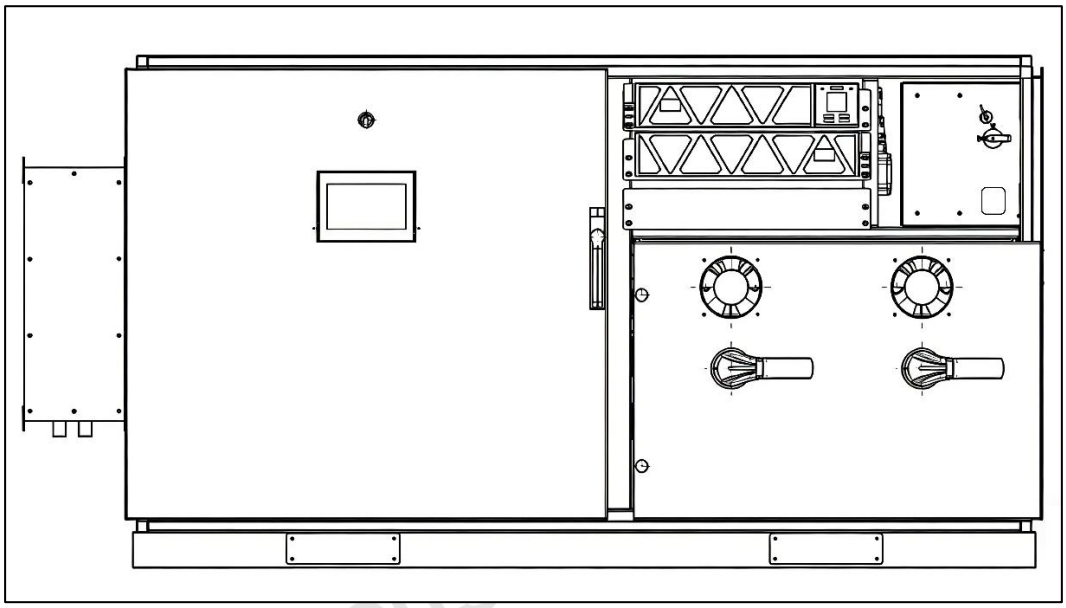


Table 3-10 Meaning of Indicator Lights

Code	Name	Indicator/ Button Color	Function Declaration
SA1	Control Selector Local/ Remote	/	BAMU selects local/remote control; Local state disables EMS control and allows host computer control; Remote state does the reverse.

3.10 Fire Protection System (FPS)

The fire protection system for energy storage containers consists of a fire detection and alarm system, fire suppression system, ventilation system, explosion venting system and Water deluge system. It monitors environmental abnormalities, controls fire-fighting equipment and provides timely early warnings, forming a crucial safety defense barrier.

3.10.1 Fire Detection and Alarm System

Each battery prefabricated cabin has a set of fire detection and alarm system, which is integrated with the fire suppression system. The fire detection and alarm system consists of the following components.

Sensors and Detectors

The energy storage container is equipped with three smoke detectors and two heat detectors inside. Two smoke detectors and two heat detectors are mounted on the ceiling of the battery compartment, and one smoke detector is installed in the electrical compartment. When the ambient temperature and smoke concentration exceed the threshold values, the detectors send

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signals to the fire alarm control panel. Upon receiving the signals, the fire alarm control panel outputs external alarm signals.

Sound and Light Alarms

Sounder beacons and alarm bells are embedded on the enclosure of the energy storage container. In the event of a fire inside the container, the fire alarm control panel activates the alarm bells and sounder beacons, which emit audible and visual signals to warn the surrounding area.

Fire Alarm Control Panel

The electrical cabinet is equipped with a dedicated Fire alarm control panel.

3.10.2 Fire Suppression System

The fire extinguishing system consists of a fire alarm control panel, four fixed aerosol fire suppression units, one release push button and one abort push button.

Operating Modes

The system supports two operation modes.

- Automatic Mode: Upon fire detection, the control panel will automatically activate the aerosol fire suppression units in accordance with preset logic sequences. The units can also be manually triggered by pressing the release push button.
- Manual Mode: During maintenance and inspection work, the system shall be switched to manual mode to prevent accidental discharge of the aerosol agent.

In the event of an unintended activation of the aerosol system, the abort push button may be pressed to terminate the discharge process.

Control Logic

The control logic of the aerosol fire extinguishing system is as follows:

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Figure 3-18 Working Principal Flow Chart of Aerosol Extinguishing System

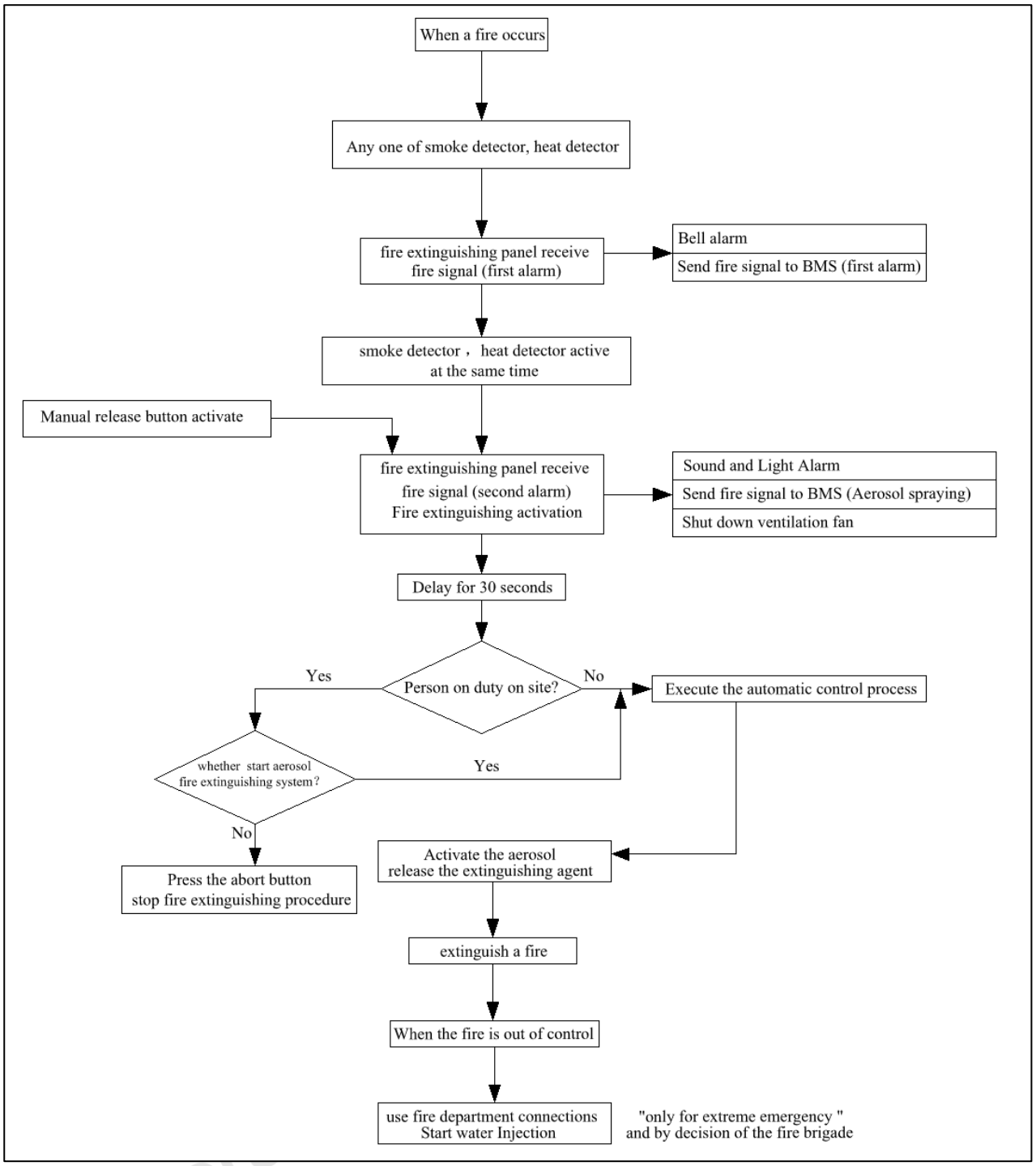


Table 3-11 Fire Protection Logic Principles Comparison

Item	Function	Action
Case 1	Smoke Detector or Heat Detector Triggered	• Bell Alarm • The first fire alarm signal to BMS
Case 2	In Battery Container Smoke Detector and Heat Detector Triggered	• Bell Alarm • Sound & Light Alarm • After the fire alarm control panel completes the 30-second countdown sequence, the aerosol fire suppression units are activated. Aerosol spraying signal to BMS QF tripping
Case3	Manual Release Button	• Bell Alarm • Sound & Light Alarm • After the fire alarm control panel completes the 30-second countdown sequence, the aerosol fire suppression units are activated.
Case 4	Abort Button	The 30-second countdown sequence running on the fire alarm control panel shall be suspended.

The current operational logic of the fire extinguishing system is as follows:

- When combustible gas exceeds the threshold,

Step 1 The contact detectors are triggered and the status is locked.

Step 2 The fire alarm control panel and combustible gas detector can only be reset by qualified personnel after confirming that the battery container is normal on-site.

Step 3 The exhaust fan will be shut down only when the combustible gas alarm is reset

- During routine inspections,
  - The fan can be manually started to verify its proper operation.
  - In emergency situations where the combustible gas detection system malfunctions, on-site personnel may assess whether it is safe to approach the container and manually activate the fan.

NOTICE

Inspection, testing, and maintenance of the aerosol fire suppression system should be in accordance with EN15276.

3.10.3 Ventilation System

Ventilation System Composition

The ventilation system is divided into a detection subsystem and a control subsystem.

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- The detection subsystem, fitted with two combustible gas detectors and two monitoring modules, is installed on the ceiling of the container.
- The control subsystem consists of an air intake louver, an exhaust fan and a fan start-stop knob.

Ventilation System Operating Modes

- Automatic Start

When the hydrogen concentration is no less than 10% LEL (Lower Explosive Limit) of hydrogen, the louver and exhaust fan operate simultaneously to keep the combustible gas concentration in the battery compartment below 25% LEL.

- Manual Start

The louver and exhaust fan can also be started manually via the fan control knob.

This design maintains combustible gas concentrations within safe thresholds and forms an important safety barrier for the whole system.

Ventilation System Operating Logic

Figure 3-19 Ventilation System Working Principal Flow Chart

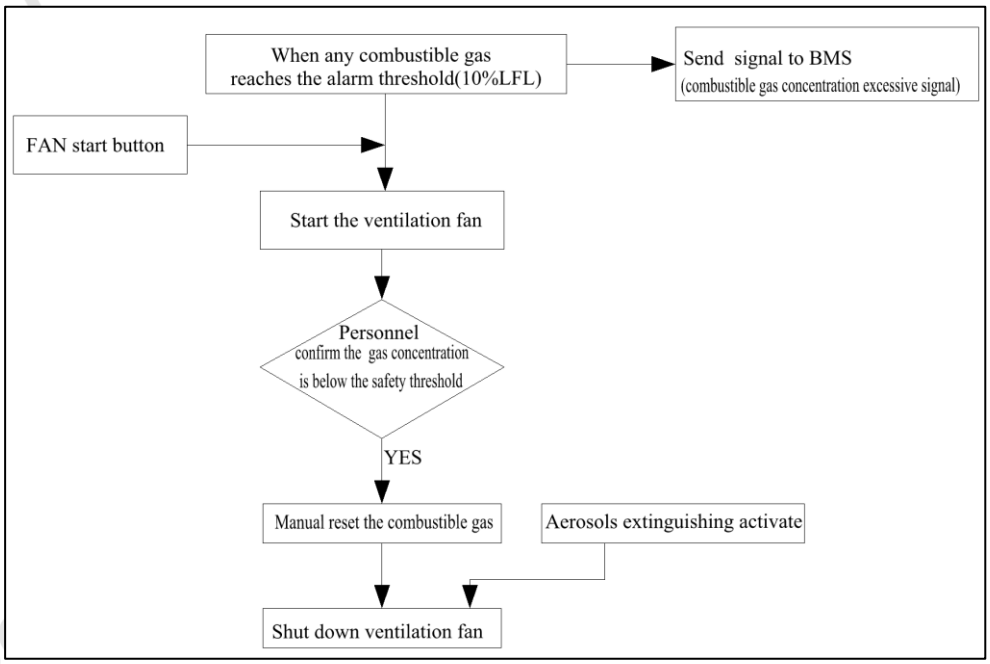


Figure 3-20 Ventilation System Logic Principles Comparison

Item	Function	Action
Case 1	H_2 Detector Warning	• The fan and louver will be activated. • Bell Alarm • Gas concentration excessive signal to BMS

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Case 2	Fan Start Button	• The fan and louver will be activated. • Firefighting fan run signal to BMS

NOTICE	• Inspection, testing and maintenance recommendations for the explosion control system should be in accordance with NFPA 69. • The installation, inspection, and maintenance of the explosion prevention system shall comply with NFPA 69 and the manufacturer's instructions.

3.10.4 Explosion Venting System (Optional)

The explosion vent is mounted on top of the battery container. In the event of an explosion, when the container pressure reaches 0.1 bar, the explosion vent opens along a predetermined area to release the pressure from the container to the surrounding area.

3.10.5 BESS Water Deluge System (Optional)

The rapid liquid injection system is the last line of defense for energy storage fire protection, which is mainly to prevent the fire spread and minimize the fire amplification and reduce the damage. Each battery container is equipped with:

- One standard water fire-fighting interface.
- Container-level piping and nozzles.

This setup enables rapid liquid injection function to ensure the safety of personnel and equipment. The prefabricated module is equipped with a reserved firewater inlet, uses one 2½" STORZ connector, conforming to EU. standard hose access requirements. If the interfaces do not match during use, Customers may select the compatible interface that meets local standards.

In case of a fire, after the site risk assessment is completed, professional firefighting personnel should decide on the operation to source water from external hydrants or fire trucks, enabling rapid cooling and extinguishing functionality , "only for extreme emergency "and by decision of the fire brigade.

3.11 Thermal Management System (TMS)

The Thermal Management System (TMS) enables optimal temperature control, enhancing product performance, extending battery life, reducing auxiliary power consumption, and increasing the overall value of the product.

This TMS is custom-designed based on the temperature characteristics of each container's components within the energy storage system: the battery container utilizes a liquid cooling system to heat and cool the batteries, while the electrical cabinet employs forced-air cooling via a air/water heat

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exchanger for heat dissipation. Both the battery and electrical cabinets are equipped with dehumidification devices to maintain the operating environment within a reasonable humidity range.

Figure 3-21 Schematic Diagram of Liquid-Cooled Piping System for ESS

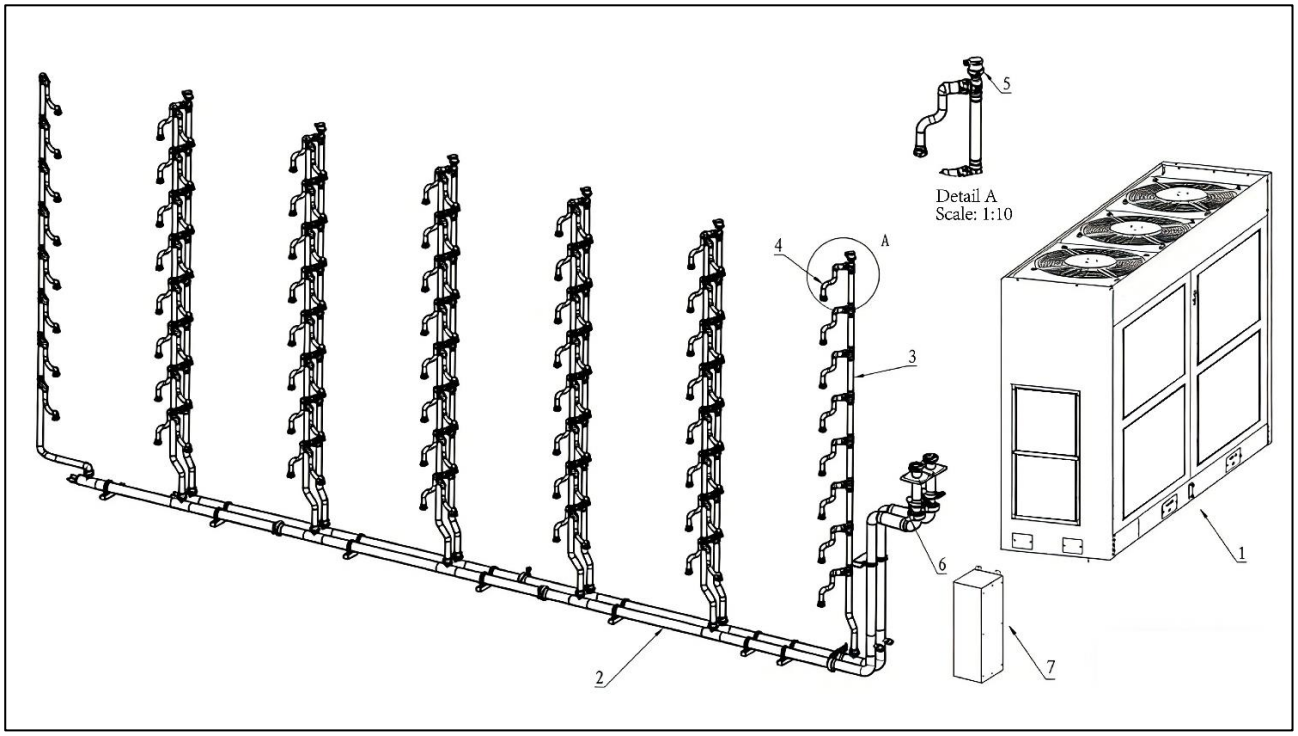


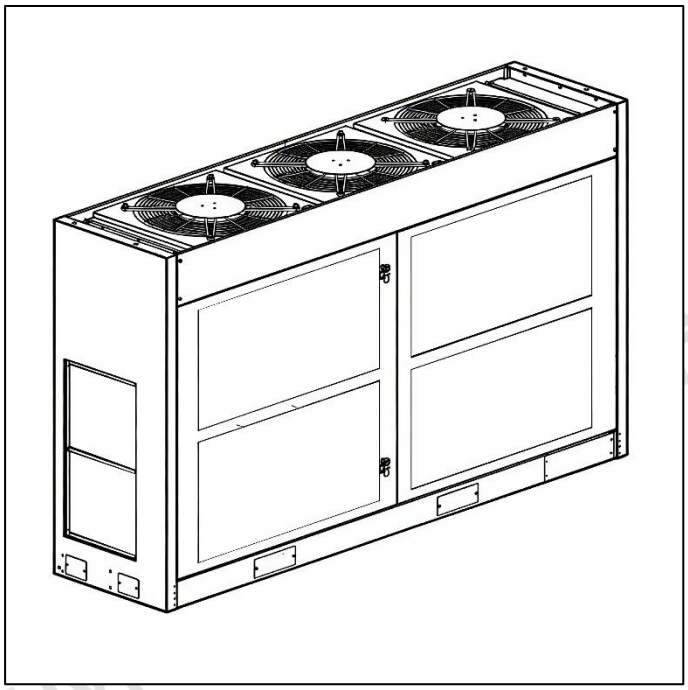
Table 3-12 Related Part Number and Part Name

No.	Name
1	Liquid-cooling Unit
2	Primary Pipe
3	Secondary Pipe
4	Tertiary Pipe
5	Automatic Exhaust Valve
6	Butterfly Valve
7	Air/Water Heat Exchanger

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3.11.2 Liquid-cooling Unit

Figure 3-22 Liquid-cooling Unit



The liquid cooling unit features an integrated design, incorporating communication with the upper-level computer, alarm functionality, and other key features. This AC-powered refrigeration unit offers high reliability, simplified installation, and does not require complex debugging.

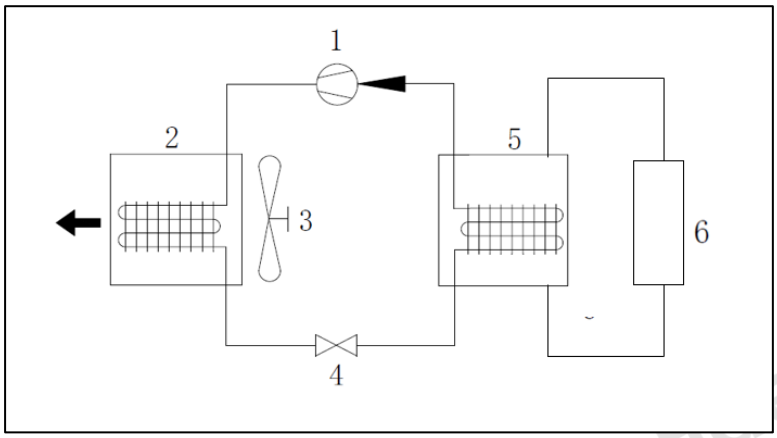
Key Characteristics of the Liquid Cooling Unit

- The unit is equipped with a dual-compressor design configuration, it achieves efficient refrigeration for low-power systems through an intelligent temperature control strategy, which optimizes compressor start-up frequency and implements a load-based frequency conversion strategy.
- The unit is equipped with a heating device; the unit activates heating up when the system temperature falls below a specified threshold.
- The unit is equipped with an automatic fluid replenishment device, which replenishes fluid automatically when it is short of fluid.
- A safety valve is integrated into the system to release pressure when it exceeds safe limits, ensuring the safety of the system's pipeline.
- The system includes liquid leakage detection functionality, issuing alarms in the event of a leak to enable early identification and resolution by personnel.

The operating system principle of the liquid-cooling unit is as follows.

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Figure 3-23 Operating System Principle of Liquid-Cooling Unit



1. Compressor 2. Condenser 3. External Circulation Fan
4. Throttle Element 5. Chiller 6. Battery Coolant Pipeline Network

3.11.1 Refrigerant Pipeline

The refrigerant (R32) within the liquid cooling unit pipeline of closed system is used for cooling the load and flows sequentially through the plate heat exchanger, compressor, condenser, and expansion valve before returning to the condenser. In the plate heat exchanger, the refrigerant absorbs heat from the load (the glycol mixture in the battery liquid cooling pipeline) and vaporizes, simultaneously generating cooling capacity.

The compressor draws in and compresses the gaseous refrigerant from the plate heat exchanger. The compressed high-temperature, high-pressure refrigerant then enters the condenser, where it is cooled by a cooling medium (such as air). The condenser liquefies the refrigerant, releasing heat. The cooled refrigerant then passes through the expansion valve, where its pressure is reduced. The cooling medium returns to the plate heat exchanger, and the gas/liquid mixture vaporizes again, absorbing heat to achieve refrigeration.

Description of Cooling Medium Pipeline

The pump draws the cooling medium (glycol mixture) from the battery cooling pipeline. After being cooled by the refrigeration system, the cooling medium is then delivered to the load. The cooling medium with high-temperature enters the plate heat exchanger for heat exchange. The cooling medium that is cooled by the plate heat exchanger continues to circulate. This heat exchange process cycles continuously.

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Table 3-13 Key Parameters of Liquid Cooling Unit

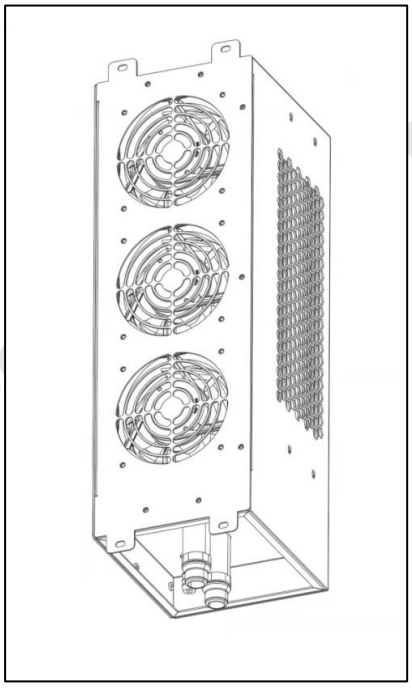
Items	Technical Parameters	Units
Rated Cooling Capacity@L45 W20	25000	W
Rated Heating Capacity	18000	W
Refrigerant	R32	
Rated Flow Rate	280	L/min
Power supply range	400±15%,50/60	V, Hz
Communication Protocol	RS485	
Dimensions	W×D×H：2250×550×1300	mm

3.11.2 Air/Water Heat Exchanger

The air/water heat exchanger provides functions such as temperature control, which can maintain the electrical cabinet temperature remains below 40℃. The key functions include:

- Cooling is activated when the ambient temperature is above 40℃, to ensure continuous and effective operation of the equipment under high-temperature conditions.
- Heating is activated when the ambient temperature is below -5℃, to prevent the startup issues caused by low temperatures.

Figure 3-24 Schematic Diagram of Heat Exchanger



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Table 3-14 Main Parameters

Technical Data	Value	Unit
Rated Cooling Capacity @L35 W20	1500	W
Air Volume, Internal Circulation	600	M³/h
Rated Voltage	24	V
IP Protection Classification	IP55	
Ambient Air Temperature	-30~55	°C
Installation Attitude	Bolt Mounted	
Dimensions(H*W*D)	200×547×168.5	mm

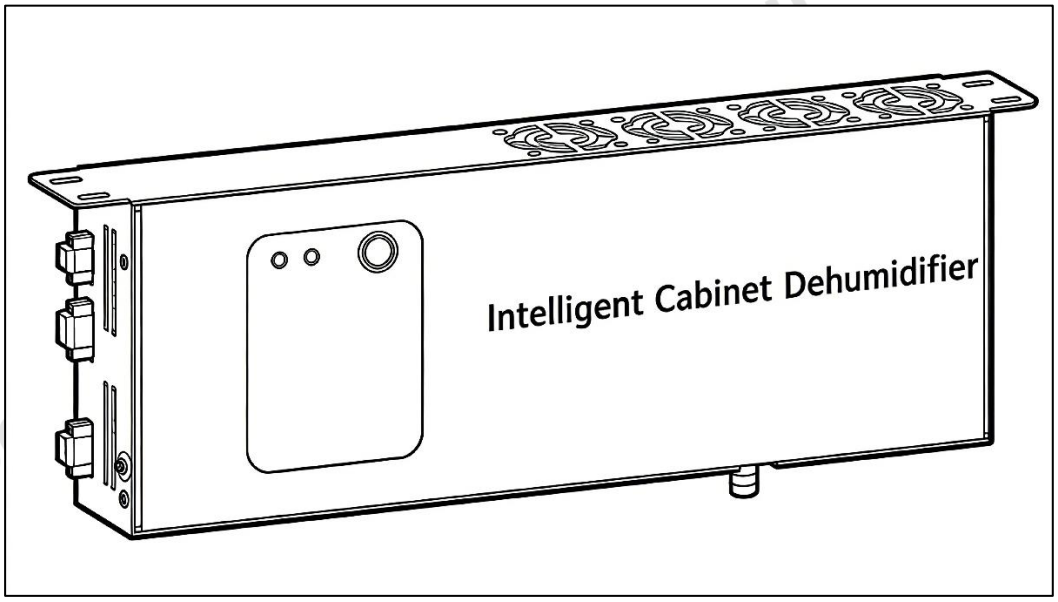
3.11.3 Dehumidifier

The dehumidifier provides functions such as humidity control, which can maintain the humidity in the battery container and electrical room within 80%RH.

The key functions are as follows:

- When the ambient humidity exceeds 80%RH, it starts dehumidification to ensure that the battery system and electrical components in the electrical room can operate effectively under appropriate humidity conditions.

Figure 3-25 Dehumidifier



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Table 3-15 Main Parameters

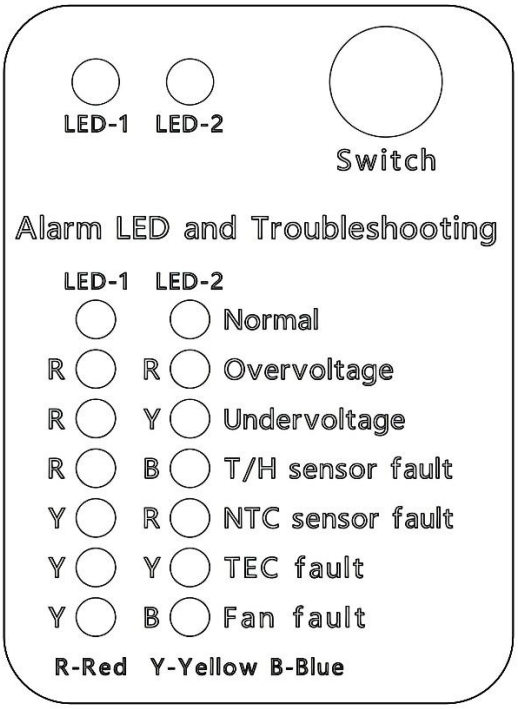
Technical Data	Value	Unit
Dehumidification Capacity	1000±10%, 35℃ 85%RH	mL/D
Rated Voltage	85V~265, 50/60	V，Hz
IP Protection Classification	IP20	
Ambient Air Temperature	5~55	℃
Installation Attitude	Door Mounted	
Dimensions(H*W*D)	142×368×50	mm

Display of Dehumidifier

Step 1 When the product is powered on, it enters the self-check mode, and the alarm indicator light and the button indicator light remain constantly on.

Step 2 After the self-check is completed, it enters the working state, and all the indicator lights turns off.

Figure 3-26 Indicator Lights



Indicator Lights of Dehumidifier

The unit is equipped with one operation indicator light (green button light) and two alarm indicator lights (tri-color lights).

- The operation indicator “Switch” shows the unit's operating status.

- During dehumidification: The operation indicator is on.
- When not dehumidifying: The operation indicator is off.

- The alarm indicators “LED-1” and “LED-2” show abnormal conditions.

In the event of an anomaly: Alarm indicator lights show the corresponding fault condition. Detailed fault descriptions are listed in the table below.

Table 3-16 Description of Alarm Indicator Lights

No.	Alarm Light 1	Alarm Light 2	Abnormalities
1	OFF	OFF	None
2	Red	Red	Overvoltage
3	Red	Yellow	Undervoltage
4	Red	Blue	Fault of Temperature and Humidity Sensor
5	Yellow	Red	Fault of NTC Sensor
6	Yellow	Yellow	Fault of refrigerator
7	Yellow	Blue	Fault of Fans

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NOTE

If multiple abnormal states exist simultaneously, they will be displayed in turns, with each state being displayed for 1 second.

3.11.4 Pipeline Network

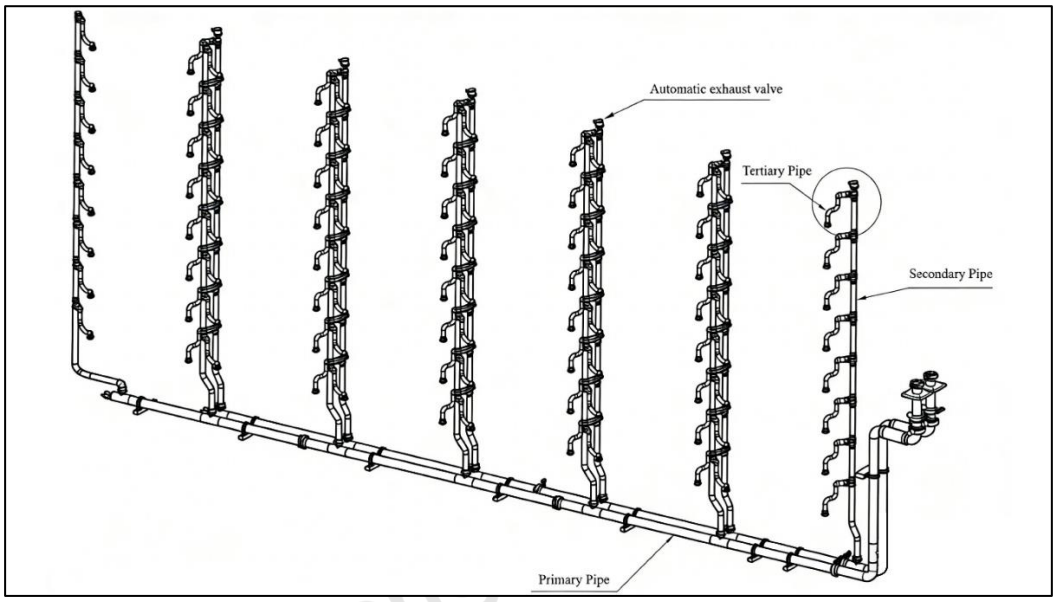
Composition of Liquid-Cooled Pipeline

The liquid cooling pipeline system consists of three hierarchical components:

- The primary pipeline,
- The secondary battery rack pipeline,
- The tertiary battery pack pipeline.

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Figure 3-27 Pipeline Network



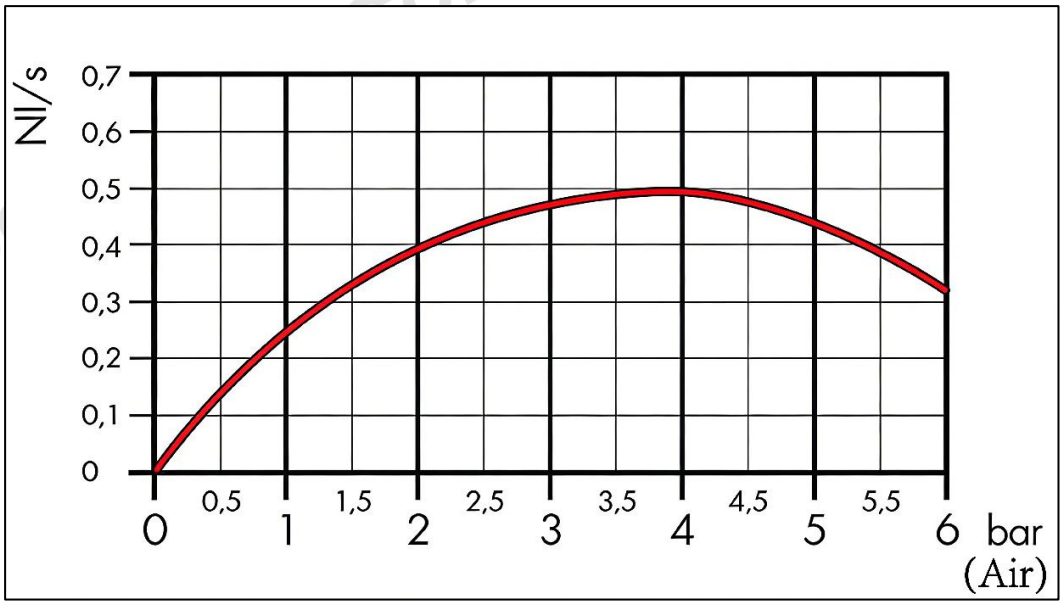
The pipeline network uniformly distributes coolant to each battery pack, ensuring consistent heat dissipation across all battery units.

Pressure Management of Liquid-Cooled Pipeline

Automatic exhaust valve is installed at the top of the secondary pipe. The valve detects whether there is gas accumulated at the high end of the pipe through an internal float or similar device.

- When the gas accumulates to a certain amount, the buoyancy of the float will be reduced, thereby opening the valve to release the gas;
- After the gas is exhausted, the liquid will flow into the valve body and the float will rise and automatically close the valve.

Figure 3-28 Curve of Pressure and Exhaust Rate



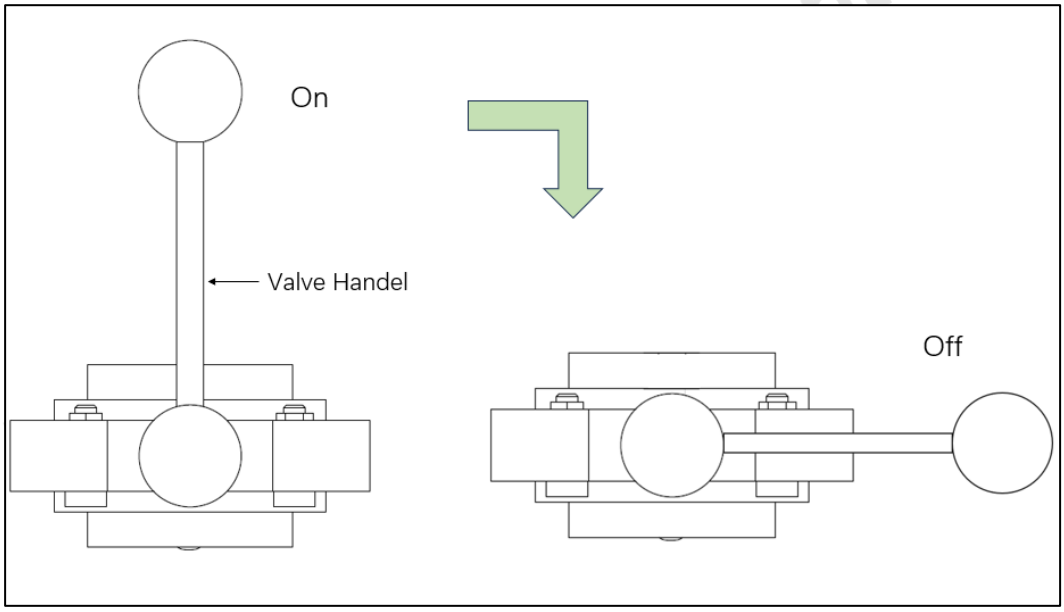
The liquid cooling system is operated within the following pressure limits:

- Maximum hydrostatic pressure:1.2bar
- Maximum working pressure: 2.5bar

Valves of Liquid-Cooled Pipeline

The butterfly valve can regulate the flow within the liquid cooling pipeline system, which can enable convenient system maintenance.

Figure 3-29 Valve Open and Closed Positions



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4 Equipment Installation

4.1 BESS Lifting

4.1.1 Basic Information of Equipment

Specifications	Value
Overall dimensions (L*W*H)	20-feet container (high box) 6058mm*2438mm*2896mm
Lifting position of the container	Corner pieces, Top lifting
Weight of container box	Approximately 7.5 tons
Net internal load capacity	Approximately 34.5 tons
The total weight of the battery prefabricated container (including batteries)	Approximately 44 tons (12 racks)

NOTICE

The lifting operation shall be conducted with a maximum load capacity of 45 tons, while taking into account the variations in the lifting process and the number of modules among different models.

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4.1.2 Lifting Precautions

⚠ CAUTION	During the lifting and transportation of the container, all standard operating procedures (SOPs) and regulations related to safety shall be strictly complied with in the country or region where the project is located.
⚠ CAUTION	The Company shall not be responsible for any personal injury or property damage resulting from the violation of relevant requirements, SOPs or regulations.
⚠ CAUTION	Throughout the entire process of BESS lifting, it is essential to strictly comply with the safe operating practice of the crane.
⚠ CAUTION	A clear safety zone of 5 to 10 meters shall be maintained around the operation area. Under any circumstances, personnel shall not stand beneath the lifting arm, the load, or any moving machinery to prevent accidents.
⚠ CAUTION	All lifting operations during adverse weather conditions shall be suspended including but not limited to heavy rain, fog, or strong wind.

4.1.3 Lifting Prerequisites

The following requirements shall be fulfilled during the lifting of the BESS container:

- All the BESS container doors shall be locked.
- A suitable crane or lifting device shall be selected based on site-specific conditions with adequate load capacity, boom length, and radius of rotation.
- Additional towing or traction equipment may be required when moving across inclined surfaces or slopes.
- The intended route shall be cleared of obstructions such as overhanging trees, power lines, or other impediments.
- The lifting operations of BESS containers should be conducted under favorable weather conditions whenever practicable and shall be suspended in adverse weather, such as heavy rain, fog, or strong wind.
- The operating location shall be secured prior to initiating the lifting process.
- During the lifting and installation operations, professional personnel shall be present to monitor the entire process.
- Warning signs or cordons shall be set up and formed to restrict access of unauthorized personnel.

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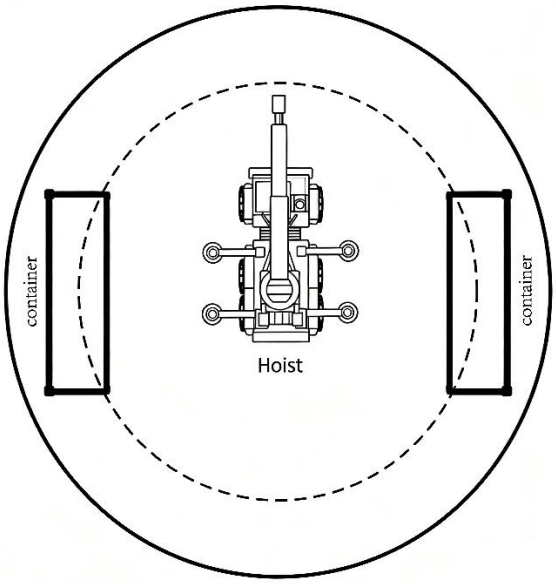
- All personnel involved in loading, unloading and bolting shall receive appropriate training, particularly about safety
- A suitable crane or lifting device shall be selected based on site-specific conditions to ensure a safety operation. The crane load-bearing capacity ≥100,000kg is preferred.
- The strength of the sling used shall support the weight of the ESS container.
- The length of the sling shall be adjusted according to specific requirements.
- All sling connections shall be secured and safe, please ensure that all sections of the sling which are connected to the corner pieces are equilong.
- The length of the sling should be adjusted appropriately based on the site-specific requirements
- The battery cabin shall not be leaning and the operation shall be smooth during the entire lifting process.
- The four telescopic pillars of the BESS shall be utilized for lifting.
- All necessary auxiliary measures shall be taken to ensure a safe and smooth lifting of the battery prefabrication module.
- The crane shall be operated strictly in accordance with the safe operating practice of the crane during the lifting process of the battery cabin.
- DO NOT stand within 5~10m of the operation area. Standing beneath the lifting arm or moving machine shall be strictly prohibited to avoid accidents.
- Lifting operations shall be suspended in adverse weather conditions, such as heavy rain, fog, or strong wind.

NOTICE

The following figure illustrates the crane operation while lifting the BESS container. The inner dotted circle in the figure indicates the working area of the crane. Standing inside of the red circle shall be strictly prohibited during the operation of the crane.

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Figure 4-1 Schematic Diagram of the Crane Operation



4.1.4 Lifting Process

All operation items shall be carried out in accordance with the following requirements during the lifting of the BESS:

Step 1 The BESS shall be lifted vertically and shall not be dragged on the ground, the roof of the lower box, or other surface. The BESS shall not be dragged or pushed on any surface.

Step 2 After the ESS is lifted to 300mm high from the support surface, the lifting shall be suspended, and the connection between the spreader and the BESS shall be inspected, and then the lifting can be only continued after secured connections are confirmed.

Step 3 During the placement of the BESS, the lowering and landing operations shall be gentle and smooth. It is strictly prohibited to place the BESS out of vertical alignment by flinging the spreader.

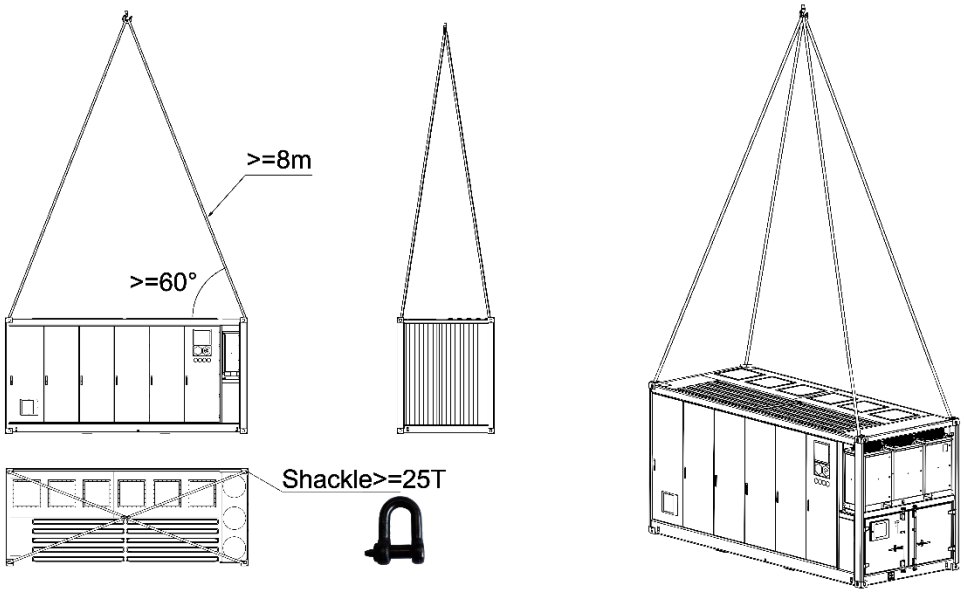
Step 4 The installation site of the BESS shall be stable, flat, well-drained, and free of obstacles or protrusions.

4.1.5 Lifting Program

Special care should be taken to protect each side of the container to be scratched. Please refer to the following figure for lifting guidance.

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Figure 4-2 Container Support and Locations of Fixing Points



NOTE	The figure above is for reference only. Please refer to the actual product for details.

Table 4-1 Part and Parameter

Part Name	Parameter
Sling	Sling ≥ 4 meters, load > 25T;

4.1.6 Bottom Fixing

Welding fixation method is recommended.

The container bottom shall be welded and secured to the foundation. After the container is welded, anti-corrosion treatment shall be applied and the bottom edges shall be sealed.

4.2 Mechanical Installation

CAUTION	During the entire process of mechanical installation, relevant standards and requirements of the project site shall be strictly followed.

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4.2.1 Pre-Installation Inspections

Inspection of Deliverables

Check the completeness of the deliverables according to the attached packing list.

Inspection Equipment

- Verify that the received container is consistent with the model that as ordered.
- Ensure that there is no damage on the BESS container and internal equipment.

⚠ CAUTION	Only an intact ESS container shall be installed.Please confirm the following items prior to installation: • The unit itself is intact. • The equipment in the ESS is intact. If any problem is recognized, contact the transporter or the Company immediately.

4.2.2 Environmental Requirements of Installation

Site Selection Requirements

When selecting the installation site, the following factors such as climate, and geological conditions (e.g. stress wave emission, underground water level, etc.) shall be taken into account with due consideration.

- The surrounding environment shall be dry and well-ventilated.
- There should be no tree around the installation site prevent obstruction from the doors of the system, and the air inlet shall not be covered by branches or leaves carried by high wind.
- The site shall be far from areas where toxic and hazardous gases are concentrated, as well as flammable, explosive and corrosive substances.
- The system shall be installed far from residential areas to avoid noise.

Foundation Requirements

⚠ CAUTION	The overall weight of the BESS is significant, and the conditions of the installation site (primarily geological limitations, environmental factors, and climatic conditions, etc.) shall be thoroughly determined prior to the construction of the foundation. The design and construction of the foundation shall be started only when the evaluation is completed.

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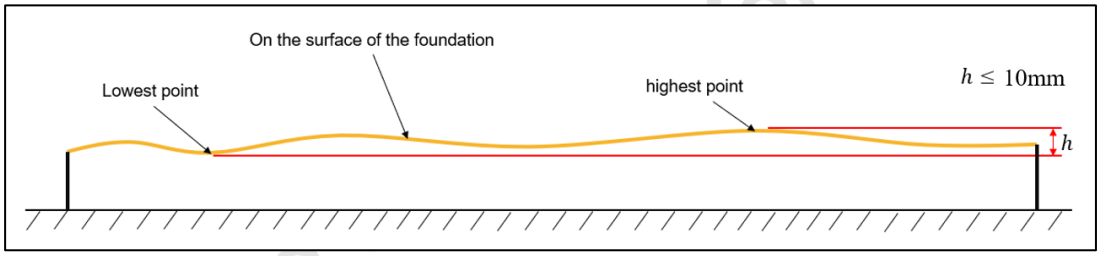
⚠ CAUTION	An improper foundation construction plan will result in difficulties or complications in the placement of the ESS container, the opening and closing of the doors, and the subsequent operations.

The foundation of the BESS container shall be designed and constructed in accordance with specific standards in advance to meet the requirements of mechanical support, cable routing, and later maintenance and overhaul.

The following requirements shall be fulfilled during the construction of the foundation:

- The soil at the installation site shall be solid. It is recommended that the relative soil compactness at the installation site shall be ≥98%. If the soil is soft, appropriate measures should be taken to ensure the stability of the foundation.
- The bottom of ditch where the foundation will be constructed shall be compacted and filled to provide adequate and effective support for the container.
- The foundation shall be constructed according to a provided plan or a protocol validated by HyperStrong. The flatness tolerance of the foundation shall be within ±5mm. Specifically, the height difference between the lowest point and the highest point of the upper surface of the foundation shall not exceed 10mm, as shown in the following figure.

Figure 4-3 Foundation Flatness Requirement



- The energy storage container shall be installed at heightened platform to prevent the foundation and interior equipment from plurosion.
- The cross-sectional area and height of the foundation shall meet the specified requirements.
- Drainage system shall be equipped with in accordance with the local geological conditions.
- A concrete foundation with sufficient cross-sectional area and height shall be constructed. The exploration unit will measure the size of the foundation based on the geological features of the site.
- Cable routing should be considered in advance during foundation construction

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⚠ CAUTION	The spoil (earth and rock excavated or dredged from foundation ditch) excavated during foundation construction shall be removed immediately to avoid affecting the subsequent lifting operations of ESS containers.

- The maintenance platform should be constructed around the foundation to facilitate later maintenance.
- According to the location and the size of upper cable entrances and exits of the energy storage containers, sufficient space shall be reserved for the cable troughs on the AC/DC side, which the cable conduits can be pre-embedded during the construction of the foundation.
- The specifications and number of tubes that will be inserted shall be determined according to the cable types and the number of inlet and outlet lines.
- A drainage system should be constructed to prevent waterlogging at the bottom of the energy storage container or interior equipment during the rainy season or heavy precipitation.
- The both ends of all pre-embedded pipes should be temporarily sealed to prevent from contaminants entering which will complicate wiring later.
- After all the cables are connected, the cable entrances, exits, and joints should be sealed with fireproofing mud or other suitable material to prevent from rodent entry.
- Grounding units shall be pre-embedded according to the relevant standards in the country or region where the project is located.

Restricted Locations

A pre-assembled battery system shall not be installed:

- In restricted locations as defined for switchboards in AS/NZS 3000;
- Within 600mm of any exit;
- Within 600mm of any vertical side of a window or building ventilation that ventilates a habitable room;
- Within 600mm of any hot water unit, air conditioning unit or any other appliance not associated with the pre-assembled battery system;
- Within 900mm below any of the items included (b), (c) and (d);
- In ceiling spaces;
- In wall cavities;
- On roofs, except where specifically determined as suitable;
- Under stairways;
- Under access walkways;
- In any evacuation route or escape route.

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Installation Space Requirements

In order to ensure optimal air intake for the air inlet, sufficient space should be reserved around the BESS container installation site. The minimum space requirement is illustrated in the figure below.

- The recommended distance between the long sides of containers is 3500mm as shown in the figure below, with a minimum limit of 3000mm (the installation base shall not exceed the outline of the container).
- The recommended distance between the short sides of two containers is 150mm, with a minimum limit of 100mm.

Figure 4-4 Individual Container Installation

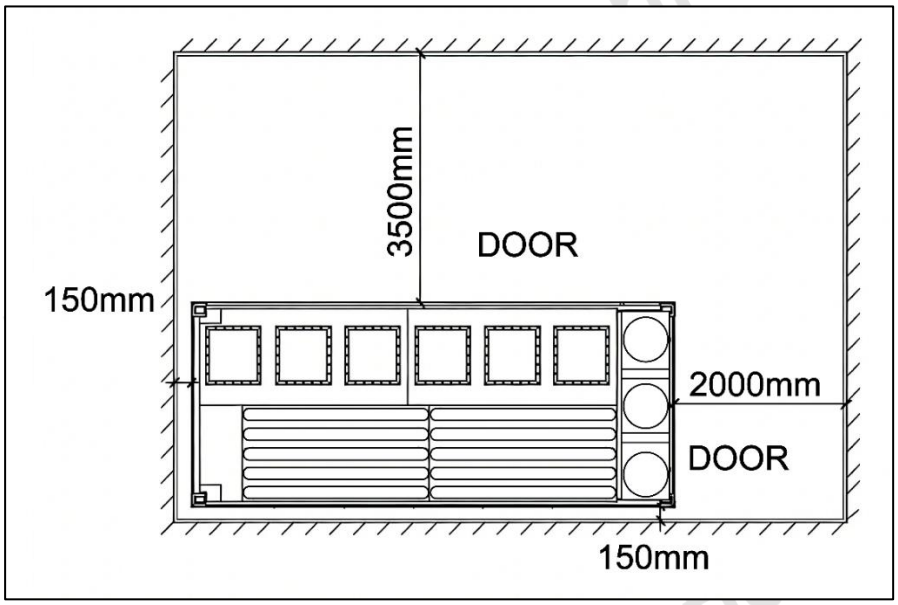
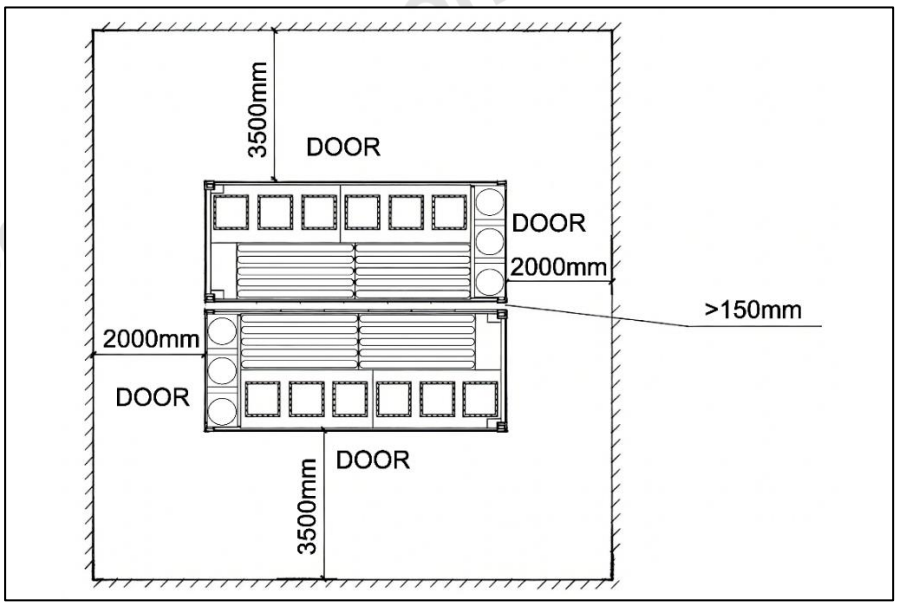


Figure 4-5 Multi Container Installation

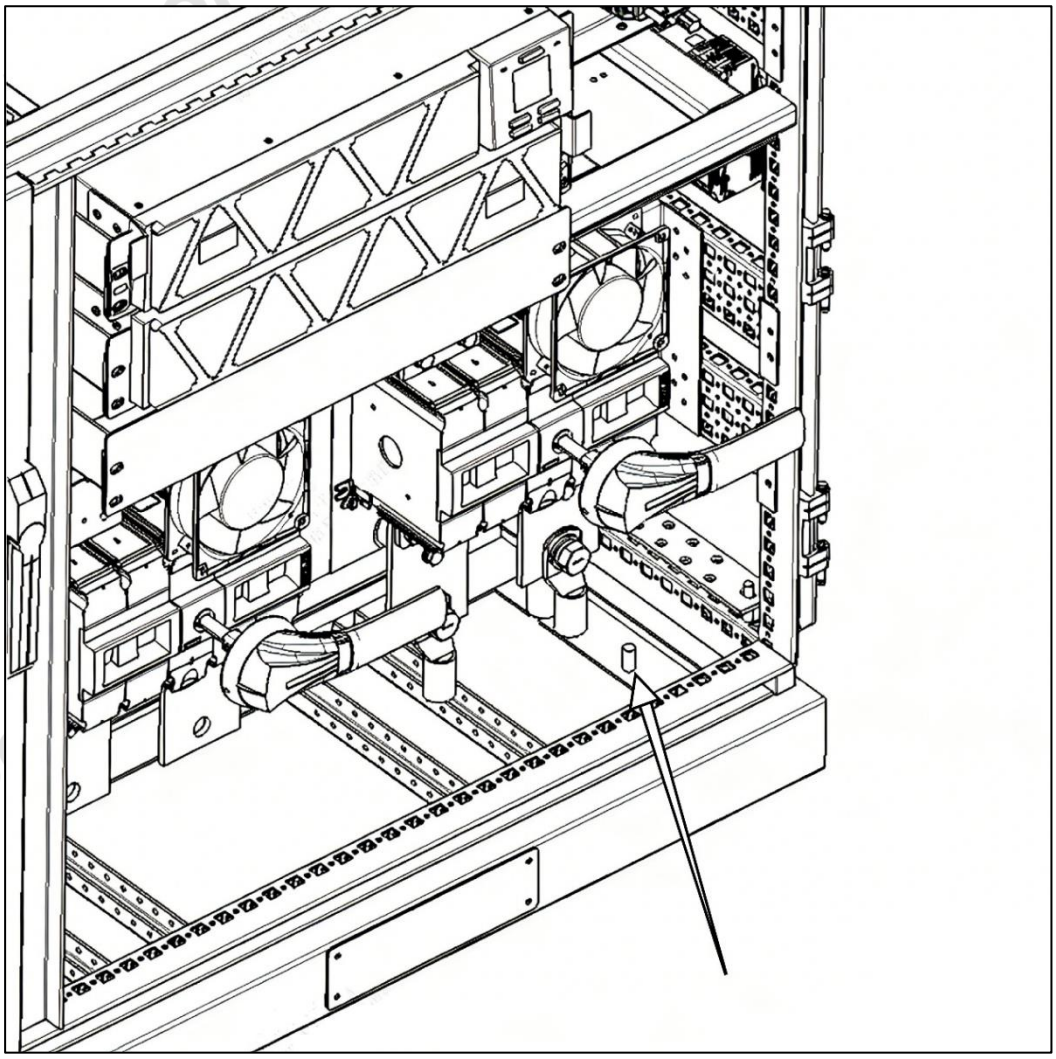


NOTE	The above figures are for reference only, please refer to the actual product for details.
NOTE	Multiple units should be placed side by side and back-to-back in accordance with the latest version of the " HyperBlock III Layout Guide " document.

4.3 SCDC Grounding

The SCDC (System Control and Distribution Cabinet) is equipped with two grounding terminals located at the bottom of the cabinet, please refer to Figure 6-6. The grounding terminals are M10 welded bolts with a height of 20mm. Grounding wire shall be configured based on project requirements and secured using M10 flat washers, M10 nuts, and M10 spring washers with an installation torque of 20Nm ±1Nm.

Figure 4-6 Grounding Diagram of SCDC



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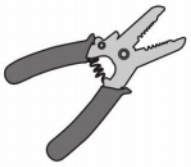
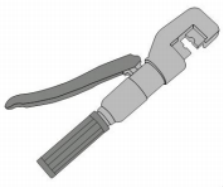
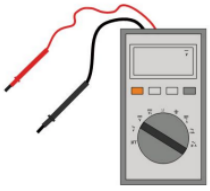
5 Product Electrical Installation

5.1 Setup Tools Preparation

Personal Protective Equipment

		
Safety gloves	Safety shoes	Safety helmet

Tool Preparation

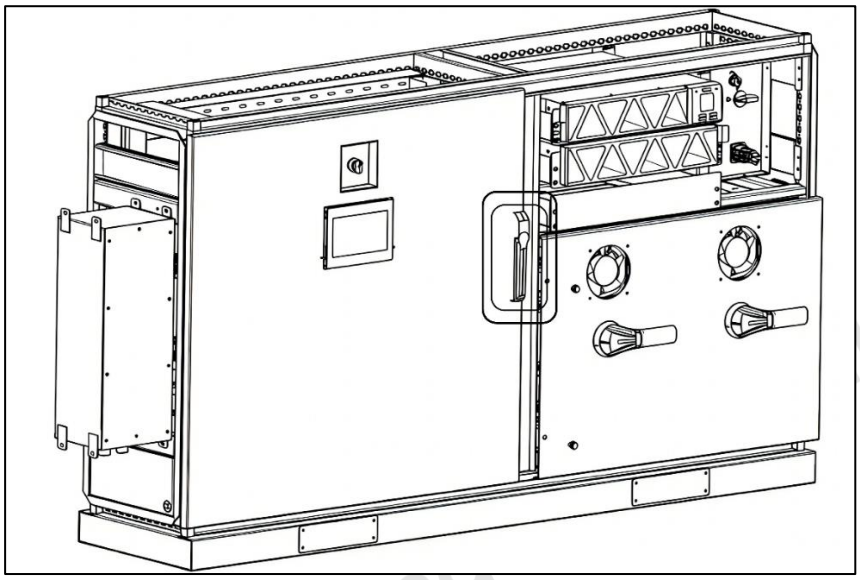
		
Screwdriver	Wire stripper	Hydraulic tong
		
Heat gun	Multimeter	M10 socket wrench

5.2 Disassembly of SCDC Guard Plate

Step 1 Open the left door of the System Control and Distribution Cabinet (SCDC);

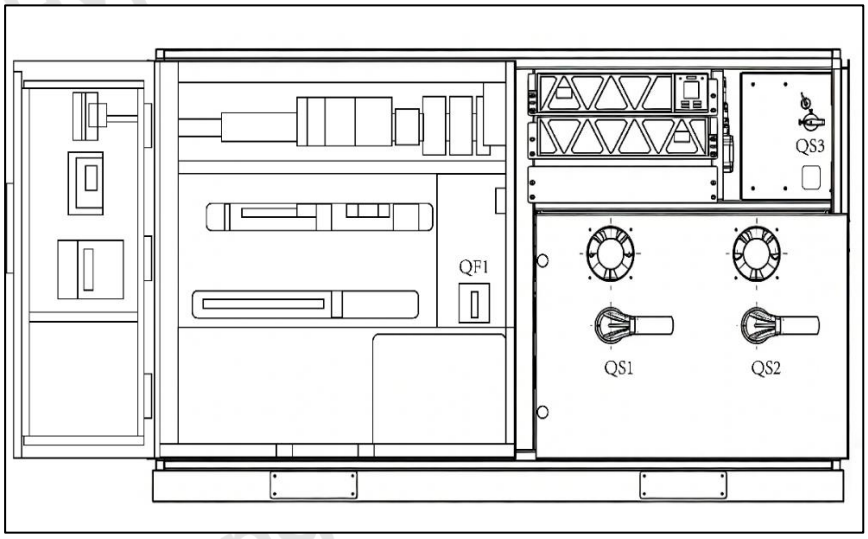
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Figure 5-1 Door Knob of SCDC



Step 2 Check the status of QF1, QS1, QS2, QS3, which shall be disconnected;

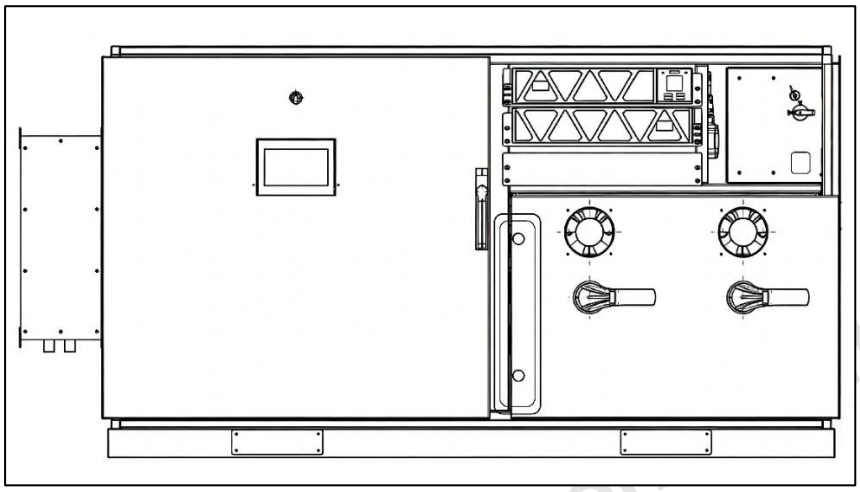
Figure 5-2 Disconnection Position for QS1, QS2, QS3 and QF1



Step 3 Remove the thumb screw of the sealing plate;

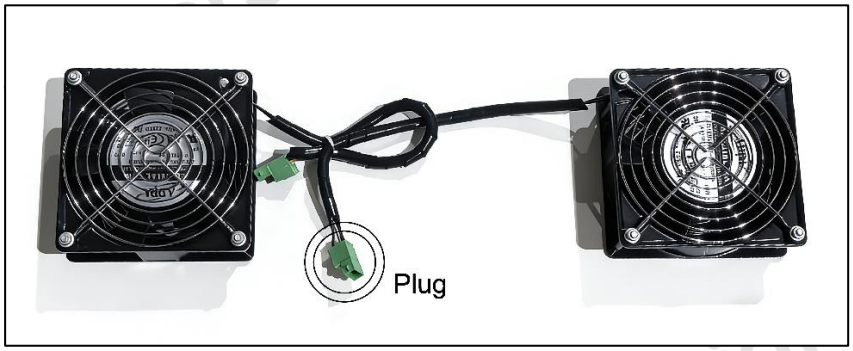
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Figure 5-3 Position of Thumb Screw



Step 4 Disconnect the plug of the fan;

Figure 5-4 Plug of Fan



Step 5 Hold the handle;pull the SCDC Guard Plate to open it;

Figure 5-5 Position of Handle

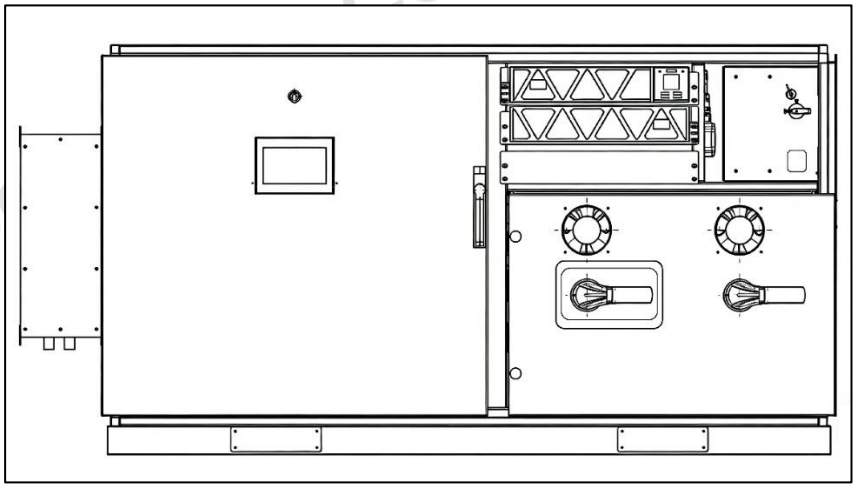
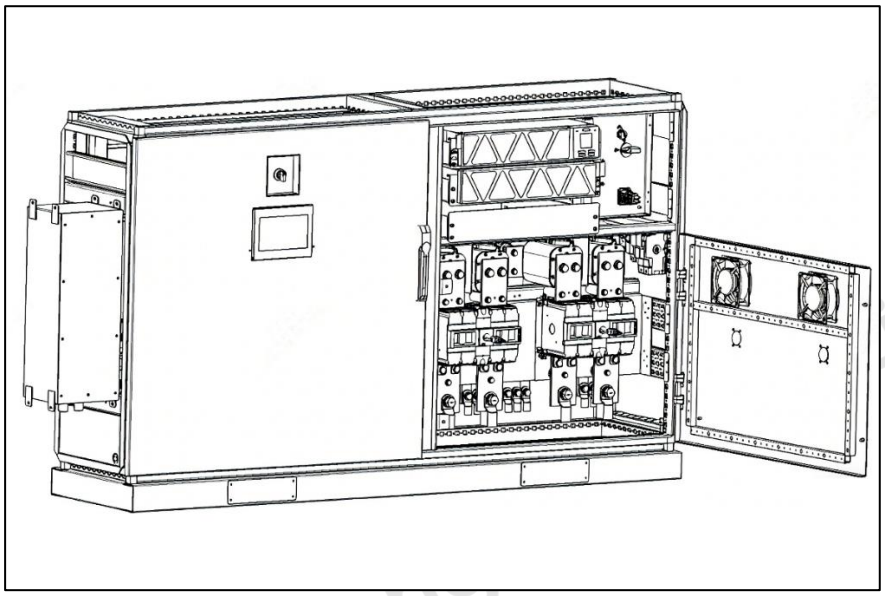
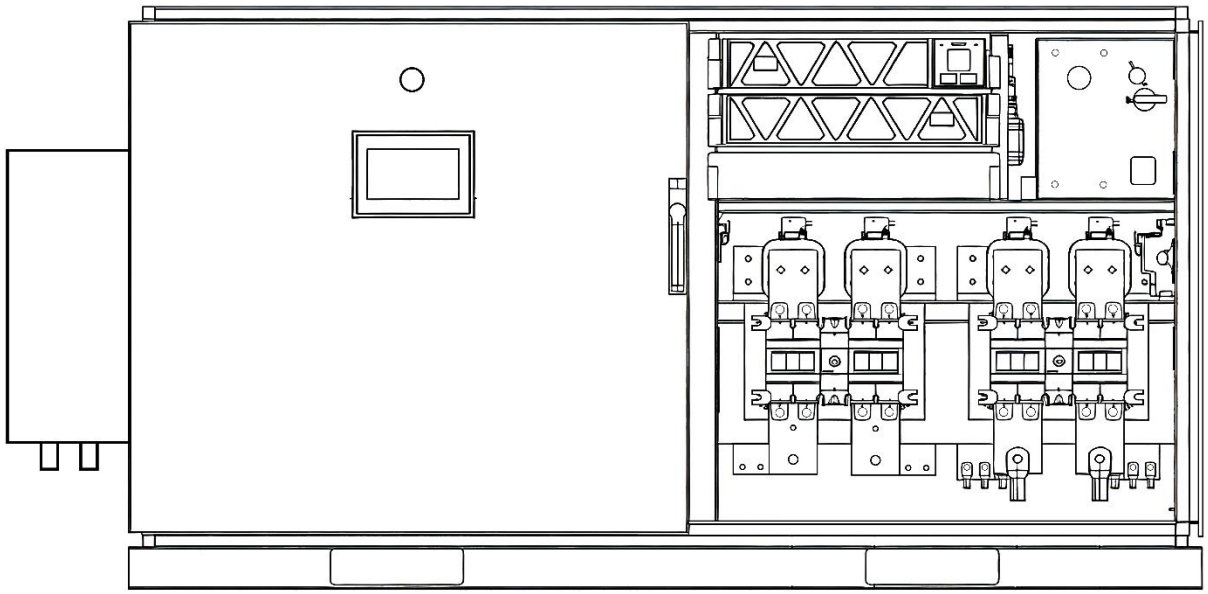


Figure 5-6 SCDC with Door Opened



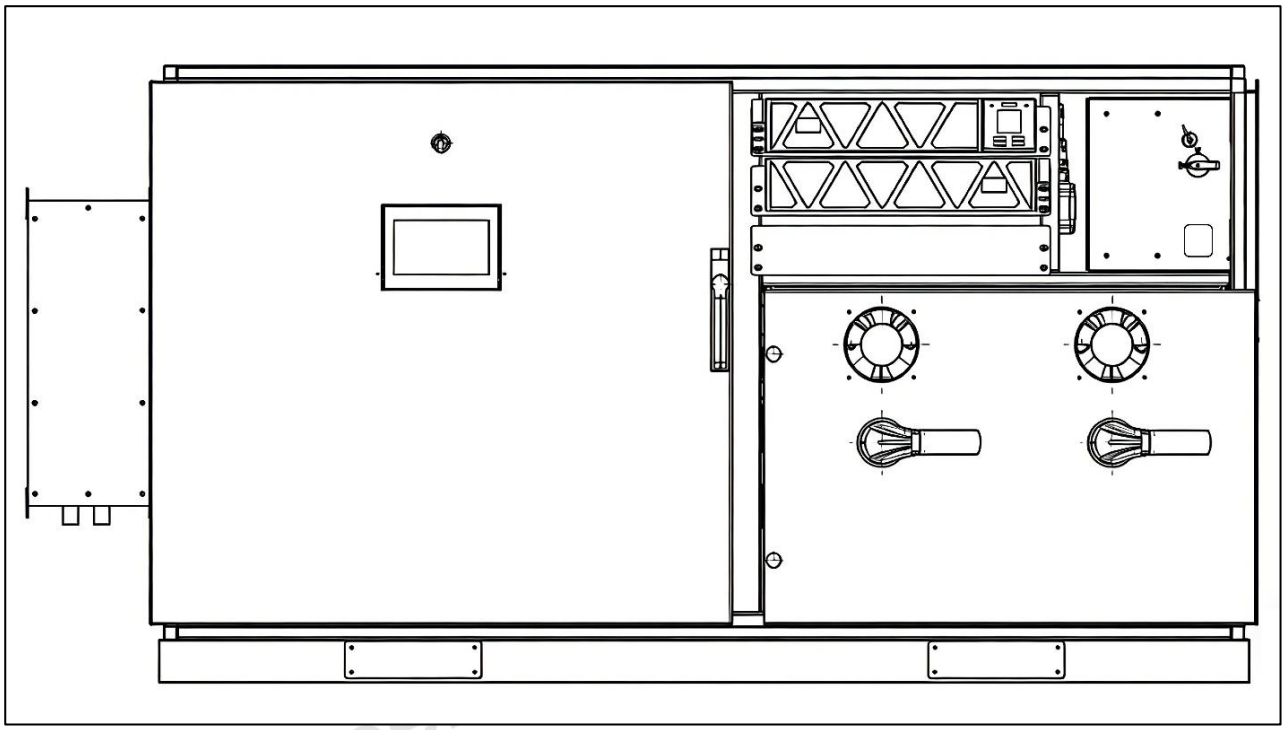
Step 6 The busbar area of the SCDC can be seen.

Figure 5-7 Busbar Area without Guard Plate



NOTICE when the works are completed, return the guard plate to the original position and re-tighten the thumb screw.

Figure 5-8 Busbar Area with Guard Plate



5.3 Key Box

The key box is located on the inside of the cabinet door in the electrical room (as shown in the figure. The key box contains the following items:

- Cabinet door key
- Cabinet door key of water-cooled unit
- Extinguishant control box door key
- Fire-fighting manual-automatic mode switching key
- Emergency release button key
- Fire reset key

For complete details, please refer to the keys of the actual product.

Figure 5-9 Schematic Diagram of the Key Box



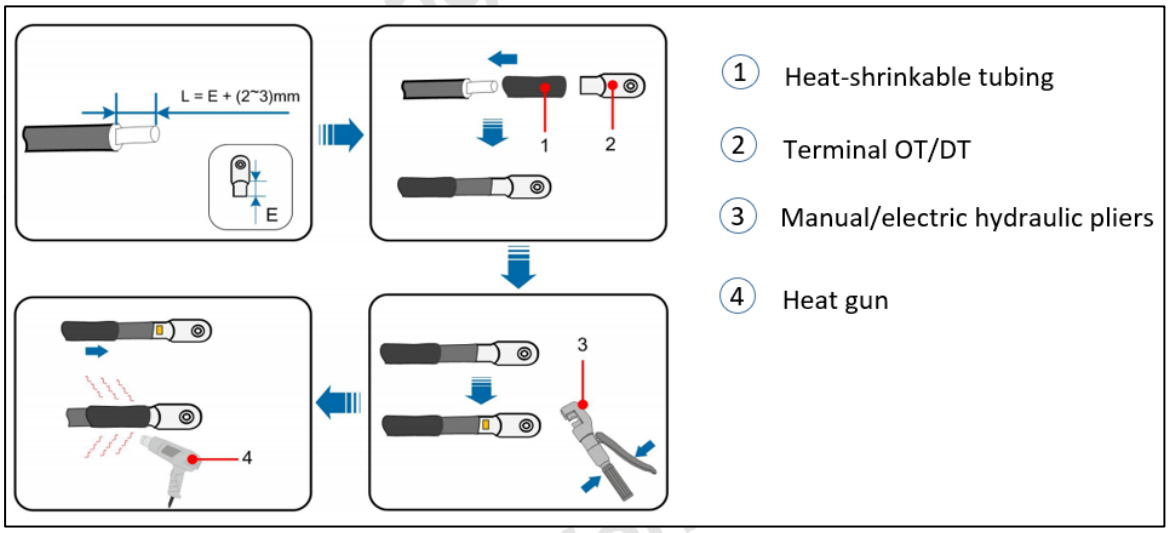
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5.4 Cable Harness Preparation

The selected cables shall meet the following conditions:

- Adequate current-carrying capacity. The factors that affect the current-carrying capacity of the conductor include but not limited to environmental conditions, material type of conductor insulation, cable routing, cable material, and cross-sectional area.
- The cable diameter shall be selected in accordance with the maximum current-carrying capacity, and there should be a margin for the length.
- All DC input cables shall have the same specification and material.
- The specifications and materials of three-phase AC output cables shall be the same.
- Flame-retardant cables shall be used.

Figure 5-10 Crimping of OT/DT Terminals



NOTICE

Armored cables need to be considered with a single-ended grounding of the shield, which can be on the BESS side or the PCS side.

5.5 DC Cable Connections

Each rack converges at the DC busbar with 50mm² (1/0 AWG) DC power cable, the DC lug is shown in the figure below.

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Figure 5-11 DC Connection Terminal

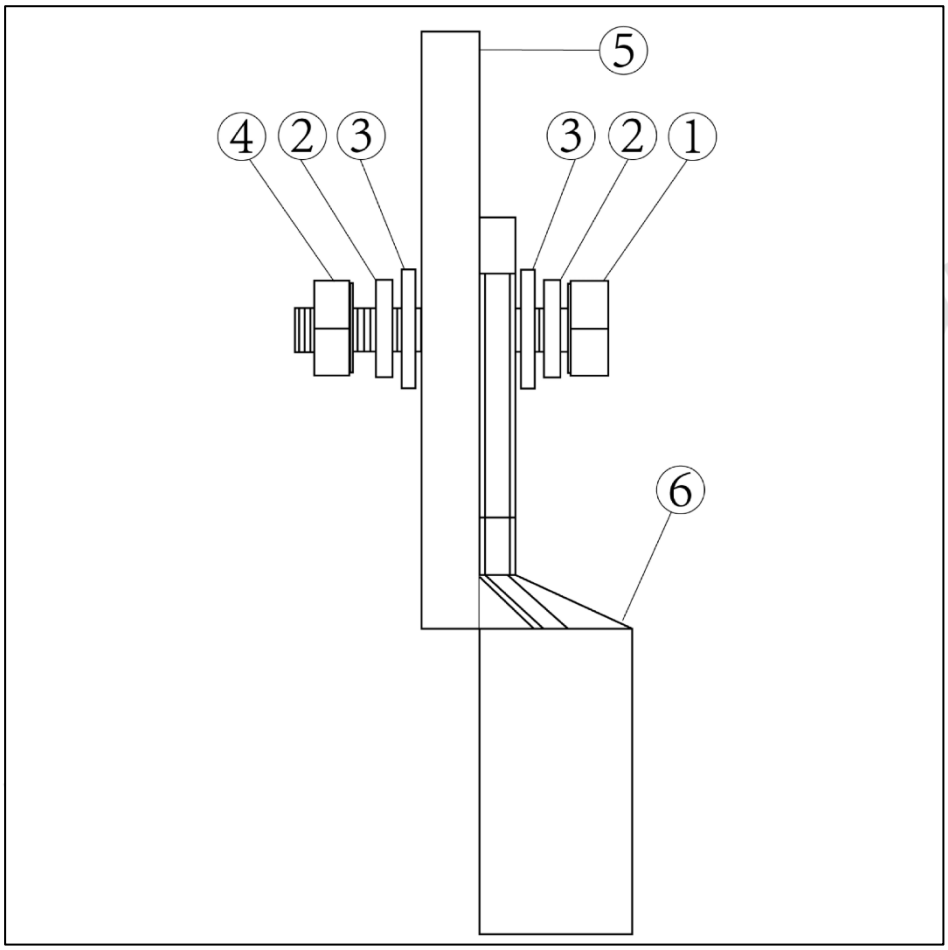


Table 5-1 Part Number and Part Name

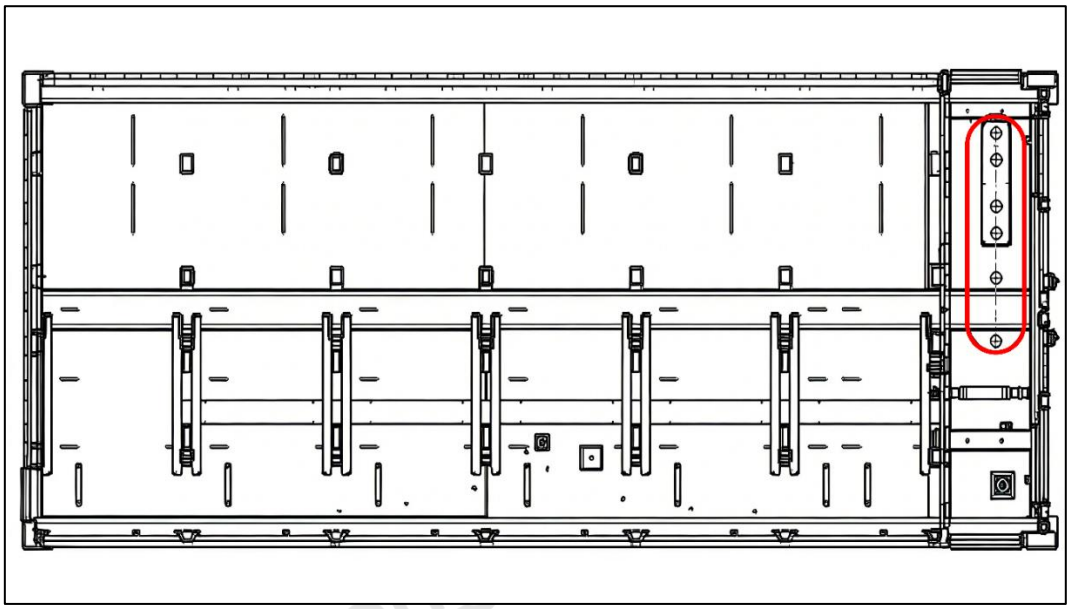
No.	Name	No.	Name
1	Bolt	4	Nut
2	Spring Washer	5	Copper Row
3	Flat Washer	6	Copper Tip

5.6 Cable Access Port

The cable inlet of the BESS container is located at the bottom of the SCDC, and the external access cables are uniformly fed from the bottom. The inlet is sealed by a tower-shaped sealing ring, which needs to be punctured during threading. After wiring, the inlet should be tightly sealed with fireproof mud or other suitable materials.

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Figure 5-12 Cable Access Port (for reference only)



NOTE	Please refer to the detailed design documents for the specific project for the exact dimensions and locations of the cable holes.

5.7 Container Grounding

There are two grounding methods: welding with grounding flat steel or fixing with a grounding cable.

5.7.1 Grounding Flat Steel

Remove the protective tape from the earthing point and weld a piece of hot-dip galvanized flat steel with a size of 50mm×100mm to the earthing point. When the welding is completed, the entire fixing surface shall be sprayed.

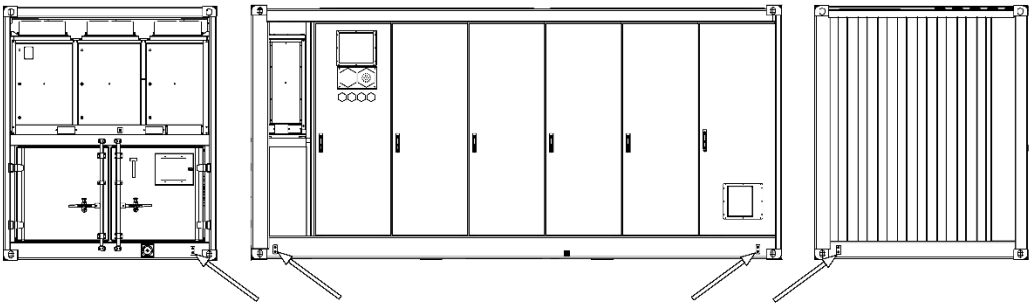
5.7.2 Grounding Cable

Reliably connect the two grounding points to the BESS grounding point with a 70mm2~95mm2 grounding cable. (Insulating tape is attached to the factory grounding point. The type needs to be removed before wiring).

Crimp the terminals and secure the cables by using M10 bolts, recommended torque range: 35-40 Nm.

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Figure 5-13 Grounding Points

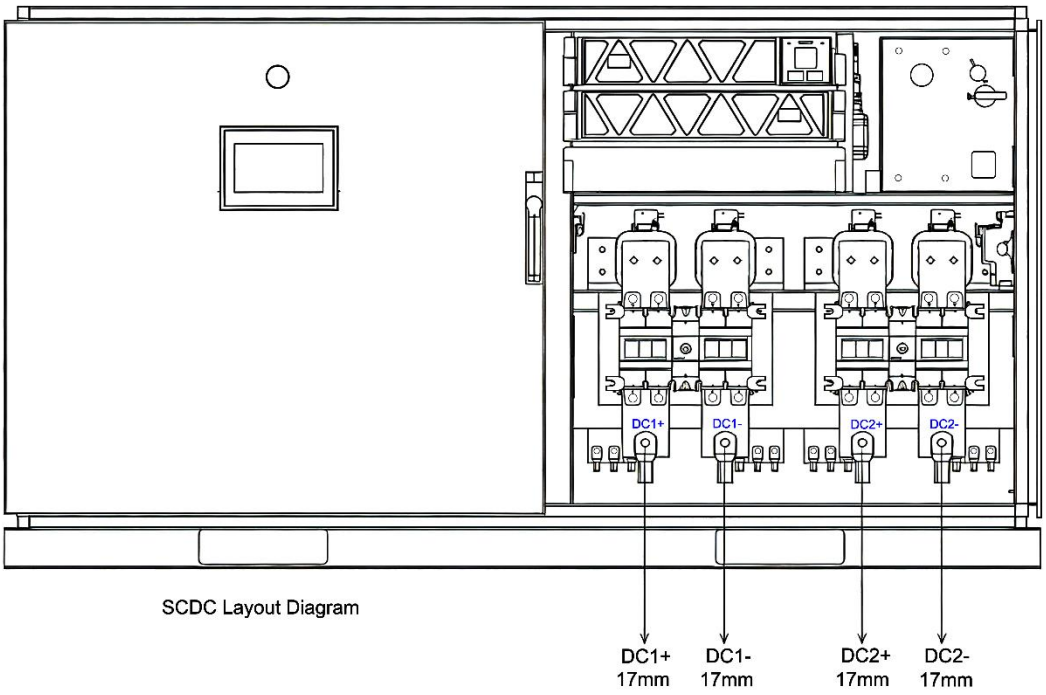


5.8 DC Input Connection

5.8.1 DC Busbar Terminals

The DC busbar terminals are shown in the figure below.

Figure 5-14 DC busbar terminals



5.8.2 DC Cables

It is recommended to select the size of 240mm² copper for single pole DC cables.

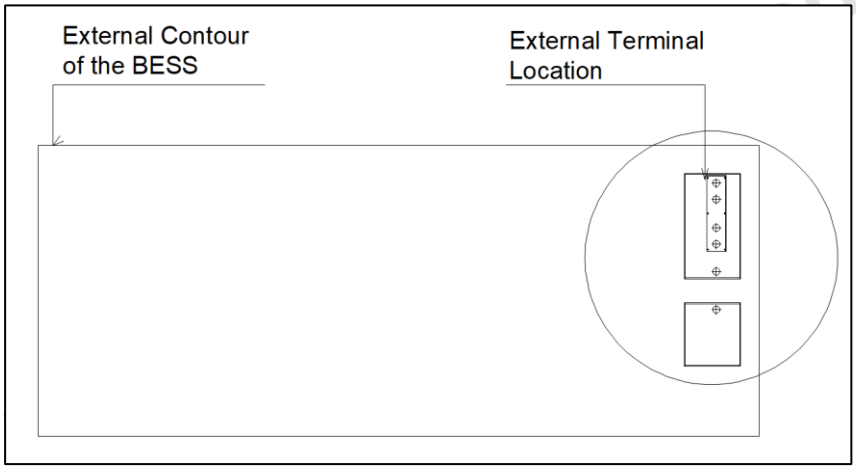
For specific cable selection and recommended quantities, please refer to the "Recommended Station-Level Cable List".

NOTE	The above figures are for reference only, please refer to the actual product for details.

5.8.3 Operating Procedures

Step 1 Insert the cable into the inlet hole and into the wiring area of the copper row, ensure that the polarity of the cable is marked.

Figure 5-15 Wiring Area(Bottom View)



Step 2 Remove the protective layer of the cable with wire strippers, and expose the copper core.

Step 3 Crimp the OT terminals.

Step 4 Secure the OT terminals to the wiring holes with M16 bolts, with a tightening torque of 32Nm~39Nm.

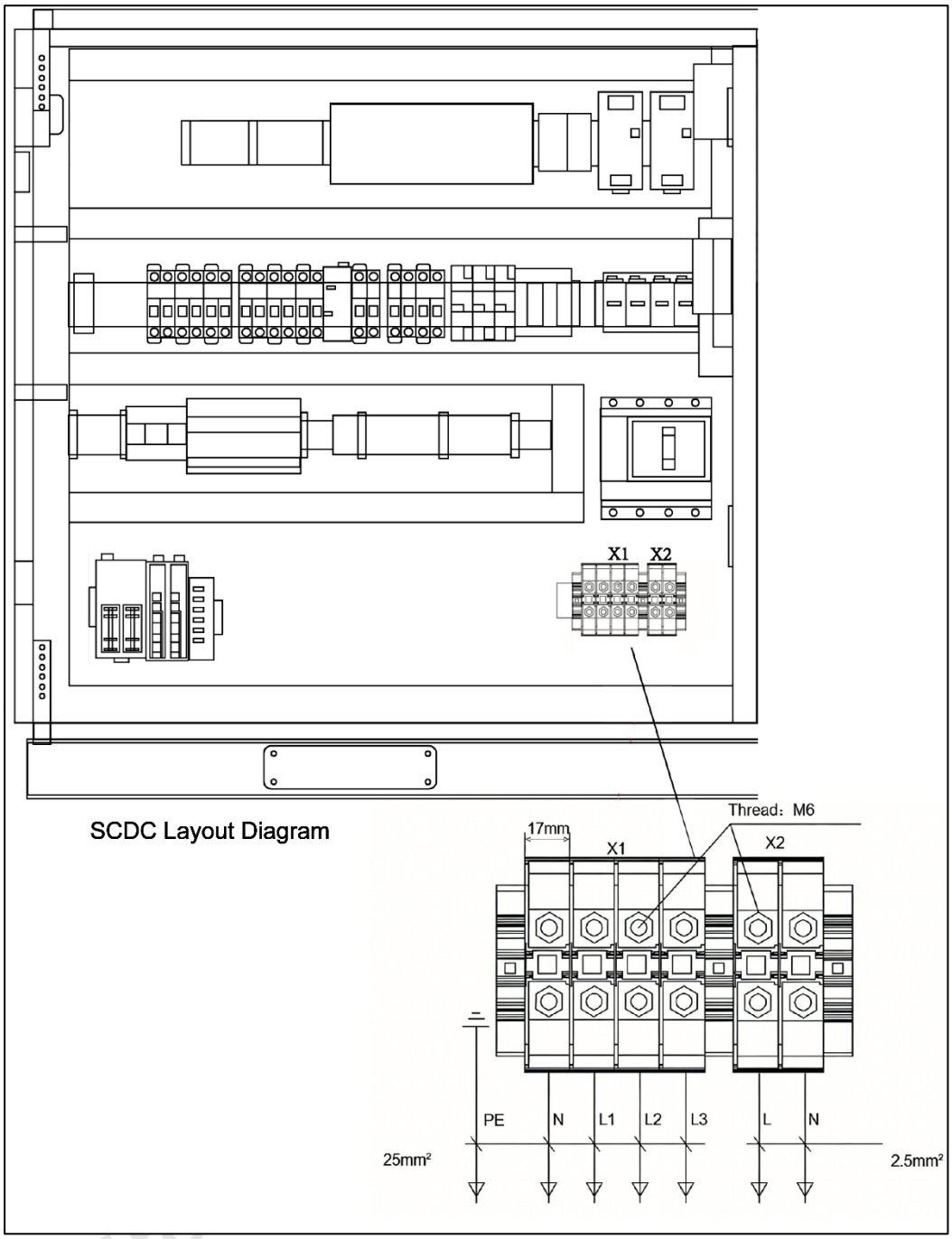
5.9 AC Input Connection

5.9.1 AC Power Input Connectors

The BESS AC power input connectors are shown in the figures below.

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Figure 5-16 AC Power Input Connector(Battery Power Version)



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Figure 5-17 AC Power Input Connector(UPS Version)

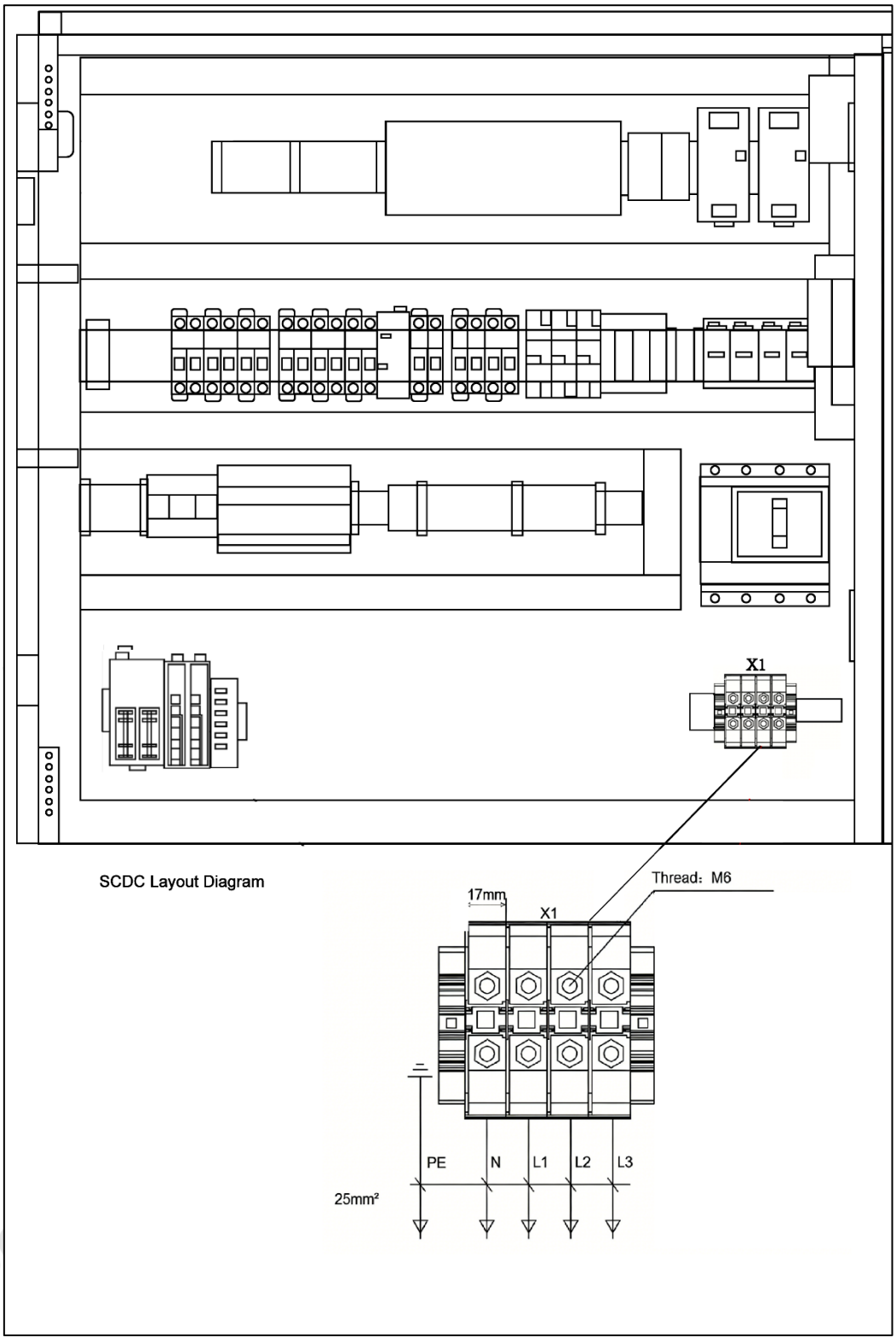


Table 5-2 Description of Part

No.	Name	Model	Picture
1	Wiring Terminals	RBO 6 (3075896)	

2	Terminal Side Panel	D-RSC 6 (3213098)	
3	Cover Plate	AP RSC-T (3059139)	

5.9.2 AC Cables

For specific cable selection and recommended quantities, please refer to the "Typical Overall Station Solution Cable List".

NOTE	The above figures are for reference only, please refer to the actual product for details.

5.9.3 Operating Procedures

Step 1 Insert the cable into the inlet hole and enter the terminal at the bottom of the SCDC's guide rail and mark the phase sequence of the cable.

Step 2 Remove the protective layer of the cable with wire strippers, and expose the copper core.

Step 3 Crimp the OT terminals.

Step 4 Remove the M6 nut supplied with the terminal and fix the OT terminal to the wiring hole with a tightening torque of 20Nm~25Nm.

NOTICE	The wiring shall be carried out in accordance with the labeled phase sequence.

5.10 Container External Communication Interfaces

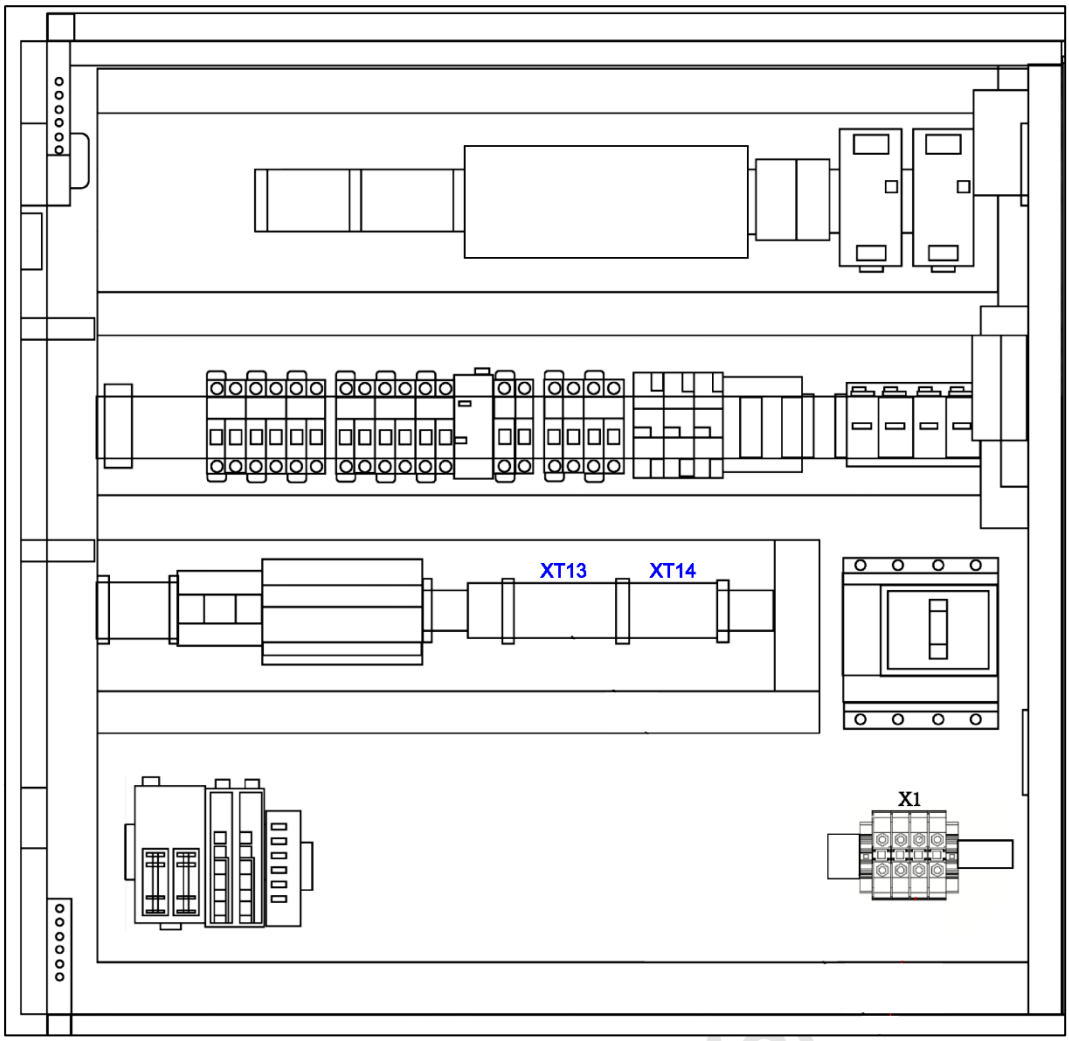
5.10.1 BESS and Fire Protection Host Communication Interfaces

Interface Position

The XT13 interface in the figure below is the communication interface between the BESS and the fire protection host.

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Figure 5-18 The Interface between Container and Station Level Fire Protection Host



NOTE	The above figures are for reference only, please refer to the actual product for details.

Connection Cable

For specific cable selection and recommended quantities, please refer to the "Typical Overall Station Solution Cable List".

Interface Terminals

Table 5-3 Description of Terminal XT13

Terminal number	Terminal position number	Definition	Remark
XT13	1	COM	Fire equipment malfunction (Output normally open passive dry contact signal)
	2	NO	

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	3	COM	Fire gas spraying (Output normally open passive dry contact signal)
	4	NO	
	5	COM	Fire alarm (Output normally open passive dry contact signal)
	6	NO	
	7	COM	Excessive concentration of combustible gases (Output normally open passive dry contact signal)
	8	NO	
	9	COM	Combustible gas malfunction (Output normally open passive dry contact signal)
	10	NO	
	11	COM	Ventilation system malfunction (Output normally open passive dry contact signal)
	12	NO	
	13	COM	Ventilation system running (Output normally open passive dry contact signal)
	14	NO	
	15	LOOT IN+	Signaling Line Circuit (SLC) Communication
	16	LOOT IN-	
	17	LOOT ING	
	18	LOOT OUT+	Signaling Line Circuit (SLC) Communication
	19	LOOT OUT-	
	20	LOOT OUTG	

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Operating Procedures

Step 1 Insert the fire fault, sprinkler, alarm, and communication cables into the inlet holes and enter the XT13 terminal wiring area inside the cabinet, upper section.

Step 2 Connect the cables into the corresponding terminal holes of XT13 according to the definition.

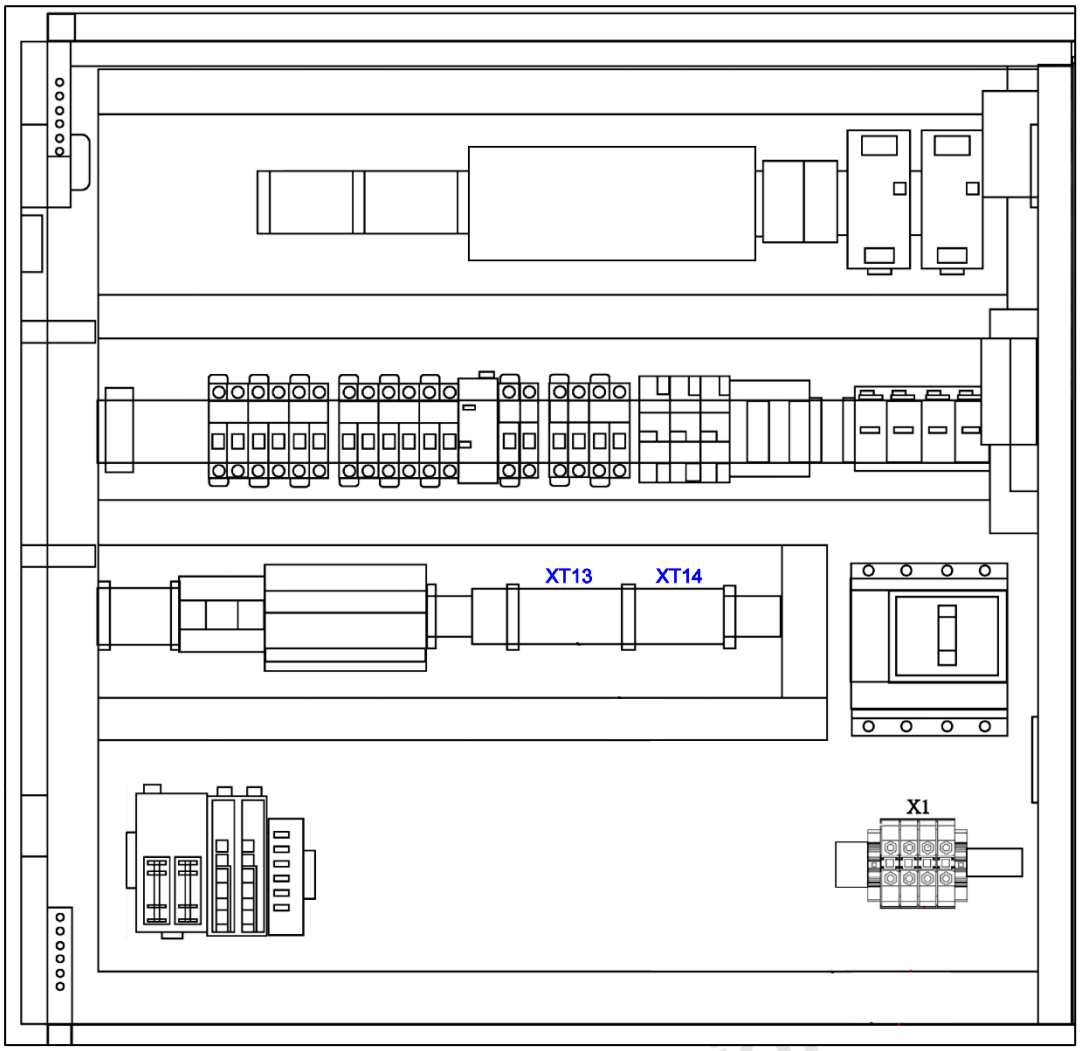
Step 3 Place the cables in the left trunking of the cabinet.

5.10.2 BESS and PCS Communication Interfaces

Interface Position

The XT14 interface in the figure below is the communication interface between the BESS and PCS.

Figure 5-19 Communication Interface between BESS and PCS



NOTE	The above figures are for reference only, please refer to the actual product for details.

Interface Terminals

Table 5-4 Description of Terminal XT14

Terminal number	Terminal position number	Definition	Remark
XT14	1	PCS1-COM	PCS emergency stop input (Output normally close passive dry contact signal)
	2	PCS1-NC	
	3	PCS2-COM	PCS emergency stop input (Output normally close passive dry contact signal)
	4	PCS2-NC	

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	5	PCS-485A	PCS communication (Reserved interface)
	6	PCS-485B	
	7	PCS-485G	

Operating Procedures

Step 1 Insert the PCS communication, PCS emergency stop input, and PCS emergency stop output cables into the inlet holes and arrange them to the XT14 terminal wiring area on the left side of the cabinet.

Step 2 Connect the cables to the corresponding terminal holes of XT14 as defined.

Step 3 Place the cables in the left wall channel of the SCDC.

5.10.3 External Ethernet Communication Interfaces

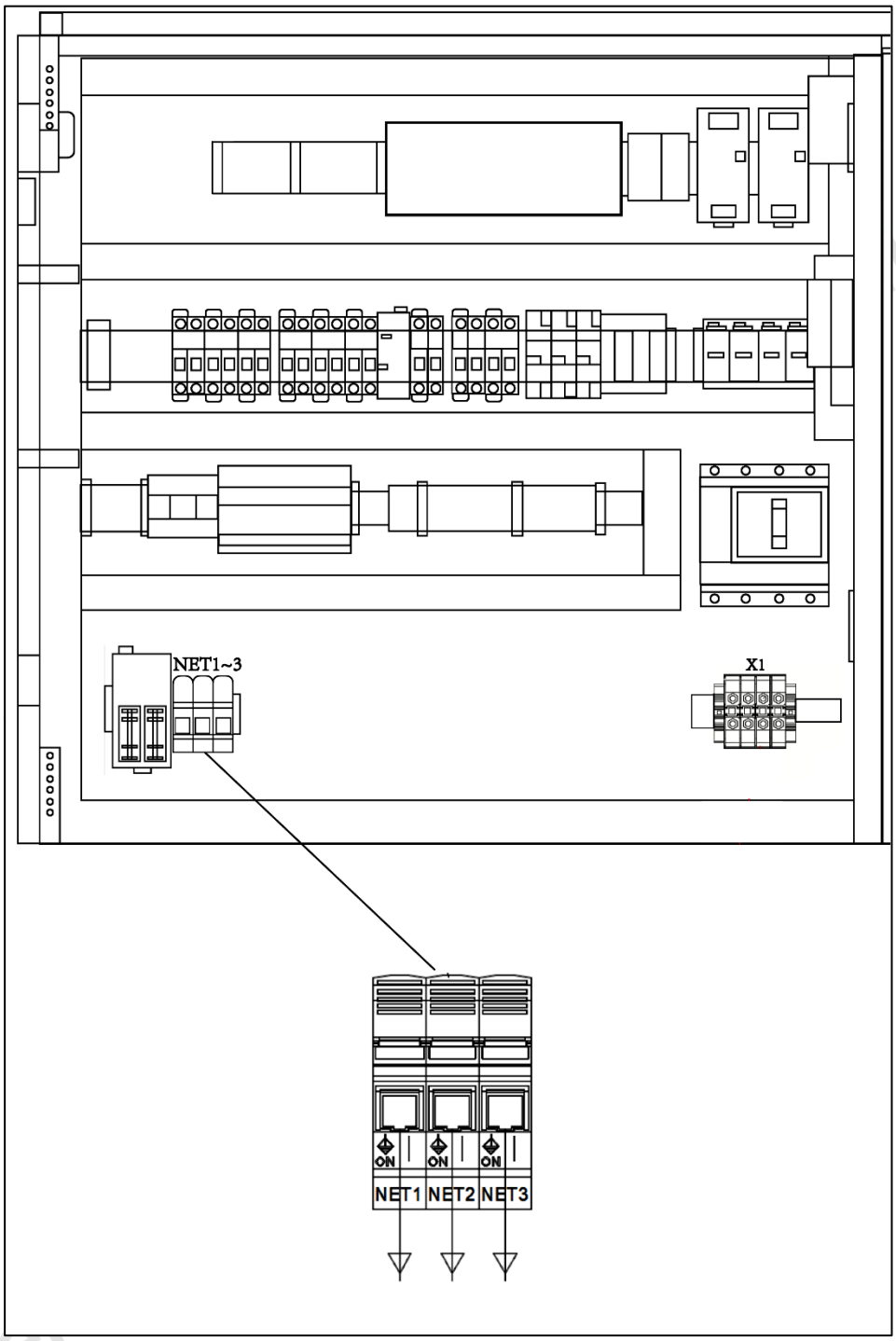
The BESS container is equipped with an A and B dual network communication system, which can support external Ethernet communication and optical fiber communication.

NET1 ~ 3

The external Ethernet communication interfaces NET1 ~ 3 are shown in the figure below.

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Figure 5-20 Location of NET1 ~ 3



NOTE	The above figures are for reference only, please refer to the actual product for details.

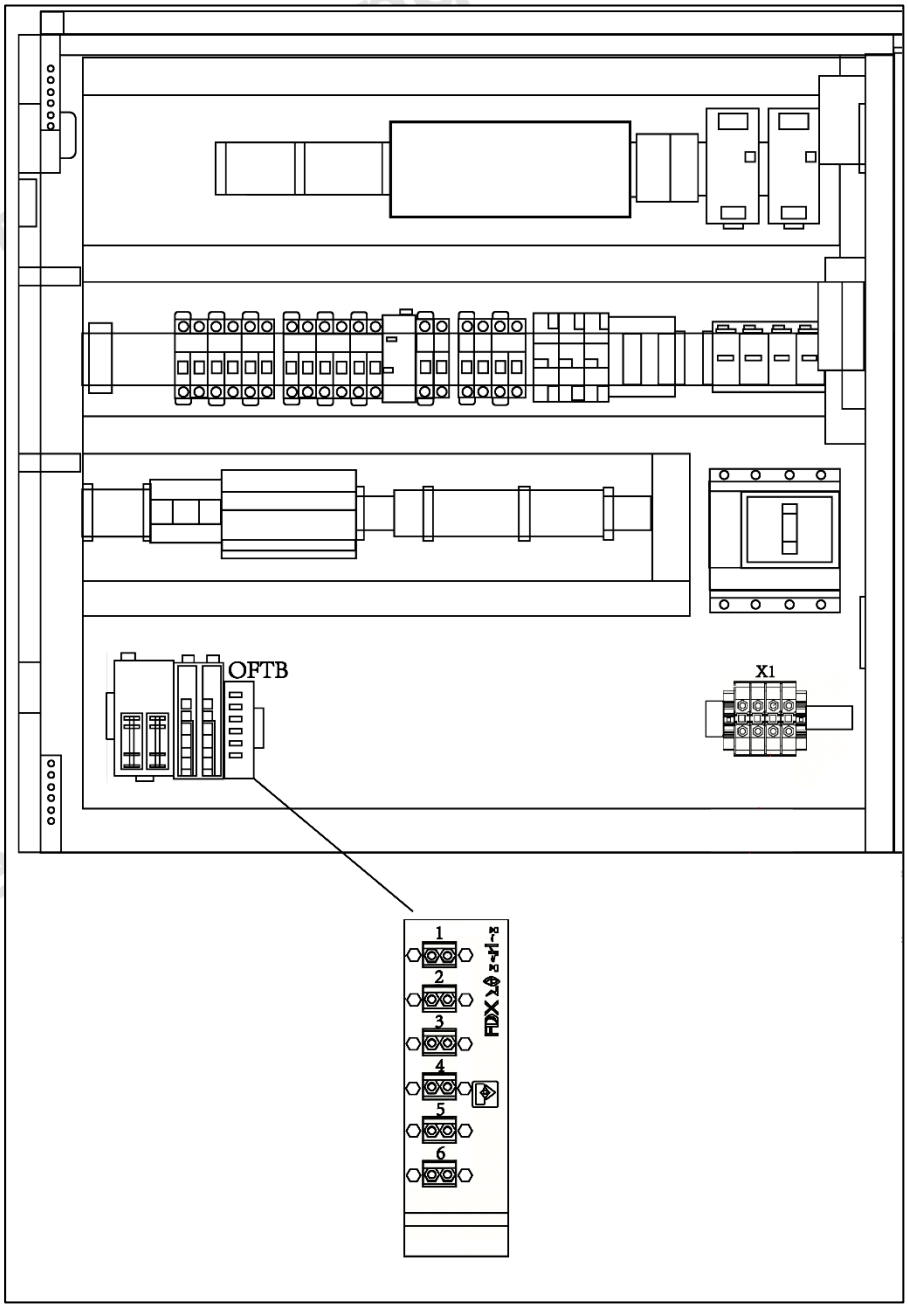
Figure 5-21 Description of NET1 ~ 3

Device Name	Interface Name	Remark
Ethernet Port	NET1	Data Network A
	NET2	Data Network B
	NET3	PCS

OFTP

The external optical fiber terminal box (OFTP) interface is shown in the following figure

Figure 5-22 Optic Fiber Communication Wiring Location



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NOTE	The above figures are for reference only, please refer to the actual product for details.

Table 5-5 Description of OFTB

Device Name	Interface Name	Remark
OFTB	1	Data Network A
	2	Data Network B
	3~6	Spare

Communication Cable Requirement

It is recommended to use network cables of category 5E or above, with a maximum length of 90 meters. If the length exceeds 90 meters, an optical fiber transceiver shall be configured for communication.

Operating Procedures

Step 1 Insert the CAT-5E cable into the inlet hole and arrange it to the wiring area on the left side in the cabinet.

Step 2 Disassemble the trunking on the left side wall in the cabinet and arrange the CAT-5E cable into it.

Step 3 Connect the Ethernet port to the external device.

5.11 Power on/off Process

CAUTION	Electrical Injuries and Arc flash.
CAUTION	• Operators shall hold the necessary legal qualifications. • Personnel should wear personal protective equipment as required during operation.
CAUTION	If any problems that may affect the battery or ESS are recognized, please contact after-sales support immediately. Unauthorized modifications to the internal equipment of the battery system shall be strictly prohibited.
CAUTION	If any abnormalities are observed in the operations of storage battery system, please stop the operation immediately and shut the system down.

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⚠ CAUTION	For BESS with long downtime, please perform a comprehensive and meticulous inspection to ensure all system indicators meet operation requirements before powering up.

5.11.1 Checks before Powering On

Before powering up, please double-check the following items:

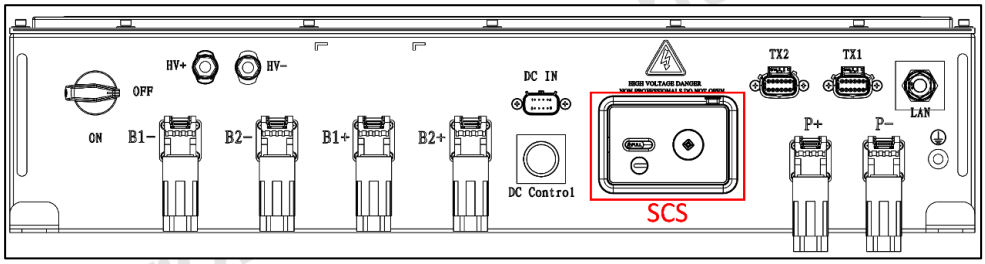
- Verify that the wiring is correct.
- Confirm that the main circuit wiring is securely tightened.
- Ensure that the protective cover in the equipment is firmly installed.
- Confirm that the emergency stop button is released.
- Check and ensure that there is no ground fault.
- Check with a multimeter and ensure that the AC and DC side voltages meet the startup conditions and confirm that there is no overvoltage risk.
- Check and ensure that there are no tools or parts left in the unit.
- Verify that all air inlets and outlets are free of foreign objects or blockages.

5.11.2 Process of Power On-Battery Power Version

Steps for powering on:

Step 1 Turn on the load switch (SCS) of the PDU (1#~6#) to "ON".

Figure 5-23 Position of SCS

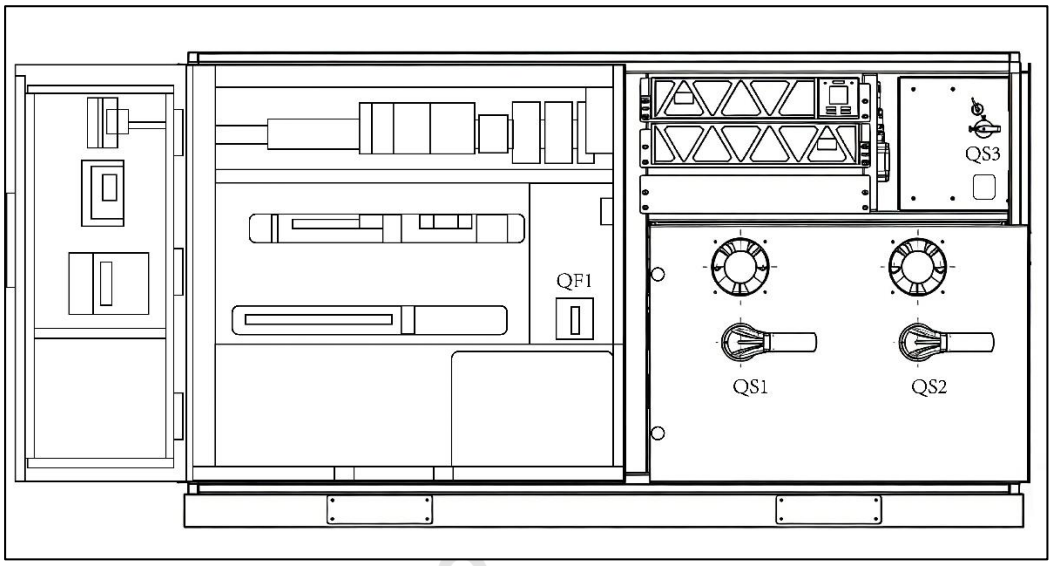


Step 2 Turn on the AC power supply main switch QF1 and the DC standby power supply switch QS3.

Step 3 Turn on the DC output main switches QS1 and QS2.

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Figure 5-24 Position of QS1 and QS2



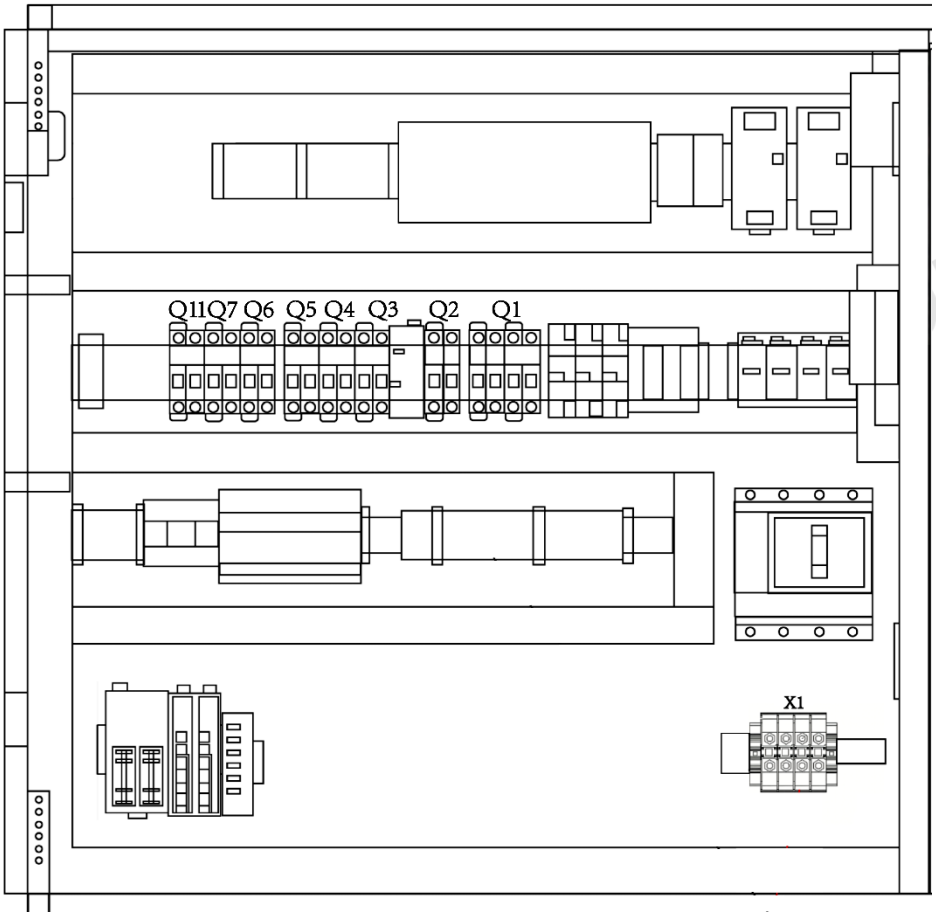
Step 4 Turn on the power supply switch of the equipment in sequence:(Push it up)

- Turn on the power supply switch Q1 of the LCS.
- Turn on the power supply switch Q2 of the Heat Exchanger and Fan.
- Turn on the power supply switch Q3 of the Socket.
- Turn on the power supply switch Q4 of the Dehumidifier.
- Turn on the power supply switch Q6 of the Ventilation System.
- Turn on the power supply switch Q7 of the SMPS.
- Turn on the power supply switch Q11 of the FSS.

The positions of Q1~Q11 is shown as below.

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Figure 5-25 Position of Q1 to Q11



The Power on operation is completed.

5.11.3 Process of Power Off-Battery Power Version

Steps for powering off:

Step 1 Turn off the power supply switch of the equipment in sequence.

- Turn off the power supply switch Q1 of the LCS.
- Turn off the power supply switch Q2 of the Heat Exchanger and Fan.
- Turn off the power supply switch Q3 of the Socket.
- Turn off the power supply switch Q4 of the Dehumidifier.
- Turn off the power supply switch Q6 of the Ventilation System.
- Turn off the power supply switch Q7 of the SMPS.
- Turn off the power supply switch Q11 of the FSS.

Step 2 Turn off the AC power supply main switch QF1 and the DC standby power supply switch QS3.

Step 3 Turn off the DC output main switches QS1 and QS2.

Step 4 Turn off the load switch (SCS) of the PDU (1#~6#) of the rack to "OFF"

The Power-off operation is completed.

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5.11.4 Process of Power On-UPS Version

> **NOTE** The positions of various switches are the same as above.

Steps for powering on

Step 1 Turn on the load switch (SCS) of the PDU (1#~6#) of the rack to "ON".

Step 2 Turn on the AC power supply main switch QF1.

Step 3 Turn on the DC output main switches QS1 and QS2.

Step 4 Turn on the power supply switch of the equipment in sequence:

- Turn on the power supply switch Q1 of the LCS.
- Turn on the power supply switch Q2 of the Heat Exchanger and Fan.
- Turn on the power supply switch Q3 of the Socket.
- Turn on the power supply switch Q4 of the Dehumidifier.
- Turn on the power supply switch Q5 of the UPS and turn on the UPS.
- Turn on the power supply switch Q6 of the Ventilation System.
- Turn on the power supply switch Q7 of the SMPS.
- Turn on the power supply switch Q11 of the FSS.

The Power on operation is completed.

5.11.5 Process of Power Off-UPS Version

Steps for powering off

Step 1 Turn off the power supply switch of the equipment in sequence.

- Turn off the power supply switch Q1 of the LCS.
- Turn off the power supply switch Q2 of the Heat Exchanger and Fan.
- Turn off the power supply switch Q3 of the Socket.
- Turn off the power supply switch Q4 of the Dehumidifier.
- Turn off the power supply switch Q5 of the UPS and turn off the UPS.
- Turn off the power supply switch Q6 of the Ventilation System.
- Turn off the power supply switch Q7 of the SMPS.
- Turn off the power supply switch Q11 of the FSS.

Step 2 Turn off the AC power supply main switch QF1.

Step 3 Turn off the DC output main switches QS1 and QS2.

Step 4 Turn off the load switch (SCS) of the PDU (1#~6#) of the rack to "OFF"

The Power-off operation is completed.

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6 Emergency Operation and Accident Handling

The faults of the ESS primarily include battery faults, control system faults, equipment faults and physical damage. For common faults, please refer to the maintenance manual. This chapter will address the emergency faults, such as fire and electric leakage.

If there is any uncertainty about identification or handling of abnormal conditions, please contact the relevant technical or after-sales service department of the Company for professional technical support.

When contacting us, please provide the following information, which will help you root the cause and clear the fault quickly and accurately:

- Equipment serial number, production date and software version
- Fault information and brief description
- Photos of faulty parts
- Information about the manufacturer, model, and configuration of the external devices connected to the energy storage system (e.g., transformers, upstream circuit breakers, and cable models).

6.1 Emergency Handling

If any of the following conditions (including but not limited to) occur, immediately stop using the ESS container and the adjacent battery container to prevent electric shock, fire, or explosion.

6.1.1 In Case of Fire

In case of electrical equipment catches fire, please immediately disconnect the power supply and use dry powder or carbon dioxide fire extinguishers to extinguish the fire.

If there is an open flame outside the ESS container, please immediately evacuate from the ESS system and contact the local fire emergency services for assistance.

After emergency troubleshooting, stand and observe for a specific period. If no abnormality occurs, proceed to the next step of troubleshooting.

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6.1.2 Typical ESS Emergencies and Anomalies

- Abnormal sound, spark, odor or other irregularities occurs during the operation or maintenance.
- Smoke or fire in the container.
- Expansion, rupture, melting, deformation of the battery pack.
- External or internal short circuit.
- Extreme weather conditions, such as rainstorms and flash floods.
- Thermal runaway occurs during use, indicated by a battery temperature exceeding 85°C.
- Structural damage that has not yet caused a severe accident.
- The coolant of the chiller leaks.
- The auxiliary power supply is abnormal or the communication is interrupted.
- Damage to the main structure of the battery container or IP protection of the battery system.
- Loose or burnt marks on electrical connections.
- Insulation fault alarm. If triggered, stop using immediately and verify whether the alarm is false.

If there are any of the above situations occur (include but not limited to the above case), which may result in the ESS dysfunction, please contact professional maintenance team or technical support immediately of the Company.

If the situation is critical, please activate the contingency plan (refer to the maintenance manual), and notify first responders or fire services.

NOTE	Before troubleshooting faults, replacing spare parts, or upgrading software, ensure that the DC and AC power supplies are disconnected. Use a multimeter to check that all devices are powered off. Ensure that the devices under maintenance cannot unexpectedly power on.

6.2 Emergency Operations

6.2.1 Power-off Process

- Power off via remote control

Step 1 The BAMU sends a shutdown command to BMS after receiving the shutdown command from EMS.

Step 2 Once it is powered off, the system will restore to the initial state.

Step 3 The main circuit of the ESS container will be disconnected, which will depressurize the high voltage in the main circuit. The RUN indicator on the cabinet door is turned off after the system is powered off.

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- Power off via local control

Step 1 Open the door of the electrical compartment of the container, and click the shutdown command on the touch screen of the SCDC.

Step 2 Confirm that the contactor of the main circuit on the touch screen has been disconnected.

Step 3 The RUN indicator on the cabinet door is turned off after the system is powered off.

6.2.2 Emergency Shutdown Operation Process

Step 1 Press the E-stop button on the ESS container or SCDC. (Refer to the ESS container appearance in Section 3.2 and the SCDC appearance in Section 3.9 for details).

Step 2 When the SCDC is in the remote-control mode, the control commands can be issued by the EMS. Please refer to related operation steps in the EMS manual.

> **NOTE** Manual emergency stop shall not be performed in case of fire.

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7 Human Machine Interface

This section organizes and elaborates on the touchscreen program structure of the ESS control system. It includes primary data display and essential function operation to assist developers and O&M personnel in understanding the system. The goal is to facilitate more effective system maintenance and operation.

7.1 Touchscreen Content and Structure

- The ESS touchscreen program consists of six key sections:
- System operation information
- Alarm information
- Event recording
- System diagram
- User management

7.2 System Settings of Touchscreen Program

The touchscreen provides information about the operation of the main equipment of the energy storage control system and facilitates the issuance of control instructions to the system. The connection between the ESS controller and the touchscreen is established via a network cable. The ESS controller transmits the real-time information of each equipment to the touchscreen; the touchscreen sends control instructions to the ESS controller, which then communicates them to the appropriate equipment.

7.3 Basic Environment Requirements for Touchscreen

Table 7-1 Equipment Environment Requirements

Requirement Items	Parameters
Altitude (m)	≤3000m
Installation Location	Inside an outdoor

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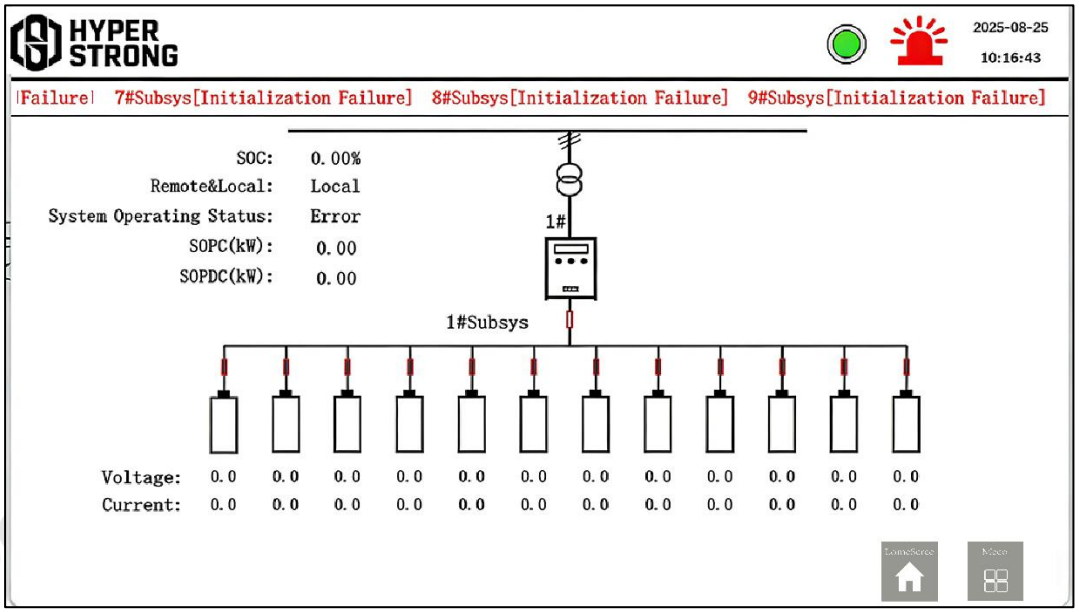
		container
Operating Ambient Temperature		-20~45℃
Seismic Resistance	Horizontal acceleration	0.2
	Vertical acceleration	0.1
Relative Air Humidity		The average daily relative humidity should be ≤ 95%. The average monthly relative humidity should be ≤ 90%. No condensation on the equipment.

7.4 Touchscreen Basic Function

7.4.1 Touchscreen Main Page

The main page of the touchscreen serves as the central hub for system operation and monitoring. It provides:

- A real-time overview of the system's performance.
- Quick access to critical features such as alarms, events, and system configurations.
- Easy navigation to other sections for detailed information or adjustments.



Once the program is in operation, the first screen displayed is the main page, which provides key information about the ESS control system. This information includes

- Subsystem SOC
- Remote local state
- Working state
- Maximum charging power and discharging power

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- Voltage and current of the rack.

The topology diagram, centrally located on the main interface, visually represents the control structure of the entire ESS. Key elements of the topology diagram include: SCDC and relay switches state:

The topology diagram, centrally located on the main interface, visually represents the control structure of the entire ESS. Key elements of the topology diagram include: SCDC and relay switches state:

- When the interface icon is shows  , it indicates that at least one relay among the main positive and main negative relays of the battery cluster is disconnected.
- When the icon changes to  , it indicates that both the main positive and main negative relays are closed.
- When all the relay switches of the subsystem are closed. the corresponding busbar will be closed, and the icon  will change to 
- The icon  indicates the basic state of the SOC of the rack,

At the top right of the main interface, the following additional information is displayed:

- Communication state of the PLC,
- Alarm display of the ESS
- Current date and time

A real-time alarm scrolling display of the ESS is located in the centre of the main interface to ensure that critical alerts are prominently visible.

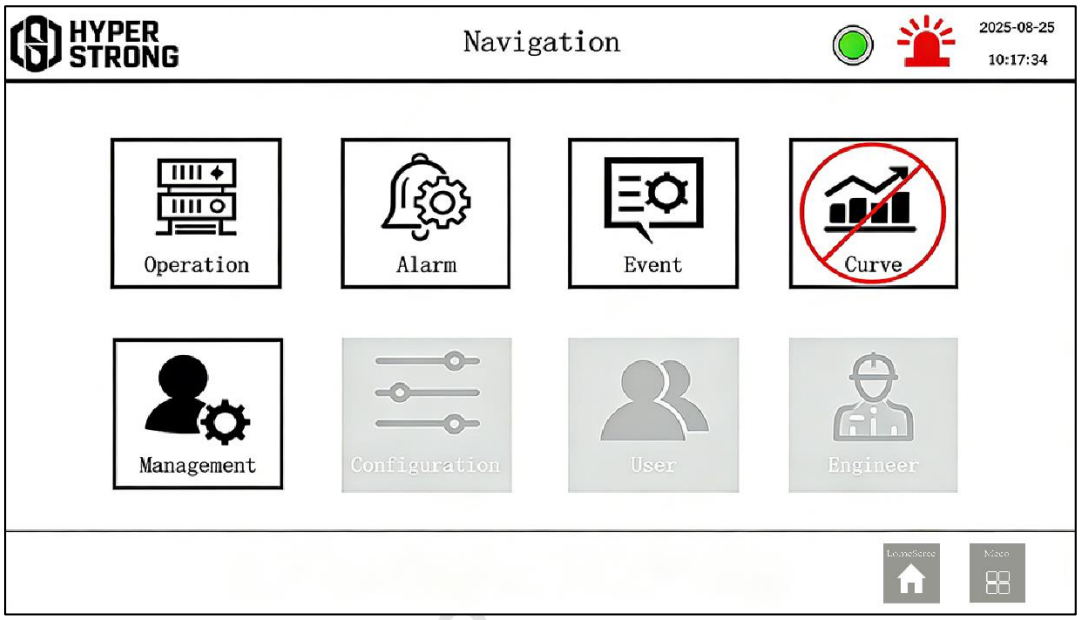
7.4.2 Navigation

At the bottom right corner of the main interface, users can switch between the Home Screen and Navigation Screen using 

or 

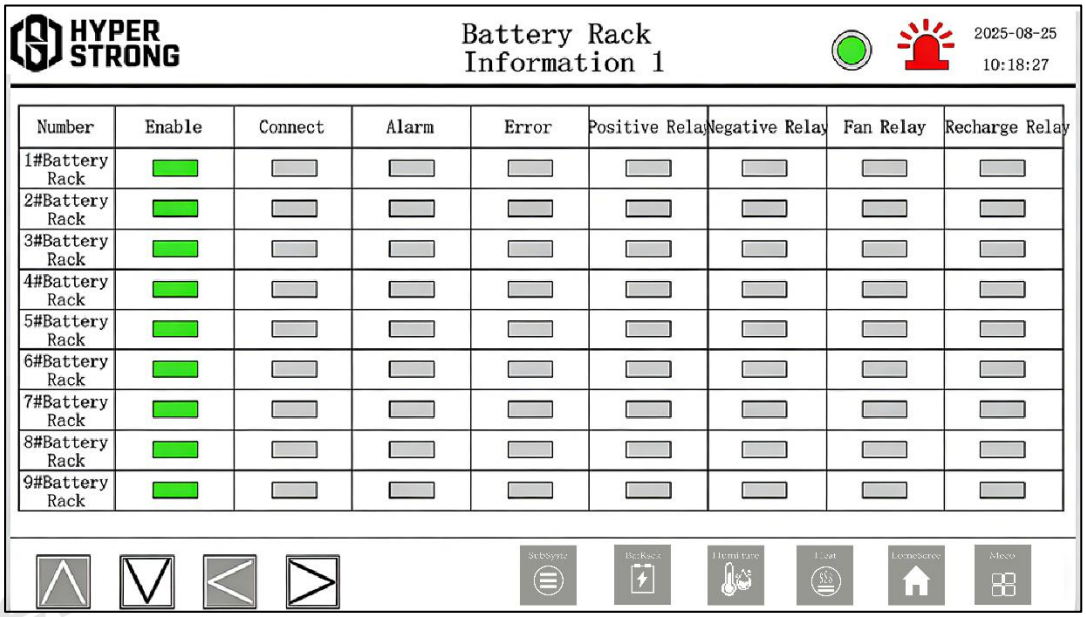
buttons.

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7.4.3 Rack Information Interface

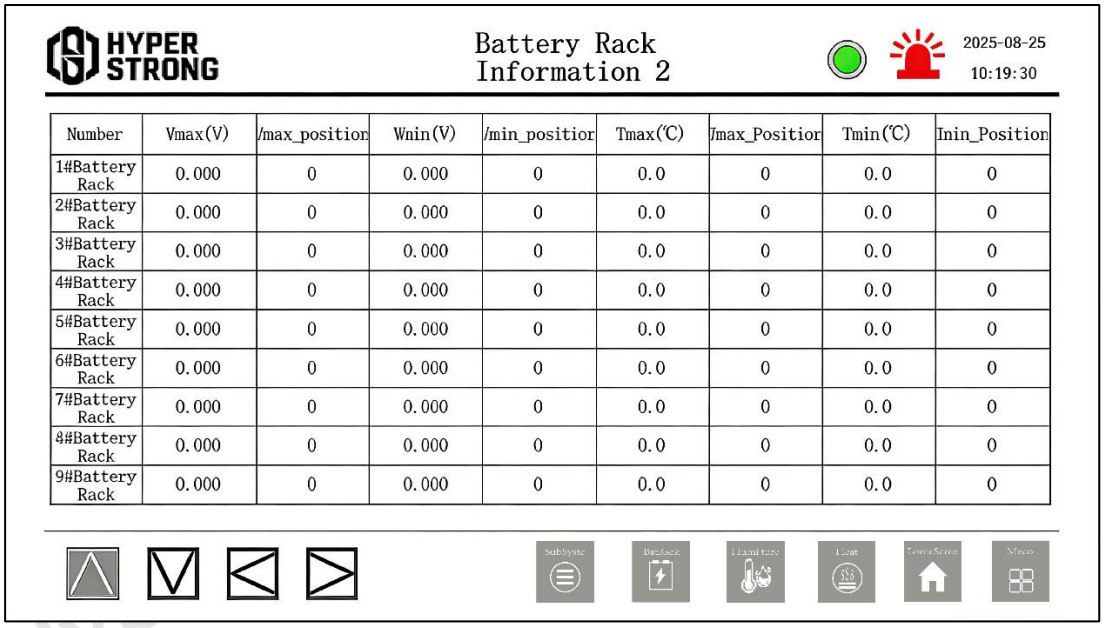
When the user clicks on "Operation" from the main page, the system transitions to the Rack Information Interface. This interface displays detailed operational data for each rack in the BCMU.



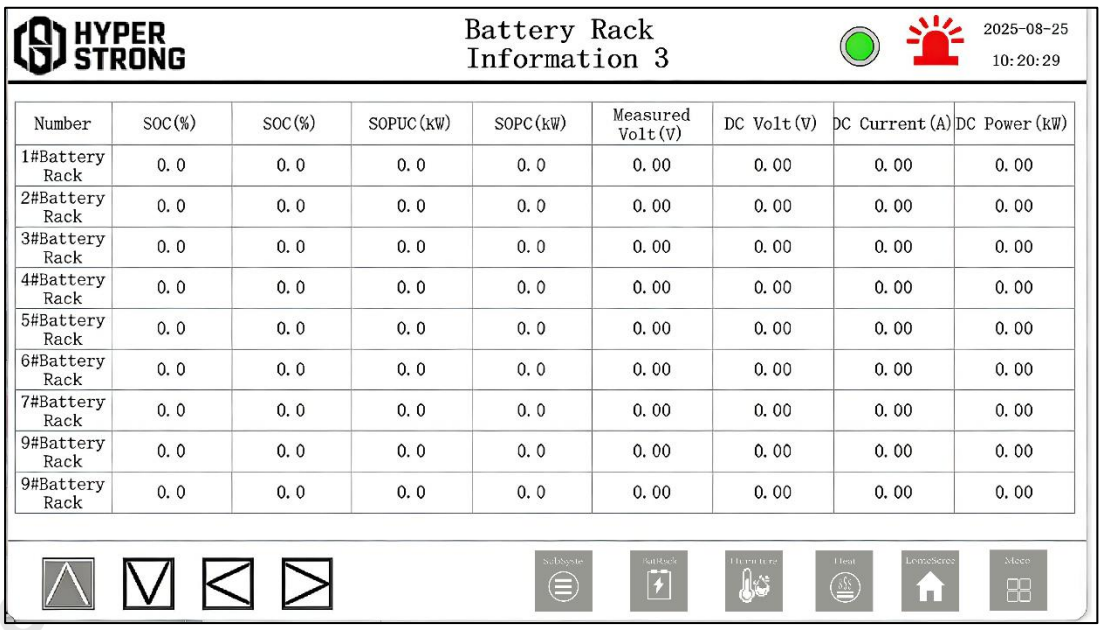
Entering the rack information interface, operators will see the current status of the racks, including “Enable”, “Connect”, “Alarm”, “Error”, “Positive Relay”, “Negative Relay”, “Fan Relay”, and “Pre-charge Relay”.

Click on >, it will enter Battery Cluster Information 2, Battery Cluster Information 3, etc

Click  to return to the return to the previous page. Operators can click  and  to enter any interface to check the operation information of each cluster of batteries



Number	Vmax(V)	/max_position	Vmin(V)	/min_position	Tmax(℃)	Tmax_Position	Tmin(℃)	Tmin_Position
1#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
2#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
3#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
4#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
5#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
6#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
7#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
8#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0
9#Battery Rack	0.000	0	0.000	0	0.0	0	0.0	0

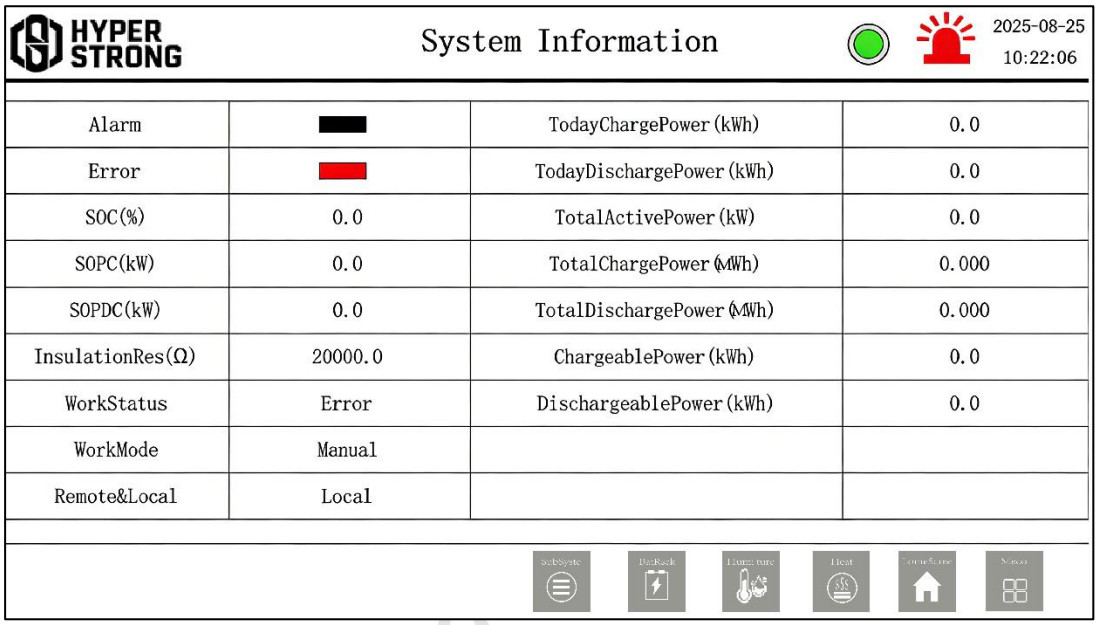


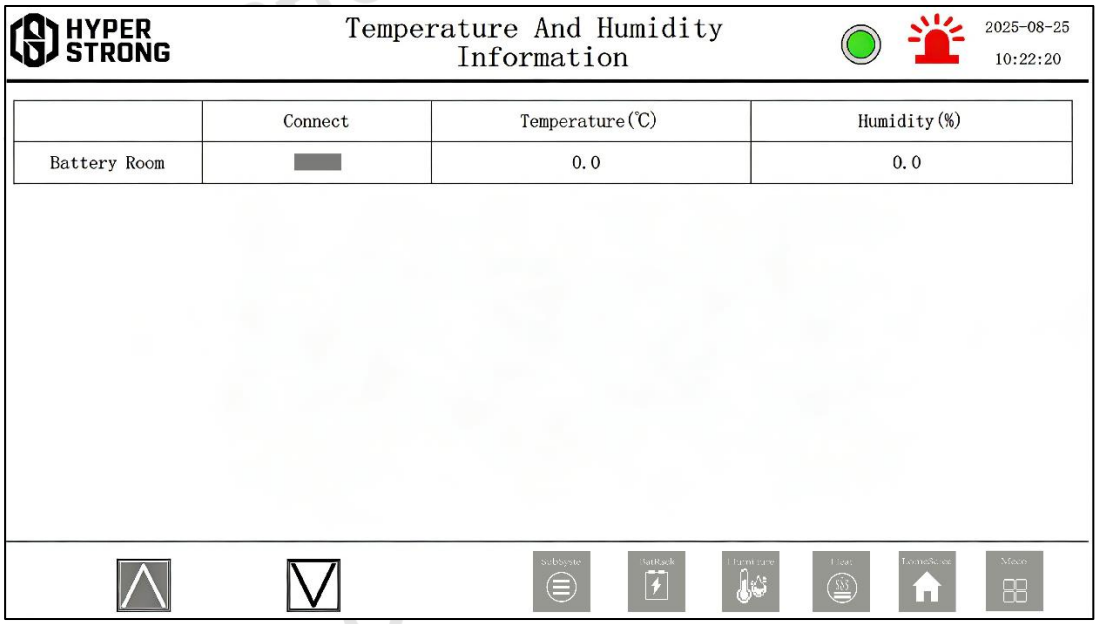
Number	SOC(%)	SOC(%)	SOPUC(kW)	SOPC(kW)	Measured Volt(V)	DC Volt(V)	DC Current(A)	DC Power(kW)
1#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
2#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
3#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
4#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
5#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
6#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
7#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
9#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00
9#Battery Rack	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.00

Click 

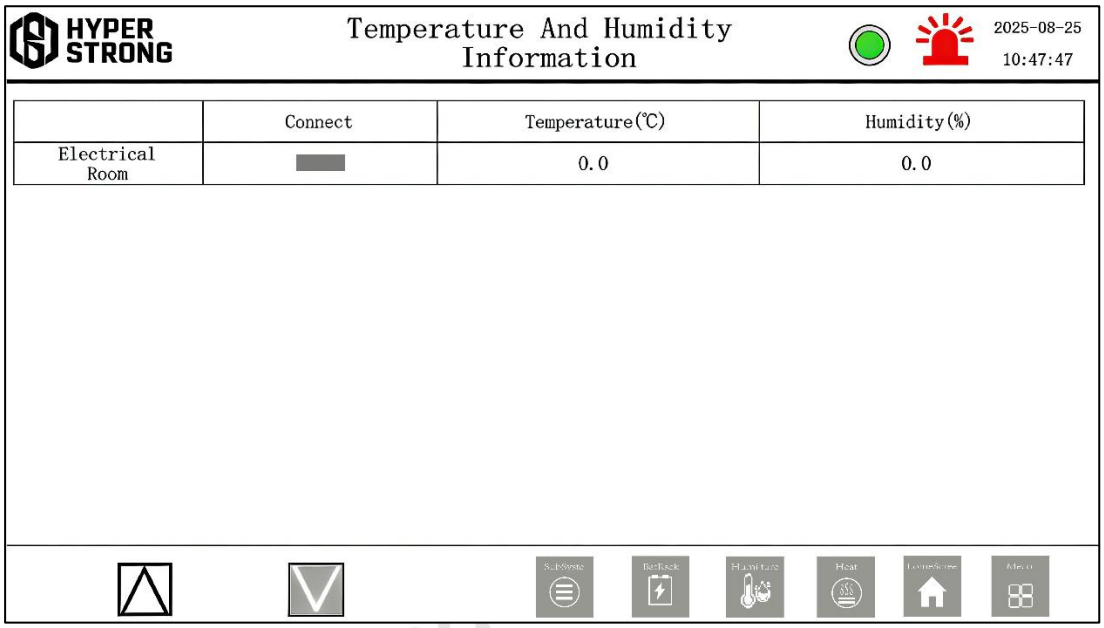
under the battery cluster information will navigate to the System Information Interface; while clicking 

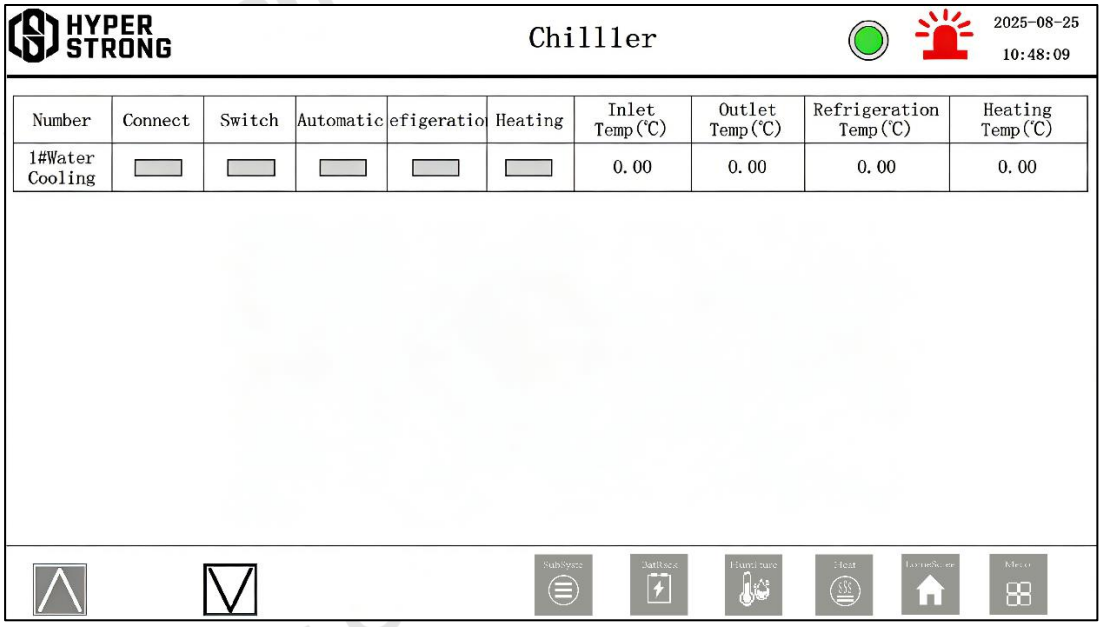
will lead to the Temperature & Humidity Information Interface. Operators can also click to review the operation information of each part of the BMS.



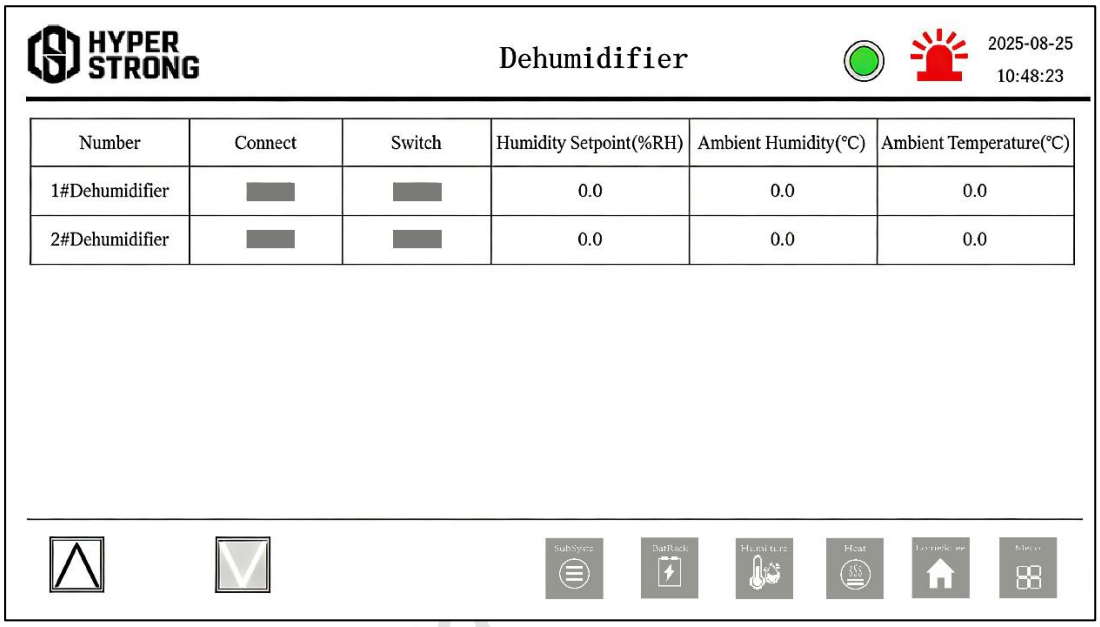


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7.4.4 Alarm Interface

In the main interface or other interfaces, click 

to enter the menu. Click “Alarm” to enter the real-time alarm interface and check real-time alarm. Click 

to enter the historical alarm information interface and check historical data. Click 

to enter the real-time alarm information interface and check real-time alarms.

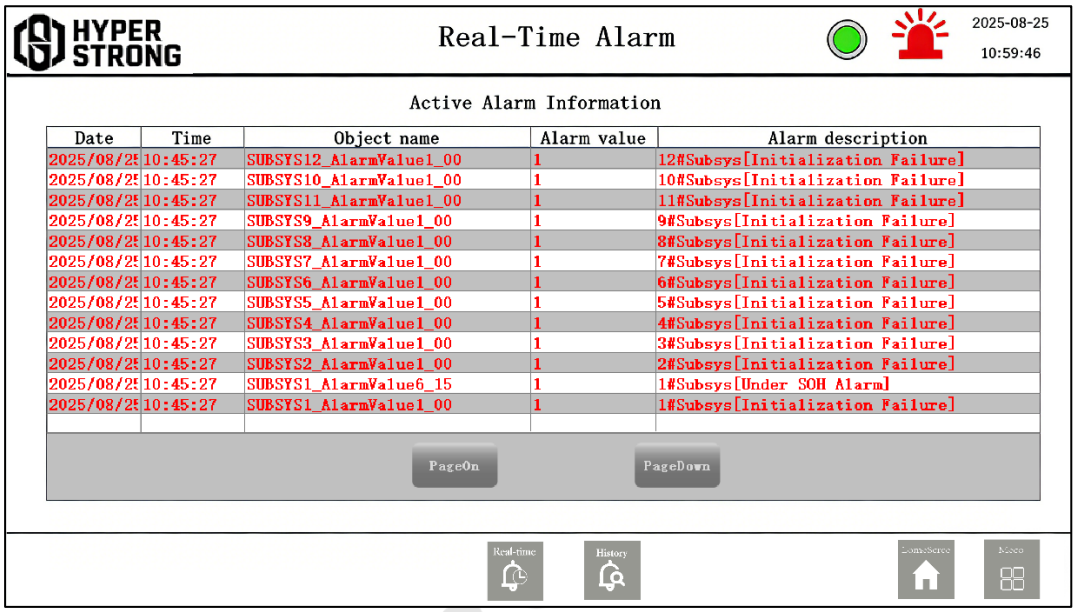
The main alarm interface information consists of two sections:

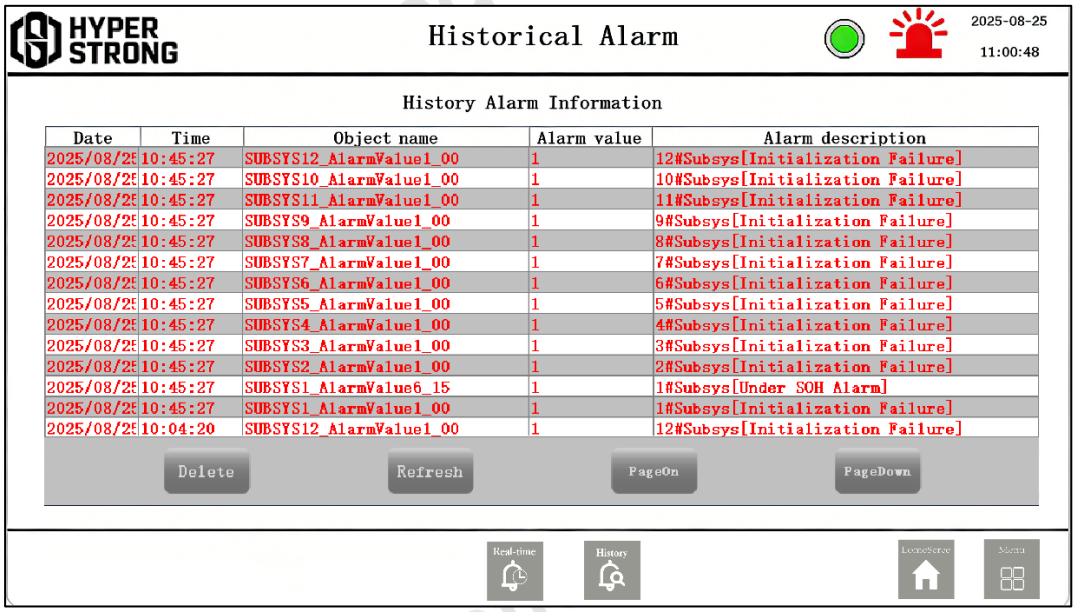
- Real-time and a historical alarm information display (the alarm information in the picture is only for reference).
- Historical Alarm information Display

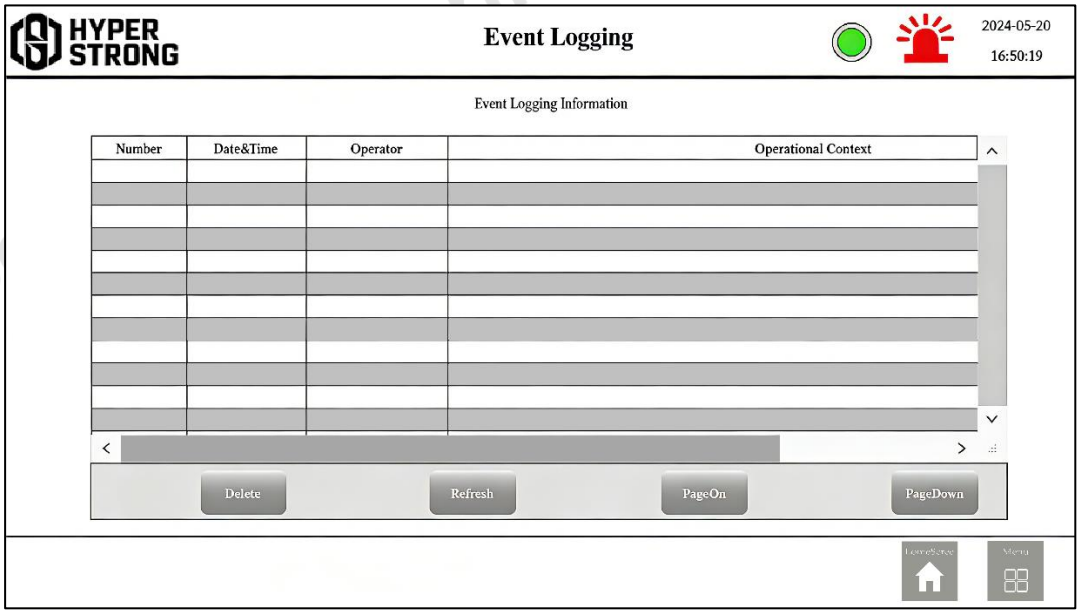
The real-time alarm interface can show up to 14 alarm messages at a time. Users can navigate through additional alarm messages by flipping up or down the page.

The historical alarm interface can also display up to 14 alarm messages at a time. Operator can perform operations such as deleting historical alarm messages, refreshing historical alarm messages, and navigating through messages by flipping up or down.

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Real-Time Alarm
2025-08-25
10:59:46
Active Alarm Information
Date Time Object name Alarm value Alarm description
2025/08/25 10:45:27 SUBSYS12_AlarmValue1_00 1 12#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS10_AlarmValue1_00 1 10#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS11_AlarmValue1_00 1 11#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS9_AlarmValue1_00 1 9#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS8_AlarmValue1_00 1 8#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS7_AlarmValue1_00 1 7#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS6_AlarmValue1_00 1 6#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS5_AlarmValue1_00 1 5#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS4_AlarmValue1_00 1 4#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS3_AlarmValue1_00 1 3#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS2_AlarmValue1_00 1 2#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS1_AlarmValue6_15 1 1#Subsys[Under SOH Alarm]
2025/08/25 10:45:27 SUBSYS1_AlarmValue1_00 1 1#Subsys[Initialization Failure]
PageOn
PageDown
Real-time
History


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Historical Alarm
2025-08-25
11:00:48
History Alarm Information
Date Time Object name Alarm value Alarm description
2025/08/25 10:45:27 SUBSYS12_AlarmValue1_00 1 12#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS10_AlarmValue1_00 1 10#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS11_AlarmValue1_00 1 11#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS9_AlarmValue1_00 1 9#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS8_AlarmValue1_00 1 8#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS7_AlarmValue1_00 1 7#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS6_AlarmValue1_00 1 6#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS5_AlarmValue1_00 1 5#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS4_AlarmValue1_00 1 4#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS3_AlarmValue1_00 1 3#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS2_AlarmValue1_00 1 2#Subsys[Initialization Failure]
2025/08/25 10:45:27 SUBSYS1_AlarmValue6_15 1 1#Subsys[Under SOH Alarm]
2025/08/25 10:45:27 SUBSYS1_AlarmValue1_00 1 1#Subsys[Initialization Failure]
2025/08/25 10:04:20 SUBSYS12_AlarmValue1_00 1 12#Subsys[Initialization Failure]
Delete
Refresh
PageOn
PageDown
Real-time
History


HYPER STRONG
Event Logging
2024-05-20
16:50:19
Event Logging Information
Number Date&Time Operator Operational Context
Delete
Refresh
PageOn
PageDown

7.4.5 Event Logging Interface

Click 

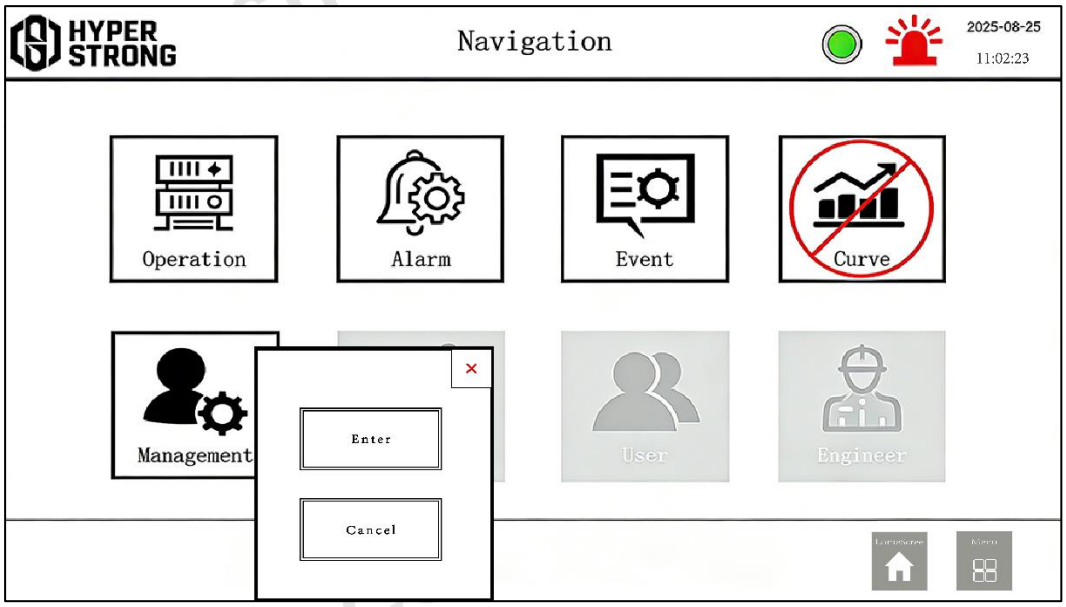
on the main page or other pages to access the Menu. Click “Event Logging” to enter Event Logging section and review operation records.

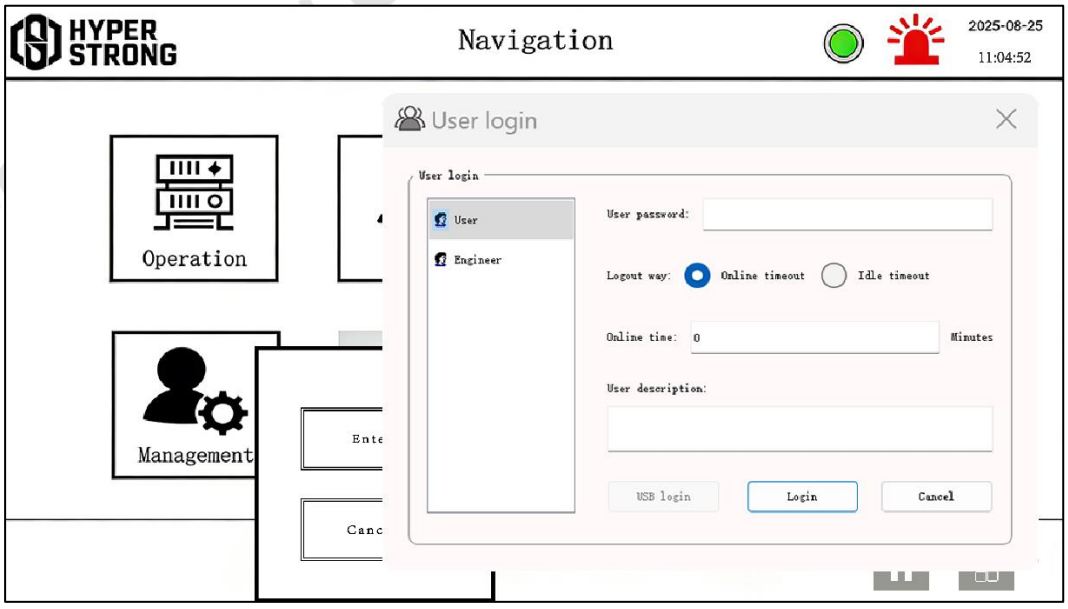
The event logging interface includes the Date & Time, Operator, and Operational Context. Additionally, operators have the option to delete records, refresh the page, and navigate up and down the page.

7.4.6 User Management Interface

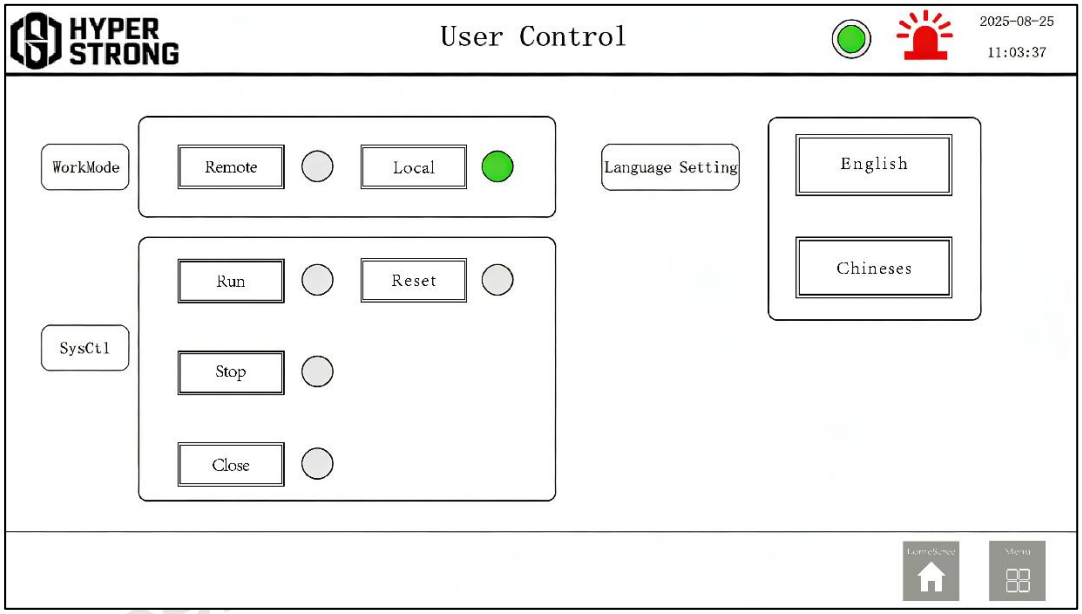
Click 

, enter the Menu page and click “User Management,” and the Login interface will pop up in the main interface or other interfaces. Select “User” or “Engineer”, enter the password, and click “Login” to complete the login. If you log in as an engineer, the “Engineer Operation” button in the menu interface becomes operational.





Click on “User” to enter User Control. This interface's main functions include subsystem run/shutdown/reset control, remote/local mode switching, and subsystem selection/ operation. (Contact the HyperStrong’s operation and maintenance personnel to obtain the password.)



The user can select the mode based on the situation and click Run depending on the operation condition; then, the corresponding subsystem will turn to the running state. Click Stop, the corresponding subsystem will be shut down. Click Close the corresponding subsystem will be turn off. If a subsystem enters a fault-recoverable state and needs to reset, click Reset, and the system reset instruction will be successfully issued.

7.5 Communications

The HMI has a network cable interface and connects to BAMU-PLC via a network cable. Modbus TCP/IP transmission protocol is used for data transmission.

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8 Appendix

8.1 Product Labels

The figures below are for reference only. Some are examples for illustration purposes. Refer to the actual product for details.

	ES Label. A circular green reflector sign with a minimum diameter of 100mm, featuring the letters “ES” on
	Restricted Access Sign
	Typical Restricted Access Sign
	Explosion Hazard Label Minimum dimensions of 175mm*175mm
	Toxic Fumes Hazard Label
	Shutdown Procedure Notice Label

Disconnection devices shall be labelled with an identification name or number as specified in the BESS wiring diagram, consistent with the shutdown procedure.
The minimum size of this sign shall ensure readability from a distance of 1m.

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All switches shall clearly and reliably indicate the isolating position of the device. The symbols “0" (off) and “I"(on) fulfil this requirement.
The isolation switch model: MVS 25 DA1 A4F2.

8.2 Contact Information

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Email:	support@hyperstrong.com
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Address:	Max-Holder-Str. 19 60437 Frankfurt am Main Germany
Email:	support@hyperstrong.com
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Email:	Service_HSNA@hyperstrong.com

8.3 Acronyms and Abbreviations

B	
BAMS	Battery Array Management System
BCMS	Battery Cluster Management System
BESS	Battery Energy Storage System
BMU	Battery Management Unit
E	
EMS	Energy Management System
ESS	Energy Storage System

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F	
FSS	Fire Suppression System
H	
HMI	Human Machine Interface
HVAC	Heating, Ventilating and Air Conditioning
P	
PACK	Battery Pack
PCS	Power Conversion System
PDU	Power Distribution Unit
R	
RACK	Battery Rack
S	
SCADA	Supervisory Control and Data Acquisition System
SCDC	System Control and Distribution Cabinet
SMPS	Switching Mode Power Supply
SOC	State Of Charge
SOE	State Of Energy
SOH	State Of Health

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